Euler–Gompertz and Bessel Values: Joint Independence, Integral Intersections, and Critical Limits

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Abstract

We develop joint independence and exact integral-intersection theorems for Euler–Gompertz, exponential-integral, and Bessel values. The functional foundation includes every depth of the Bessel primitive tower, at all nonzero algebraic scales modulo sign, jointly with the entire exponential-integral towers. Continued phase embeddings and local pole charts classify full rational fields even when an exponential-integral argument and a Bessel scale have dependent phases: the integral intersection is the polynomial ring in the entire integral, with Laurent polynomial coefficients in the matching exponential. Arbitrarily many positive derivative-ratio scales can also be retained with one harmonic Bessel coordinate. Complex independent-scale intersections yield nonintegrality for Mahler combinations. Further consequences include exact unit and radical criteria, a conjugation half-defect theorem, and a common density-one set of integer arguments on which six coordinate families are jointly independent; every integer-order second-kind Bessel value at those arguments is transcendental. Uniform critical asymptotics and explicit reconstruction results complete the synthesis. The arithmetic nature of Euler's and Euler–Gompertz's constants, and full independence of the original four constants, remain unresolved.

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1 Introduction and common notation

The starting point is the four-constant field

$$H_4 = k(\gamma, \delta, \kappa, C), \qquad 3 \le \operatorname{trdeg}_k H_4 \le 4, \qquad k = \overline{\mathbb{Q}}. \tag{1}$$

Here γ is Euler's constant and

$$\delta = \int_0^\infty \frac{e^{-u}}{1+u}\,du, \qquad I = I_0(2), \quad K = K_0(2), \quad \kappa = K/I, \quad C = I_1(2)/I.$$

The lower bound in (1) is proved below, not assumed as a consequence of separate transcendence statements. Algebraic independence always means the absence of a nonzero polynomial relation over the stated coefficient field. In particular, a lower bound of three does not identify a prescribed independent triple and does not decide the individual arithmetic nature of γ or δ.

1.1 Functions, normalizations, and coefficient rings

For integers $r \ge 1$, define the entire functions

$$\operatorname{Ein}_r(z) = \sum_{n\ge1} \frac{(-1)^{n-1} z^n}{n^r n!}, \qquad \operatorname{Ein} = \operatorname{Ein}_1, \qquad \eta = \operatorname{Ein}(1) = \gamma + \delta/e.$$

The classical tail integral is denoted by $E_1^{\mathrm{tail}}(z)$; on compatible branches, $\operatorname{Ein}(z) = \gamma + \log z + E_1^{\mathrm{tail}}(z)$. Thus Ein_1 and E_1^{tail} are different functions. We use the standard modified Bessel functions I_ν, K_ν and the classical Bessel functions J_ν, Y_ν. Define the harmonic series

$$Y(t) = \sum_{n\ge0} \frac{H_n (t/2)^{2n}}{(n!)^2}, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}, \quad H_0 = 0. \tag{2}$$

This Y is not the classical second-kind Bessel function Y_0. For nonzero algebraic t, put

$$\beta(t) = \frac{Y(t)}{I_0(t)}, \qquad C(t) = \frac{I_1(t)}{I_0(t)}.$$

The nonvanishing needed in these definitions is proved before the complex joint-value theorem. At positive real t, the connection and Wronskian identities are

$$\beta(t) = \gamma + \mu(t), \qquad \mu(t) = \log(t/2) + \frac{K_0(t)}{I_0(t)}, \tag{3}$$

$$I_0(t)K_1(t) + I_1(t)K_0(t) = \frac{1}{t}. \tag{4}$$

We abbreviate $\beta = \beta(2) = \gamma + \kappa$. The following three combinations have an exact relation that will be retained throughout:

$$A = \eta - \beta = \delta/e - \kappa, \qquad U = 3\eta - 2\beta, \qquad V = \beta, \qquad U = 3A + V. \tag{5}$$

At the imaginary scale $2i$, set

$$\nu = -\frac{\pi Y_0(2)}{2J_0(2)}, \qquad C_J = \frac{J_1(2)}{J_0(2)}; \qquad \beta(2i) = \gamma + \nu, \quad C(2i) = iC_J.$$

The branch-free series definitions make these signs unambiguous. The primitive coordinate and three distinct limits are

$$B = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n(n!)^2}, \qquad \mathcal{L} = B - \gamma,$$
$$L_+ = \gamma + 2\kappa, \qquad L_- = \ell = \gamma - 2\kappa.$$

In particular $\mathcal{L} \neq L_+$; their separate critical approximation formulas are proved in Section 32.

Write $K_{\exp} = k(e^\alpha : \alpha \in k)$. Every assertion over this infinite field means its corresponding assertion for each finite set of coefficients. The ring $\mathbf{E}$ consists of values of E-functions at algebraic points, and

$$\overline{\mathbf{E}} = \{z \in \mathbb{C} : z \text{ satisfies a monic polynomial over } \mathbf{E}\}.$$

This is an integral closure in $\mathbb{C}$, not the algebraic closure of $\mathrm{Frac}(\mathbf{E})$. Since $k \subset \mathbf{E}$, exclusion from $\overline{\mathbf{E}}$ implies transcendence over k. The proof of exclusion is stronger than such an individual transcendence assertion.

1.2 The closures proved by the synthesis

The five predecessor notes are consolidated by subject, with the functional inputs proved first and repeated arguments replaced by internal references. The additional results proved in this integrated manuscript are the following.

1. At every depth and every nonzero algebraic scale modulo sign, Bessel primitives can be adjoined jointly to the full exponential-integral and Bessel family over $K_{\exp}$ (Theorem 5.3). The conjugation principle also bounds the independence defect after a real coefficient-field extension of transcendence degree d by $\lfloor d/2 \rfloor$ (Theorem 17.1).

2. At arbitrary positive algebraic a, t,

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(t), C(t)) = k[e^a, e^{-a}, \mathrm{Ein}(a)]$$

   (Theorem 27.1). In particular, the previously open full matched-phase intersection at $(a, t) = (1, 2)$ is closed. Theorem 29.1 strengthens this to the entire derivative family:

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(s), C(t) : t > 0 \text{ algebraic}) = k[e^a, e^{-a}, \mathrm{Ein}(a)]$$

   for arbitrary positive algebraic a, s. This includes dependent scales such as 2 and 4 and arbitrary rational expressions in the generators.

3. At complex $\mathbb{Q}$-linearly independent algebraic scales, the full rational field in all $\beta(t_i), C(t_i)$ has integral intersection exactly k (Theorem 22.2). This includes the pair of scales a, ia and its gamma-free combinations.

4. At nonzero $\mathbb{Q}$-linearly independent algebraic arguments, the rational field in $e^{a_i}, \mathrm{Ein}(a_i)$ and additional independent exponential units, whose exponents are jointly $\mathbb{Q}$-linearly independent with the a_i, has exactly the Laurent–polynomial intersection of Theorem 23.1. The one-point consequence is

$$\overline{\mathbf{E}} \cap k(e, \gamma + \delta/e) = k[e, e^{-1}, \gamma + \delta/e].$$

5. A single set of at least $N - \pi(N) - 1$ integers in $[1, N]$ supports joint independence of all six coordinates in Theorem 35.4. At every retained integer, all integer-order K_r and πY_r values are transcendental over $K_{\exp}$. Every positive algebraic argument has a six-coordinate rank bound of at least five, without an exceptional argument.

6. If $\operatorname{trdeg}_k H_4 = 3$, for algebraic $r > 0$ with $r \neq 1$, the geometric family $K_0(2r^n)$ loses at most one coordinate over $K_{\exp}H_4$, and integer arguments have the improved lower bound $N - \pi(N)$. These are conditional strengthenings of the unconditional density-one theorem.

7. The exact intersections give unit and radical classifications (Section 31), including explicit integral units that are not themselves E-values. The upper-half-plane Ein_r family remains independent after adjoining γ, all real-argument Ein_r values, and all complex Bessel and primitive coordinates to the exponential coefficient field.

The distinction between pairwise distinct scales and $\mathbb{Q}$-linearly independent scales remains important. The joint-value theorem needs only distinctness modulo sign. Independent phase variables classify the general multiscale Bessel rational field; synchronized local charts remove that restriction for the one-harmonic-coordinate families identified above. Several harmonic coordinates at dependent scales remain a separate question.

1.3 External inputs and status

The analytic and differential-algebraic foundations are the classical Bessel differential Galois group, Picard–Vessiot theory, ordinary-point Siegel–Shidlovskii specialization [4, 5], and the phase-embedding construction of Fischler–Rivoal [3]. The latter construction and the continuation step used here are stated and justified explicitly in the integrality part. Standard connection identities are recorded in [2]. Mahler's classical one-point transcendence theorem [16] is part of the prior context. The predecessor notes and supplied corpus are research manuscripts; the present text proves extensions of their identified arguments and makes no claim of a comprehensive historical priority determination. The source concordance at the end records which results were merged, strengthened, or retained as separate applications.

Part I
Foundations

2 Polyexponential towers and their solvable closure

For integers $r \geq 1$, set

$$\mathrm{Ein}_r(z) := \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n^r n!}, \qquad \mathrm{Ein}_0(z) = 1 - e^{-z}.$$

Thus Ein_1 in this section denotes the entire complementary exponential integral, not the classical tail integral $\int_z^\infty e^{-u} du/u$. Write $D = d/dz$ and $\vartheta = zD$; then $\vartheta \,\mathrm{Ein}_r(\lambda z) = \mathrm{Ein}_{r-1}(\lambda z)$. For every fixed r and algebraic λ, this is an E-function: a power of a denominator of λ times $\operatorname{lcm}(1, \ldots, N)^r$ clears its first N normalized Taylor coefficients, with exponential growth in N.

We first prove the primitive-tower lemma and identify its solvable Picard–Vessiot closure. We then establish the full Bessel product group at complex scales, before applying the classical specialization theorem. The classical inputs are Picard–Vessiot theory and Siegel–Shidlovskii specialization [5, 4].

Lemma 2.1 (Primitive tower and its logarithm)**.** *Let $\lambda_1, \dots, \lambda_s$ be distinct nonzero algebraic numbers, let $\nu_1, \dots, \nu_d$ be a $\mathbb{Z}$-basis of their generated additive group, and put*

$$K_0 = \mathbb{C}(z)(e^{-\nu_1 z}, \dots, e^{-\nu_d z}), \qquad K_q = K_0(\mathrm{Ein}_r(\lambda_i z) : 1 \le i \le s,\ 1 \le r \le q).$$

For every $q \ge 0$ the constant field is $\mathbb{C}$, $1/z \notin D(K_q)$, and

$$\mathrm{trdeg}_{\mathbb{C}(z)} K_q = d + qs.$$

Moreover $\log z$ is transcendental over every K_q.

Proof. Write $X_j = e^{-\nu_j z}$ and $\lambda_i = m_i \cdot \nu$. The rationally independent exponential rates identify K_0 with the fraction field of the Laurent polynomial ring $\mathbb{C}(z)[X_1^{\pm 1}, \dots, X_d^{\pm 1}]$, with $DX_j = -\nu_j X_j$. There is no irreducible nonunit P in this ring with $P \mid DP$. Indeed, if $DP = AP$, comparison of Newton polytopes gives $A \in \mathbb{C}(z)$; comparison of two nonzero coefficients a_u, a_v of P would give

$$\frac{(a_u/a_v)'}{a_u/a_v} = (u - v) \cdot \nu.$$

A rational function has no nonzero constant logarithmic derivative, so P would have a single Laurent monomial and be a unit. Consequently differentiation of a genuine nonunit denominator raises its pole order and cannot yield a Laurent polynomial. If

$$Dg = z^{-1} \sum_u c_u X^u, \qquad g \in K_0, \quad c_u \in \mathbb{C},$$

then g is a Laurent polynomial. Coefficient comparison gives $a_u' - (u \cdot \nu) a_u = c_u/z$ with $a_u \in \mathbb{C}(z)$. For $u = 0$ a rational derivative has zero residue. For $u \ne 0$ a pole of a_u would create a higher-order pole, whereas absence of a pole at zero makes the left side regular there. Hence all $c_u = 0$. In particular $1/z$ is not exact in K_0, and no nonzero constant combination of $(1 - X^{m_i})/z$ is exact there.

We use the following elementary primitive-field observation. If $a_1, \dots, a_s \in F$ have no nonzero constant combination in $D(F)$, and algebraically independent primitives u_i satisfy $Du_i = a_i$, then

$$g \in F(u_1, \dots, u_s), \quad Dg \in F \quad \Longrightarrow \quad g = f + \sum_i b_i u_i \quad (f \in F,\ b_i \in \mathbb{C}).$$

To see this, a normalized nonconstant Darboux polynomial in the u_i would have derivative zero; its highest homogeneous part has constant coefficients, and comparison of the next degree would make a nonzero constant combination of the a_i exact in F. Thus no such polynomial exists and differentiation excludes nonconstant denominators in g. The same comparison of the top two homogeneous degrees of a polynomial g leaves degree at most one and proves the displayed assertion. The observation also shows that no new constants are introduced.

Kolchin–Ostrowski applied to $D\,\mathrm{Ein}_1(\lambda_i z) = (1 - X^{m_i})/z$ gives independence of the first primitive layer. The displayed observation and the Laurent argument show that $1/z$ is still not exact in K_1. Inductively, the last layer of K_q/K_{q-1} admits every translation $\mathrm{Ein}_q(\lambda_i z) \mapsto \mathrm{Ein}_q(\lambda_i z) + h_i$. If $Dg = \sum_i c_i \,\mathrm{Ein}_q(\lambda_i z)/z$ with $c \ne 0$ and $g \in K_q$, subtracting a translate of this identity would make a nonzero constant multiple of $1/z$ exact in K_q, a contradiction. Kolchin–Ostrowski therefore adds s independent primitives at the next layer. If $Dg = 1/z$ in K_{q+1}, the primitive-field observation writes $g = f + \sum_i b_i \,\mathrm{Ein}_{q+1}(\lambda_i z)$. Translating the preceding layer gives the same contradiction when $b \ne 0$; when $b = 0$ it contradicts the preceding layer directly. This proves the induction. All fields are realized as meromorphic functions on a fixed slit plane, so their constants are indeed $\mathbb{C}$.

Finally, a primitive algebraic over a differential field of characteristic zero lies in that field up to a constant: take the normalized field trace of the primitive. Applied to $D\log z = 1/z$, this would contradict the nonexactness just proved. Thus adjoining $\log z$ raises the transcendence degree by one. □

Lemma 2.2 (Explicit solvable Picard–Vessiot closure)**.** *For $R \geq 1$, the field $L = K_R(\log z)$ is a Picard–Vessiot extension of $\mathbb{C}(z)$ with solvable differential Galois group and*

$$\operatorname{trdeg}_{\mathbb{C}(z)} L = d + Rs + 1.$$

Proof. For each i, consider the order-$R+2$ operator

$$\mathcal{L}_i = (D + \lambda_i)D\vartheta^R.$$

A fundamental solution basis is

$$1, \log z, \dots, (\log z)^R, \operatorname{Ein}_R(\lambda_i z).$$

Indeed, $D\vartheta^R \operatorname{Ein}_R(\lambda_i z) = \lambda_i e^{-\lambda_i z}$, and $\operatorname{Ein}_R(\lambda_i z)$ cannot be a constant-coefficient polynomial in $\log z$, as its Rth ϑ-derivative is $1 - e^{-\lambda_i z}$. The entries of a fundamental matrix and their derivatives generate exactly

$$\mathbb{C}(z)\big(e^{-\lambda_i z}, \log z, \operatorname{Ein}_1(\lambda_i z), \dots, \operatorname{Ein}_R(\lambda_i z)\big).$$

Their compositum for all i is L, so it is the PV field of the direct sum of these operators.

Its Galois group preserves the flag of polynomials in $\log z$ and acts triangularly on each displayed solution basis. More explicitly, if $e^{-\lambda_i z} \mapsto c_i e^{-\lambda_i z}$, then

$$\operatorname{Ein}_R(\lambda_i z) \mapsto c_i \operatorname{Ein}_R(\lambda_i z) + (1 - c_i)\frac{(\log z)^R}{R!} + \text{a polynomial in } \log z \text{ of degree below } R.$$

Also $\log z \mapsto \log z + c$. Hence the group is a subgroup of a product of triangular groups and is solvable. Its dimension is the transcendence degree given by Lemma 2.1. □

3 Complex Bessel scales and joint specialization

Throughout this section $K = \mathbb{C}(z)$ and $D = d/dz$. All functions are realized as meromorphic functions on a fixed slit plane, with one choice of $\log z$; the constant field is $\mathbb{C}$. For $t \neq 0$ write

$$f_t(z) = I_0(tz), \qquad y_t(z) = Y(tz), \qquad Y(w) = \sum_{n \geq 0} \frac{H_n (w/2)^{2n}}{(n!)^2}, \quad H_0 = 0.$$

The scale t may be complex. Coefficient comparison gives

$$(D^2 + z^{-1}D - t^2)f_t = 0, \qquad (D^2 + z^{-1}D - t^2)y_t = 2f_t'/z.$$

Consequently

$$g_t(z) = y_t(z) - f_t(z)\log z \tag{6}$$

is a second homogeneous solution. Its Wronskian with f_t is

$$f_t g_t' - f_t' g_t = -1/z. \tag{7}$$

Indeed, Abel's identity makes the Wronskian a constant times $1/z$, and $f_t = 1 + O(z^2)$, $g_t = -\log z + O(z^2 \log z)$ determine the constant. This normalization avoids any branch convention for $K_0(tz)$ or $\log t$.

Lemma 3.1 (The complex-scale product group)**.** *Let $t_1, \dots, t_m \in \mathbb{C}^*$ satisfy $t_i \neq \pm t_j$ for $i \neq j$. The compositum of the solution fields of $D^2 + z^{-1}D - t_j^2$ has differential Galois group $\mathrm{SL}_2(\mathbb{C})^m$ over K.*

Proof. The individual order-zero Bessel group is SL_2 by the classical Bessel theorem [5], unchanged by the nonzero linear scale change. The full group G is therefore a subdirect algebraic subgroup of SL_2^m. Its identity component projects onto every factor: the image is connected and normal, with finite quotient in the connected group SL_2.

The Lie algebra of G° is a subdirect subalgebra of $\mathfrak{sl}_2^m$. Its solvable radical maps to a solvable ideal of each factor and is therefore zero. Each simple ideal maps either to zero or isomorphically onto a factor, and exactly one simple ideal maps onto each factor, since distinct ideals commute. If G is proper, a simple ideal must map onto two different factors. Their adjoint representations are then isomorphic as representations of the connected group G°. By the PV correspondence, the two adjoint differential modules become isomorphic over the finite algebraic fixed field of G°.

This is impossible for two scales $a \neq \pm b$. The rank-two system matrix is

$$A_t(z) = \begin{pmatrix} 0 & 1 \\ t^2 & -1/z \end{pmatrix}.$$

The adjoint connection has distinct leading eigenvalues $0, 2t, -2t$ at infinity. It can be formally diagonalized over $\mathbb{C}((z^{-1}))$: at each successive order, the off-diagonal coefficients are determined by division by nonzero eigenvalue differences. After a finite algebraic extension, choose a place over infinity and embed its completion in $\mathbb{C}((z^{-1/e}))$. An entry h of a differential intertwiner between the two formally diagonalized modules satisfies

$$h' = (\mu - \lambda + O(z^{-1}))h, \quad \lambda \in \{0, 2a, -2a\}, \quad \mu \in \{0, 2b, -2b\}.$$

For a nonzero Puiseux Laurent series h, its logarithmic derivative is $O(z^{-1})$. Thus a nonzero entry requires $\lambda = \mu$. An invertible intertwiner requires equality of the two eigenvalue multisets, forcing $a = \pm b$, a contradiction. Hence the group is the full product. □

Corollary 3.2 (Distinguished columns and nonvanishing)**.** *For nonzero algebraic $t_1, \dots, t_m$ distinct modulo sign, the $2m$ values $I_0(t_j), I_1(t_j)$ are algebraically independent over $\overline{\mathbb{Q}}$. In particular $I_0(t) \neq 0$ at every nonzero algebraic t, so $C(t) = I_1(t)/I_0(t)$ and $\beta(t) = Y(t)/I_0(t)$ are defined.*

Proof. The product group moves the first solution column of each rank-two fundamental matrix independently through all nonzero linear combinations of its two columns. A polynomial relation among f_{t_j}, f'_{t_j} would therefore vanish on a product of Zariski-dense punctured affine planes. The nonzero Wronskians make the changes of coordinates invertible, so no such relation exists. These $2m$ functions form a first-order E-function system with only the finite singularity $z = 0$. Siegel–Shidlovskii at $z = 1$ proves independence of $f_{t_j}(1), f'_{t_j}(1)$; the latter derivatives are $t_j I_1(t_j)$, with nonzero algebraic factors t_j. Nonvanishing follows from independence. □

Theorem 3.3 (Joint theorem at arbitrary complex algebraic scales)**.** *Let $\lambda_1, \dots, \lambda_s$ be distinct nonzero algebraic numbers, $R \geq 1$, and let $t_1, \dots, t_m$ be nonzero algebraic numbers distinct modulo sign. Put $d = \dim_{\mathbb{Q}} \langle \lambda_1, \dots, \lambda_s \rangle$. Then*

$$\mathrm{trdeg}_{\overline{\mathbb{Q}}}\, \overline{\mathbb{Q}}(e^{\lambda_i}, \mathrm{Ein}_r(\lambda_i), I_0(t_j), I_1(t_j), Y(t_j) : i \leq s, r \leq R, j \leq m) = d + Rs + 3m.$$

The $Rs+3m$ nonexponential coordinates are algebraically independent over the field of the displayed exponential coordinates. Moreover, after choosing one representative from each nonzero algebraic sign class, all the coordinates

$$\mathrm{Ein}_r(\lambda) \quad (r \geq 1,\ \lambda \in \overline{\mathbb{Q}}^{\times}), \qquad I_0(t), C(t), \beta(t)$$

are jointly algebraically independent over

$$K_{\exp} = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}}),$$

in the finite-subset sense. The exponential coordinates themselves are not part of an independent family over $K_{\exp}$.

Proof. The primitive-tower and solvable-closure lemmas (Lemmas 2.1 and 2.2) give a solvable PV field

$$S = K\big(e^{\lambda_i z}, \mathrm{Ein}_r(\lambda_i z), \log z : i \leq s, r \leq R\big)$$

of transcendence degree $d + Rs + 1$ over K. Those proofs allow arbitrary complex algebraic λ_i. Let B be the full Bessel PV field from Lemma 3.1. A quotient common to its connected perfect group SL_2^m and the solvable group of S is trivial. Goursat's lemma therefore makes the group of BS the direct product, and

$$\mathrm{trdeg}_K\, BS = 3m + d + Rs + 1.$$

More explicitly, before specialization put

$$F = K\big(e^{\lambda_i z}, \mathrm{Ein}_r(\lambda_i z), I_0(t_j z), I_1(t_j z), Y(t_j z) : i \leq s, r \leq R, j \leq m\big).$$

Equations (6) and (7) recover the second Bessel solutions and their derivatives after adjoining $\log z$; conversely all displayed functions lie in BS. Thus $BS = F(\log z)$. It follows that $\mathrm{trdeg}_K F \geq 3m + d + Rs$; counting generators with the exponential character lattice gives the reverse inequality.

All these entire functions are E-functions. Include 1, both exponential signs, the complete polyexponential chains, and $f_{t_j}, f'_{t_j}, y_{t_j}, y'_{t_j}$ in one first-order system. Its matrix has no finite pole except $z = 0$. The Wronskian calculation also gives

$$I_1(w)Y(w) - I_0(w)Y'(w) = \frac{1 - I_0(w)^2}{w}.$$

Corollary 3.2, proved independently before forming quotients, ensures that this identity expresses $Y'(t_j)$ in the other value coordinates. Hence the generated value field is exactly the one in the statement. Siegel–Shidlovskii at the ordinary point $z = 1$ gives its transcendence degree [4]. The exponential value field has transcendence degree d by Lindemann–Weierstrass: distinct integer characters of a rationally independent algebraic exponent basis give distinct algebraic exponents. This proves the asserted relative independence. The transformations $(I_0, I_1, Y) \leftrightarrow (I_0, C, \beta)$ are birational by nonvanishing.

Finally, any relation over $K_{\exp}$ uses only finitely many algebraic exponents in its coefficients. Add them to the finite λ list (omitting zero), enlarge R if necessary, and apply the finite theorem. This proves the infinite-base statement. Distinctness modulo sign is necessary: $I_0(-t) = I_0(t)$, $Y(-t) = Y(t)$, and $C(-t) = -C(t)$. □

4 Simultaneous primitives at every Bessel scale

Write $k = \overline{\mathbb{Q}}$ and $K_{\exp} = k(e^{\alpha} : \alpha \in k)$. For $t \neq 0$, let

$$H_t(z) = -\sum_{n\geq 1} \frac{(tz/2)^{2n}}{2n(n!)^2}, \qquad H_t'(z) = \frac{1 - I_0(tz)}{z}. \tag{8}$$

Thus $H_{-t} = H_t$. The conditions on distinct scales below are therefore necessary even before considering algebraic independence.

Lemma 4.1 (No exact linear combination of Bessel derivatives)**.** *Let $K = \mathbb{C}(z)$. Let P/K be the full Bessel PV field at nonzero scales $s_1, \dots, s_n$ distinct up to sign, with group $G_B = \mathrm{SL}_2^n$. Let S/K be a solvable PV field containing $\log z$, and put $L = PS$. Suppose $\mathrm{Gal}(L/K) = G_B \times G_S$. Choose any subset of scales $t_1, \dots, t_m$ among the s_j. If $c_1, \dots, c_m \in \mathbb{C}$ are not all zero, no $u \in L$ satisfies*

$$u' = \sum_{j=1}^{m} c_j \frac{I_0(t_j z)}{z}. \tag{9}$$

Proof. For each active index j with $c_j \neq 0$, let V_j be the solution space of $y'' + z^{-1}y' - t_j^2 y = 0$. These are pairwise nonisomorphic irreducible G_B-modules: the jth SL_2 factor acts by its standard representation, and all the other factors act trivially. The orbit of the right side of (9) spans

$$V = \bigoplus_{c_j \neq 0} V_j/z.$$

Indeed each nonzero component generates its irreducible summand, and the summands are nonisomorphic.

If u existed, the vector space

$$U = \mathrm{span}_{\mathbb{C}}\{1, \sigma(u) : \sigma \in G_B \times G_S\}$$

would be finite-dimensional, with differentiation giving the exact sequence $0 \to \mathbb{C} \to U \to V \to 0$. Any finite-dimensional Galois-stable subspace of a PV field is a rational representation of its differential Galois group. Complete reducibility for G_B supplies a complement U_0 to the constants. It is unique since $\mathrm{Hom}_{G_B}(V, \mathbb{C}) = 0$. The commuting group G_S preserves U_0 and fixes it pointwise: it fixes V, and differentiation is injective on U_0. Thus $U_0 \subset L^{G_S} = P$.

Projecting U_0 onto any active standard summand yields a row (u_f, u_g) of primitives of $(f/z, g/z)$ for a fundamental pair f, g of the selected Bessel equation. This row transforms in the same way as (f, g). Hence

$$(a, b) = (u_f, u_g) \begin{pmatrix} f & g \\ f' & g' \end{pmatrix}^{-1}$$

is invariant under the entire group G_B, so $a, b \in K$. Differentiation gives

$$a' + t_j^2 b = 1/z, \qquad a + b' - b/z = 0,$$

and therefore

$$-b'' + b'/z - b/z^2 + t_j^2 b = 1/z. \tag{10}$$

A rational b cannot have a nonzero finite pole: the highest-order pole from $-b''$ could not cancel. Thus it is a Laurent polynomial. If its lowest nonzero term is dz^r with $r \leq 0$, the left side has leading term $-(r-1)^2 dz^{r-2}$, impossible on the right side. If instead $r \geq 1$, the left side is holomorphic at zero, since its potential z^{-1} term at $r = 1$ has coefficient zero. This also contradicts (10). The identically zero choice of b is plainly impossible. This proves the lemma. □

Theorem 4.2 (Joint independence of all selected Bessel primitives)**.** *Under the hypotheses of Lemma 4.1, the functions $H_{t_1}(z), \dots, H_{t_m}(z)$ are algebraically independent over L.*

Proof. Put $u_j = \log z - H_{t_j}(z)$, so $u_j' = I_0(t_j z)/z \in L$. The field $M = L(u_1, \dots, u_m)$ is a PV extension of L for the triangular system whose fundamental solution matrix is

$$\begin{pmatrix} 1 & u_1 & \cdots & u_m \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

All fields are realized as fields of analytic germs on a common simply connected domain, so their constant field is $\mathbb{C}$. The PV group is an algebraic subgroup of $\mathbb{G}_a^m$, acting by translations of the u_j.

In characteristic zero, algebraic subgroups of $\mathbb{G}_a^m$ are linear subspaces. If this group were proper, some nonzero vector $(c_1, \dots, c_m)$ would annihilate all its translations. The Galois correspondence would then give $\sum_j c_j u_j \in L$, contradicting Lemma 4.1 after differentiation. Consequently the group is $\mathbb{G}_a^m$, and $\operatorname{trdeg}_L M = m$. Since $\log z \in L$, the same joint independence holds for the H_{t_j}. □

Theorem 4.3 (All-scale arithmetic primitive extension)**.** *Let $s_1, \dots, s_n \in k^\times$ be distinct up to sign, and select any subset $t_1, \dots, t_m$ of these scales. Let $\lambda_1, \dots, \lambda_q \in k^\times$ be distinct, with positive integers $R_1, \dots, R_q$. Then the $\sum_i R_i + 3n + m$ numbers*

$$\operatorname{Ein}_r(\lambda_i) \quad (1 \le i \le q,\ 1 \le r \le R_i),$$

$$I_0(s_j), \quad C(s_j) = \frac{I_1(s_j)}{I_0(s_j)}, \quad \beta(s_j) = \frac{Y(s_j)}{I_0(s_j)} \quad (1 \le j \le n), \qquad H_{t_l}(1) \quad (1 \le l \le m)$$

are algebraically independent over $K_{\exp}$. Here $Y(s) = \sum_{n \ge 0} H_n (s/2)^{2n}/(n!)^2$ is the harmonic Bessel series, with $H_0 = 0$.

Proof. Any relation over $K_{\exp}$ uses only finitely many exponential values. Include their corresponding functions in the solvable PV field of the all-order polyexponential construction, together with $\log z$. The Bessel product field and this solvable field have a direct-product group: a connected product of SL_2 groups has no nontrivial solvable quotient. The previously established E-function coordinates lie in their composite L. Theorem 4.2 adds exactly m independent functions to these coordinates.

For algebraic t, the coefficient of $z^{2n}/(2n)!$ in H_t is $-\binom{2n}{n}(t/2)^{2n}/(2n)$; its conjugate-growth and common-denominator bounds are exponential. Together with (8), this verifies that H_t is an E-function. Appending its first-order equation to the old system introduces no finite singularity other than zero. Siegel–Shidlovskii specialization at $z = 1$ therefore preserves the functional transcendence-degree increase. Theorem 3.3 supplies the independence of all the other coordinates, and passage to the displayed Bessel quotients is birational because $I_0(s_j) \neq 0$ at nonzero algebraic s_j by Corollary 3.2. Since the finite set of exponential coefficients was arbitrary, the assertion holds over $K_{\exp}$. □

Equivalently, choose one representative for every sign class in $k^\times/\{\pm 1\}$. The entire union of the polyexponential values, Bessel triples, and primitive values indexed as above is independent over $K_{\exp}$, in the sense of every finite subfamily. One must not include both $H_t(1)$ and $H_{-t}(1)$ as distinct coordinates.

4.1 Critical limits and matched ordinary-Bessel combinations

For $\lambda > 0$ put

$$B(\lambda) = \sum_{n\geq 1} \frac{(-1)^{n-1}\lambda^n}{n(n!)^2} = 2H_{2i\sqrt{\lambda}}(1).$$

The critical Newton–Laguerre formula is

$$\mathcal{L}(\lambda) = B(\lambda) - \gamma - \log\lambda, \qquad R_N(\lambda/N) - \log(\lambda/N) \longrightarrow \mathcal{L}(\lambda).$$

Here $R_N(x) = -\sum_{n=1}^{N} L_n(x)/n$, with $L_n(x) = \sum_{j=0}^{n}(-1)^j\binom{n}{j}x^j/j!$. The exact finite-sum identity and dominated-convergence proof are given in Proposition 32.3. For positive algebraic λ define

$$j_\lambda = J_0(2\sqrt{\lambda}), \qquad c_\lambda = \frac{J_1(2\sqrt{\lambda})}{J_0(2\sqrt{\lambda})}, \qquad \nu_\lambda = -\frac{\pi Y_0(2\sqrt{\lambda})}{2J_0(2\sqrt{\lambda})},$$

$$\beta_\lambda := \beta(2i\sqrt{\lambda}) = \gamma + \frac{1}{2}\log\lambda + \nu_\lambda = \frac{\displaystyle\sum_{n\geq 0}(-1)^n H_n \lambda^n/(n!)^2}{j_\lambda}. \tag{11}$$

The factor $1/2$ in front of $\log\lambda$ is essential.

Corollary 4.4 (Four independent coordinates per critical scale)**.** *For distinct positive algebraic $\lambda_1, \ldots, \lambda_m$, the $4m$ numbers*

$$j_{\lambda_j}, \quad c_{\lambda_j}, \quad \beta_{\lambda_j}, \quad B(\lambda_j) \qquad (1 \leq j \leq m)$$

are jointly independent over $K_{\exp}$. One may replace $B(\lambda_j)$ by

$$Q_{\lambda_j} := B(\lambda_j) - \beta_{\lambda_j} = \mathcal{L}(\lambda_j) + \tfrac{1}{2}\log\lambda_j - \nu_{\lambda_j}. \tag{12}$$

In particular, the Q_{λ_j} are jointly independent over the field generated by $K_{\exp}$, all the old poly-exponential values, and all the old Bessel coordinates, including those at their own scales. Their series-and-quotient expression $B(\lambda) - \beta_\lambda$ contains no explicit γ or logarithm.

Proof. The scales $2i\sqrt{\lambda_j}$ are nonzero algebraic and distinct up to sign. Apply Theorem 4.3, using $I_0(2i\sqrt{\lambda}) = j_\lambda$ and $C(2i\sqrt{\lambda}) = ic_\lambda$. The substitution $B(\lambda_j) \mapsto B(\lambda_j) - \beta_{\lambda_j}$ is an invertible linear change over the old coordinate field. Extra old coordinates in any proposed relation are finite in number and may be included in the same theorem. □

Corollary 4.5 (A logarithm-cancelled one-defect family)**.** *Let*

$$W_\lambda := \mathcal{L}(\lambda) - 2\nu_\lambda = \gamma + B(\lambda) - 2\beta_\lambda \qquad (\lambda \in k \cap \mathbb{R}_{>0}).$$

Among all the numbers

$$\gamma, \quad \delta, \quad \kappa, \quad \gamma - 2\kappa, \quad \nu_1, \quad \gamma - 2\nu_1, \quad W_\lambda \quad (\lambda \in k \cap \mathbb{R}_{>0}),$$

at most one is algebraic over $K_{\exp}$, and hence at most one is algebraic over k.

Proof. Put $\eta = \mathrm{Ein}(1)$, $\beta_K = \gamma + \kappa$, and $\beta_J = \gamma + \nu_1$. The previously known part follows from the joint independence of η, β_K, β_J over $K_{\exp}$: normalize each displayed form as a nonzero $K_{\exp}$-multiple of γ plus a linear combination of these three independent coordinates. No two normalized forms have the same coefficient vector. Two algebraic values would give a nonzero linear combination of independent coordinates algebraic over $K_{\exp}$.

For a pair involving W_λ, use its expression $\gamma + B(\lambda) - 2\beta_\lambda$ and eliminate γ in the same way. Against any old form, the coefficient of $B(\lambda)$ is nonzero. For two distinct W_λ, W_μ, their difference has nonzero coefficients of the independent primitive coordinates $B(\lambda)$ and $B(\mu)$. Theorem 4.3 excludes algebraicity in every case. Possible equality of β_λ with an already selected Bessel coordinate causes no difficulty: the new primitive coefficient still does not vanish. □

The unshifted limits $\mathcal{L}(\lambda)$ for $\lambda \neq 1$ should not be inserted into a one-defect count without accounting for their $\log \lambda$ terms. The combinations in (12) and Corollary 4.5 make that normalization explicit.

Corollary 4.6 (A relation over the enlarged old field forces rank four)**.** *Let $H_4 = k(\gamma, \delta, \kappa, C)$, and let*

$$F_{\mathrm{old}} = K_{\exp}\big(\mathrm{Ein}_r(\lambda),\ I_0(t),\ C(t),\ \beta(t) : \lambda, t \in k^\times,\ r \geq 1\big),$$

with duplicate sign-related Bessel coordinates omitted. If $\mathrm{trdeg}_k H_4 = 3$, the entire family $\{H_t(1) : t \in k^\times/\{\pm 1\}\}$ is algebraically independent over H_4F_{old}. Consequently any nonzero polynomial relation among finitely many of these primitive values, with coefficients in H_4F_{old}, would imply $\mathrm{trdeg}_k H_4 = 4$. The same criterion applies to finitely many distinct Q_λ or W_λ at positive algebraic scales.

Proof. The field

$$F_0 = k(e, \eta, I_0(2), C, \beta(2)) \subset H_4(e, I_0(2))$$

has transcendence degree five by the basic joint theorem. If $\mathrm{trdeg}_k H_4 = 3$, the field on the right has transcendence degree at most five. Therefore $H_4(e, I_0(2))/F_0$ is algebraic, so $H_4F_{\mathrm{old}}/F_{\mathrm{old}}$ is algebraic as well. Theorem 4.3 gives joint independence of the primitive family over F_{old}; algebraic extension of the coefficient field preserves this independence. The known bound $3 \leq \mathrm{trdeg}_k H_4 \leq 4$ proves the criterion. Finally Q_λ and W_λ are invertible affine transforms of $B(\lambda) = 2H_{2i\sqrt{\lambda}}(1)$ over the indicated enlarged coefficient field. □

No such relation is asserted to exist. The criterion identifies what an additional relation would accomplish; it does not furnish one.

5 Bessel primitive towers of every depth

The primitive extension admits arbitrary finite depths at all selected scales. The extra logarithmic weights require a nonexactness argument; independence of the first primitives alone would not justify this step. For $t \neq 0$ and integers $r \geq 1$ define

$$H_{t,r}(z) = -\sum_{n \geq 1} \frac{(tz/2)^{2n}}{(2n)^r (n!)^2}, \qquad H_{t,0}(z) = 1 - I_0(tz). \tag{13}$$

With $\theta = z\,d/dz$, these functions satisfy $\theta H_{t,r} = H_{t,r-1}$. In particular $H_{t,1} = H_t$ in (8), and $H_{-t,r} = H_{t,r}$.

5.1 Nonexactness with polynomial logarithmic weights

Lemma 5.1 (Weighted Bessel nonexactness)**.** *Let $K = \mathbb{C}(z)$, let P/K be the full Bessel Picard–Vessiot field at nonzero scales $s_1, \dots, s_n$ distinct modulo sign, and let S/K be a solvable Picard–Vessiot field containing $l = \log z$. Assume that $L = PS$ has differential Galois group $\mathrm{SL}_2^n \times \mathrm{Gal}(S/K)$. For any polynomials $A_1(T), \dots, A_n(T) \in \mathbb{C}[T]$, not all zero, there is no $u \in L$ satisfying*

$$u' = \sum_{i=1}^{n} A_i(l) \frac{I_0(s_i z)}{z}. \tag{14}$$

Proof. Write $G_B = \mathrm{SL}_2^n$. The group G_B fixes S and hence l. For each active index i, let V_i be the standard two-dimensional solution space of the corresponding Bessel equation. The orbit of the right side of (14) under G_B spans the direct sum

$$\bigoplus_{A_i \neq 0} A_i(l) V_i / z.$$

The active summands are nonisomorphic irreducible G_B-modules. If u existed, the span of 1 and its G_B-orbit would be finite-dimensional: differentiation maps this span into the displayed finite-dimensional space, with kernel consisting of constants. Complete reducibility supplies, for each active scale $t = s_i$, a pair of primitives (u_f, u_g) transforming like a fundamental Bessel solution pair (f, g) and satisfying

$$u_f' = A_i(l) f / z, \qquad u_g' = A_i(l) g / z.$$

Here finite-dimensional Galois-stable subspaces of a Picard–Vessiot field carry rational representations. The invariant row

$$(a, b) = (u_f, u_g) \begin{pmatrix} f & g \\ f' & g' \end{pmatrix}^{-1}$$

belongs to $L^{G_B} = S$. Differentiating gives

$$a' + t^2 b = A_i(l)/z, \qquad a + b' - b/z = 0,$$

and therefore

$$\mathcal{D}_t b := -b'' + b'/z - b/z^2 + t^2 b = A_i(l)/z. \tag{15}$$

We first reduce b from S to $K(l)$. Every automorphism of $S/K(l)$ changes b by a solution of $\mathcal{D}_t h = 0$. These homogeneous solutions are precisely zV_i, since $\mathcal{D}_t(zf) = -z(f'' + z^{-1} f' - t^2 f)$. They lie in P, while the difference also lies in S. The product-group hypothesis gives $P \cap S = K$; no nonzero member of zV_i belongs to K, because the standard SL_2 module has no invariant vector. Thus every such difference is zero. The fixed-field theorem, applied to the Picard–Vessiot extension $S/K(l)$, implies $b \in K(l)$.

In fact $b \in K[l]$. To prove this, suppose that an irreducible monic $q(l) \in K[l]$ occurs in its denominator. The derivation is $D = d/dz$ with $Dl = 1/z$. If $q \mid Dq$, monicity implies $\deg_l Dq < \deg_l q$ and hence $Dq = 0$. If $d = \deg_l q > 0$, the coefficient of l^{d-1} would then give $q_{d-1}' = -d/z$, which has no rational solution. Therefore $q \nmid Dq$. If b has a pole of order $h > 0$ along q, then b'' has a pole of order $h + 2$ there, with nonzero leading coefficient proportional to $h(h+1)(Dq)^2$. The other terms on the left of (15) have pole orders at most $h + 1$, whereas its right side is polynomial in l. This is impossible, so there is no such denominator.

Write $b = \sum_{j=0}^{M} b_j(z) l^j$, with $b_M \neq 0$. The operator $\mathcal{D}_t$ does not increase logarithmic degree. If $M > \deg A_i$, its highest coefficient gives $\mathcal{D}_t b_M = 0$, contradicting the absence of a nonzero rational

homogeneous solution just proved. If $M < \deg A_i$, degrees already contradict (15). In the remaining case the highest coefficient gives

$$\mathcal{D}_t b_M = c/z, \qquad c \neq 0.$$

The rational obstruction in the proof of Lemma 4.1 excludes this equation: a rational solution has no nonzero finite poles and is thus a Laurent polynomial; its lowest power either gives an impossible pole of order at least two at zero, or leaves the left side holomorphic there. Scaling by c gives exactly the same obstruction as (10). This contradiction proves the lemma. □

Theorem 5.2 (Functional independence of every selected depth)**.** *Under the hypotheses of Lemma 5.1, choose nonnegative integers $R_1, \dots, R_n$. The $\sum_i R_i$ functions*

$$H_{s_i,r}(z), \qquad 1 \le r \le R_i,$$

are jointly algebraically independent over L.

Proof. For each active scale $t = s_i$, put

$$u_{t,0} = I_0(tz), \qquad u_{t,r} = \frac{l^r}{r!} - H_{t,r}(z) \quad (r \ge 1).$$

Then $\theta u_{t,r} = u_{t,r-1}$. Define

$$v_{t,j} = \sum_{h=0}^{j} (-1)^h \frac{j!}{(j-h)!} l^{j-h} u_{t,h+1}, \qquad 0 \le j < R_i. \tag{16}$$

Direct differentiation telescopes and gives $v'_{t,j} = l^j I_0(tz)/z$. Conversely,

$$u_{t,r} = \sum_{j=0}^{r-1} \frac{(-1)^j l^{r-1-j}}{j!(r-1-j)!} v_{t,j}. \tag{17}$$

Thus the two displayed families generate the same field over L.

Adjoining the $v_{t,j}$ is a Picard–Vessiot primitive extension whose group is an algebraic subgroup of the additive group $\mathbb{G}_a^N$, where $N = \sum_i R_i$. All functions may be realized as analytic germs on a common simply connected domain, so no constants beyond $\mathbb{C}$ are introduced. If this subgroup were proper, a nonzero constant linear form in the $v_{t,j}$ would be fixed and hence would belong to L. Its derivative would contradict Lemma 5.1. Consequently the group is $\mathbb{G}_a^N$ and the $v_{t,j}$ are independent over L. The invertible triangular transformations (16)–(17), followed by subtraction of $l^r/r! \in L$, prove the theorem. □

5.2 Arithmetic specialization and the infinite old field

Choose one representative of every nonzero algebraic sign class. Define the coefficient field

$$\begin{aligned} F_{\mathrm{old}} = K_{\exp}\big(&\mathrm{Ein}_r(\lambda),\ I_0(t), C(t), \beta(t) : \\ &r \ge 1,\ \lambda \in k^\times, t \in k^\times/\{\pm 1\}\big). \end{aligned} \tag{18}$$

The notation for t means the chosen representatives. This field contains the old exponential-integral and Bessel coordinates, but does not include any of the new primitive coordinates as additional generators.

Theorem 5.3 (All-scale, all-depth arithmetic extension)**.** *The values*

$$H_{t,r}(1), \qquad t \in k^\times/\{\pm 1\}, \quad r \geq 1,$$

are jointly algebraically independent over F_{old}. Equivalently, every finite set of old nonexponential coordinates from Theorem 3.3, together with any finite set of these primitive coordinates, is algebraically independent over $K_{\exp}$.

Proof. First choose finitely many old coordinates, primitive scales and depths, and exponential values. Enlarge the Bessel list to contain every selected primitive scale. Include the finite set of exponential functions and the required polyexponential chains, together with $\log z$, in the solvable Picard–Vessiot field S from the proof of Theorem 3.3. The Bessel product field P and S have direct-product group, since a connected product of SL_2 groups has no nontrivial solvable quotient. Theorem 5.2 therefore adds exactly $\sum_i R_i$ independent functions to the entire old-coordinate field, which is a subfield of PS.

Each $H_{t,r}$ is an E-function. Its coefficient at degree $2n$ in exponential-series normalization is

$$-\binom{2n}{n}\frac{(t/2)^{2n}}{(2n)^r}.$$

For fixed algebraic t and fixed r, all conjugates have exponential growth, and common denominators through degree N are bounded exponentially: apart from the fixed algebraic denominators of powers of $t/2$, it is enough to use $\mathrm{lcm}(1,\dots,N)^r$. The equations

$$H'_{t,1} = (1 - I_0(tz))/z, \qquad H'_{t,r} = H_{t,r-1}/z \quad (r \geq 2)$$

append the full primitive chains to the old first-order E-system without introducing any finite singularity other than zero. Siegel–Shidlovskii specialization at the ordinary point $z = 1$ preserves the resulting functional transcendence degree [4]. As in Theorem 3.3, the old derivative coordinates are recovered from its displayed coordinates, and the passage to $C(t), \beta(t)$ is birational because $I_0(t) \neq 0$ by Corollary 3.2. This proves the finite arithmetic assertion, including relative independence over the finite exponential field.

A relation over F_{old} would use only finitely many of its generators and finitely many exponential coefficients. Include all of them in the finite construction just given and clear denominators. The resulting relation contradicts that finite independence statement. This proves the theorem over the entire field (18). □

Corollary 5.4 (Higher ordinary-Bessel primitive values)**.** *For positive algebraic λ and integers $r \geq 1$, put*

$$B_r(\lambda) = \sum_{n\geq 1} \frac{(-1)^{n-1}\lambda^n}{n^r(n!)^2} = 2^r H_{2i\sqrt{\lambda},r}(1).$$

All the $B_r(\lambda)$ are jointly independent over F_{old}. The same holds after replacing each selected value by $B_r(\lambda) - \beta(2i\sqrt{\lambda})$.

Proof. Distinct positive λ give distinct scales modulo sign. Apply Theorem 5.3; multiplication by 2^r and subtraction of an old coordinate are invertible affine changes over F_{old}. □

Corollary 5.5 (A one-defect family of every depth)**.** *Put $T_{r,\lambda} = B_r(\lambda) - \gamma$. Among the numbers*

$$\gamma,\ \delta,\ \kappa,\ \gamma - 2\kappa, \nu,\ \gamma - 2\nu,$$
$$T_{r,\lambda} \quad (r \geq 1,\ \lambda \in k \cap \mathbb{R}_{>0}),$$

at most one is algebraic over $K_{\exp}$. This does not establish the individual transcendence of a prescribed $T_{r,\lambda}$.

Proof. Write $\eta = \mathrm{Ein}(1)$, $\beta_K = \beta(2)$ and $\beta_J = \beta(2i)$. Each of the six old expressions is a nonzero $K_{\exp}$-multiple of γ plus a linear combination of η, β_K, β_J. After normalizing the coefficient of γ to one, their coefficient vectors are distinct. If two values were algebraic over $K_{\exp}$, eliminating γ would make a nonzero linear combination of these independent old coordinates algebraic over $K_{\exp}$, which is impossible. For any pair involving a new $T_{r,\lambda}$, the same elimination has a nonzero coefficient of at least one primitive coordinate. Corollary 5.4 excludes algebraicity also in this case. □

Corollary 5.6 (Primitive towers under the original rank-three alternative)**.** *If $\mathrm{trdeg}_k H_4 = 3$, then $H_4F_{\mathrm{old}}/F_{\mathrm{old}}$ is algebraic. Over H_4F_{old} the entire family $\{B_r(\lambda)\}$ is jointly independent, and so is the entire family $\{T_{r,\lambda}\}$. More generally, one may choose either $B_r(\lambda)$ or $T_{r,\lambda}$ at each index and retain joint independence.*

If, instead, any one $T_{r,\lambda}$ is algebraic over F_{old}, then γ is transcendental over F_{old} and the six values $\gamma, \delta, \kappa, C, \nu, C_J$ are algebraically independent over $K_{\exp}$.

Proof. The field $H_4(e)$ contains $k(e, \eta, \beta_K, C)$, whose four generators are independent over k by Theorem 3.3 and Lindemann–Weierstrass. Under the rank-three hypothesis, $H_4(e)$ has transcendence degree at most four, so it is algebraic over that subfield. In particular γ is algebraic over F_{old}. Moreover $H_4F_{\mathrm{old}} = F_{\mathrm{old}}(\gamma)$, because $\delta = e(\eta - \gamma)$ and $\kappa = \beta_K - \gamma$. Algebraic extension of a coefficient field preserves algebraic independence; Corollary 5.4 and translation by γ now prove the first assertions. Both versions of a single coordinate are not to be included simultaneously.

If $T_{r,\lambda}$ is algebraic over F_{old} while γ were also algebraic there, their sum $B_r(\lambda)$ would be algebraic, contrary to Corollary 5.4. Thus γ is transcendental over F_{old}. Adjoin it to the five independent old coordinates $\eta, \beta_K, C, \beta_J, C_J$ over $K_{\exp}$ and use

$$\delta = e(\eta - \gamma), \qquad \kappa = \beta_K - \gamma, \qquad \nu = \beta_J - \gamma.$$

This is an invertible affine change over $K_{\exp}$ and proves the six-value assertion. No algebraicity premise for such a witness is established here. □

Part II
Arithmetic consequences and reconstruction

Arithmetic notation. In this part $k = \overline{\mathbb{Q}}$, $H = H_4 = k(\gamma, \delta, \kappa, C)$, $\eta = \mathrm{Ein}(1) = \gamma + \delta/e$, and $\mu(t) = \beta(t) - \gamma$ for positive algebraic t. Thus $\mu(t) = \log(t/2) + K_0(t)/I_0(t)$ and $\mu(2) = \kappa$. All appearances of Ein_r refer to the entire polyexponential tower, not the classical exponential-integral tail.

6 Uniform codimension one and a global relation ideal

Theorem 6.1. *For distinct positive algebraic $t_1, \dots, t_k$, put*

$$H_T = \overline{\mathbb{Q}}(\gamma, \delta, \mu(t_1), C(t_1), \dots, \mu(t_k), C(t_k)).$$

Then

$$\mathrm{trdeg}_{\overline{\mathbb{Q}}} H_T \geq 2k + 1. \tag{19}$$

Thus the deficit from algebraic independence of its $2k+2$ displayed generators is at most one, independently of k.

Proof. The rational change of coordinates $(I_0(t_j), I_1(t_j)) \mapsto (I_0(t_j), C(t_j))$ is invertible because $I_0(t_j) > 0$. Theorem 3.3 therefore makes $e, \eta, I_0(t_j), C(t_j), \beta(t_j)$ algebraically independent. Discard the k values $I_0(t_j)$. The remaining $2k+2$ independent numbers belong to $H_T(e)$, by the connection identity $\beta(t) = \gamma + \mu(t)$ and $\eta = \gamma + \delta/e$. Hence $\operatorname{trdeg} H_T + 1 \geq \operatorname{trdeg} H_T(e) \geq 2k+2$. □

Corollary 6.2. *The ideal of polynomial relations over $\overline{\mathbb{Q}}$ among the displayed generators of H_T is either zero or generated by one irreducible polynomial. In particular, two relatively prime nonzero polynomials cannot both vanish at that tuple.*

Proof. The evaluation kernel in the polynomial ring in $2k+2$ variables over $\overline{\mathbb{Q}}$ is prime. Its height is the number of variables minus the transcendence degree of the image's fraction field, and is at most one by Theorem 6.1. A nonzero height-one prime in a polynomial ring over a field, a unique factorization domain, is principal and is generated by an irreducible polynomial. □

Theorem 6.3 (The entire positive algebraic family)**.** *Consider the indexed family*

$$\gamma,\ \delta,\quad \mu(t),\ C(t) \qquad (t \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}). \tag{20}$$

Either it is algebraically independent, or every polynomial relation among its members is a multiple of one fixed irreducible polynomial P. Deleting any variable that genuinely occurs in P leaves an algebraically independent family. Moreover one can choose the deleted member among γ, δ, and the $\mu(t)$, retaining every $C(t)$.

Proof. Every polynomial involves only finitely many variables. If there is a nonzero relation, factor it in a finite polynomial ring and choose an irreducible factor P that vanishes at the corresponding tuple. Irreducibility is preserved on adjoining extra indeterminates. Given another relation Q, put all arguments occurring in P or Q in a finite set T, and include both coordinates for each argument. Corollary 6.2 applies in this larger ring. Its prime kernel has height one and contains the irreducible element P, so it equals (P). Therefore P divides Q.

If X occurs in P with positive degree, no nonzero polynomial independent of X can be a multiple of P, since degrees in X add under multiplication in a polynomial ring over a domain. Thus deleting X leaves no relations. Finally, all the $C(t)$ form an algebraically independent family by Theorem 3.3 and the same coordinate change as above. Consequently P cannot involve only C-coordinates. □

Remark 6.4. The exceptional member in Theorem 6.3 is not identified. In particular, the theorem is consistent with γ being algebraic: in that case it is γ that must be deleted. "One relation" means one generator of the relation ideal; if $P = 0$ is a relation, its infinitely many polynomial multiples are also relations.

Corollary 6.5 (If the original four are dependent)**.** *Let $H = \overline{\mathbb{Q}}(\gamma, \delta, \kappa, C)$. If $\operatorname{trdeg}_{\overline{\mathbb{Q}}} H = 3$, then:*

1. *for every finite set of distinct positive algebraic $t_j \neq 2$, all the numbers $\mu(t_j), C(t_j)$ are algebraically independent over H;*
2. *e and $I_0(2)$ are algebraically independent over H.*

Proof. Adjoining k new pairs to the original pair gives transcendence degree at least $2(k+1)+1 = 2k+3$ by Theorem 6.1; this is the largest possible increase $2k$ over a field of transcendence degree three. For the second assertion, Theorem 3.3 shows that $H(e, I_0(2))$ contains five independent values. Adjoining two elements to H can increase its transcendence degree by at most two, so the relative increase is exactly two. □

7 An affine family of independent triples

Theorem 7.1. *For all but at most one $\lambda \in \overline{\mathbb{Q}}$, the three numbers*

$$C, \qquad \gamma + \kappa, \qquad \gamma + \lambda\delta \tag{21}$$

are algebraically independent over $\overline{\mathbb{Q}}$.

Proof. Set $\mathcal{F} = \overline{\mathbb{Q}}(C, \gamma + \kappa)$. Theorem 3.3 gives $\operatorname{trdeg}_{\overline{\mathbb{Q}}} \mathcal{F} = 2$, while $\mathcal{F}(\gamma, \delta) = H$ has transcendence degree at least three by the bound $\operatorname{trdeg}_k H_4 \geq 3$. An exception in (21) means that $\gamma + \lambda\delta$ is algebraic over $\mathcal{F}$. If distinct λ, λ' were exceptions, their difference would make δ algebraic over $\mathcal{F}$, and then γ would also be algebraic over $\mathcal{F}$. This contradicts $\operatorname{trdeg} H \geq 3$. □

For example, at least two of the following three triples are independent:

$$(C, \gamma + \kappa, \gamma), \quad (C, \gamma + \kappa, \gamma + \delta), \quad (C, \gamma + \kappa, \gamma + 2\delta).$$

The theorem does not determine whether $\lambda = 0$ is exceptional, so it does not prove the transcendence of γ.

Proposition 7.2 (A sufficient new input for full independence)**.** *An algebraic dependence of $e, I_0(2)$ over $H = \overline{\mathbb{Q}}(\gamma, \delta, \kappa, C)$ would imply $\operatorname{trdeg}_{\overline{\mathbb{Q}}} H = 4$. Alternatively, transcendence of γ over $\overline{\mathbb{Q}}(e, \eta, C, \gamma + \kappa)$ would imply the same conclusion.*

Proof. The first assertion is the contrapositive of Corollary 6.5(2), since $3 \leq \operatorname{trdeg} H \leq 4$. For the second, Theorem 3.3 makes the four generators of $\mathcal{F}_0 = \overline{\mathbb{Q}}(e, \eta, C, \gamma + \kappa)$ independent. If γ is transcendental over $\mathcal{F}_0$, then $\mathcal{F}_0(\gamma) = H(e)$ has transcendence degree five, forcing $\operatorname{trdeg} H = 4$. □

Neither extra input in Proposition 7.2 is proved here. They specify what would suffice, not an available route around the remaining arithmetic problem.

8 Upper-half-plane values independent over the target field

Lemma 8.1 (One real adjunction)**.** *Let $\mathcal{F} \subset \mathbb{C}$ be stable under complex conjugation. Suppose $u_1, \ldots, u_m, \overline{u_1}, \ldots, \overline{u_m}$ are algebraically independent over $\mathcal{F}$. For every real number c, the $u_1, \ldots, u_m$ are algebraically independent over $\mathcal{F}(c)$.*

Proof. Otherwise their transcendence degree over $\mathcal{F}(c)$ is at most $m - 1$. Conjugation preserves $\mathcal{F}(c)$, so the conjugate tuple has the same upper bound. The union then has transcendence degree at most $2m - 2$ over $\mathcal{F}(c)$ and at most $2m - 1$ over $\mathcal{F}$, contradicting the assumed independence of its $2m$ members. □

Theorem 8.2 (An unconditional relative family)**.** *The family*

$$\{\mathrm{Ein}_r(\lambda) : r \geq 1, \ \lambda \in \overline{\mathbb{Q}}, \ \operatorname{Im}\lambda > 0\}$$

is jointly algebraically independent over $H = \overline{\mathbb{Q}}(\gamma, \delta, \kappa, C)$.

Proof. Let $\mathcal{F}$ be the field generated over $K_{\exp}$ by all $\mathrm{Ein}_r(x)$ with nonzero real algebraic x and all $I_0(t), C(t), \beta(t)$ with positive algebraic t. It is stable under complex conjugation. For any finite subfamily in the upper half-plane, the conjugate arguments lie in the lower half-plane and do not meet the upper-half-plane or real arguments. The indexed pairs (r, λ) and their conjugate pairs

are distinct. Theorem 3.3 shows that the values and their conjugates are independent over $\mathcal{F}$. By Lemma 8.1, the upper-half values are independent over $\mathcal{F}(\gamma)$.

Now $e, \eta = \mathrm{Ein}_1(1), C, \beta(2) \in \mathcal{F}$ and

$$\delta = e(\eta - \gamma), \qquad \kappa = \beta(2) - \gamma.$$

Therefore $H \subset \mathcal{F}(\gamma)$, which gives the conclusion. The argument is finite at every stage; it makes no assumption about the arithmetic nature of γ or about $\mathrm{trdeg}_{\overline{\mathbb{Q}}} H$ beyond the established setup. □

Corollary 8.3 (A concrete infinite sequence)**.** *All the numbers* $\mathrm{Ein}_r(i)$*,* $r = 1, 2, \ldots$*, are jointly algebraically independent over* H*. Consequently, for every* N*,*

$$\mathrm{trdeg}_{\overline{\mathbb{Q}}} H(\mathrm{Ein}_1(i), \ldots, \mathrm{Ein}_N(i)) = \mathrm{trdeg}_{\overline{\mathbb{Q}}} H + N \in \{N+3, N+4\}.$$

In particular $\mathrm{Ci}(1) - i\,\mathrm{Si}(1)$ *is transcendental over* H*.*

Proof. Take $\lambda = i$. With the standard real sine and cosine integrals [2, §6.6],

$$\mathrm{Ein}_1(i) = \int_0^1 \frac{1 - \cos u}{u}\, du + i \int_0^1 \frac{\sin u}{u}\, du = \gamma - \mathrm{Ci}(1) + i\,\mathrm{Si}(1).$$

Since $\gamma \in H$, translation and sign change give the last assertion. □

Remark 8.4. The last corollary does not assert separate algebraic independence of $\mathrm{Ci}(1)$ and $\mathrm{Si}(1)$ over H. Likewise the proof does not allow an arbitrary additional real normalization such as $\sqrt{\pi}$: its one-real-adjunction allowance has already been used for γ.

9 Conjugation transfer to Euler–Gompertz integrals

Write $k = \overline{\mathbb{Q}}$, $\eta = \mathrm{Ein}(1) = \gamma + \delta/e$, $\kappa = K_0(2)/I_0(2)$, and $C = I_1(2)/I_0(2)$. For positive algebraic t, put

$$C(t) = \frac{I_1(t)}{I_0(t)}, \qquad \beta(t) = \gamma + \log(t/2) + \frac{K_0(t)}{I_0(t)}.$$

For any finite collection of distinct positive algebraic scales, Theorem 3.3 gives the joint independence needed below. Set

$$F = k(e, I_0(t_j), C(t_j), \beta(t_j) : 1 \le j \le s).$$

This field is stable under complex conjugation: its displayed generators are real, and conjugation preserves k setwise. It does not fix k pointwise, which is not needed. The input makes η transcendental over F.

Lemma 9.1 (Finite reconstruction and conjugation)**.** *Let* u, v *be real numbers in* $k(\gamma, \delta)$ *such that* $k(\gamma, \delta)/k(u, v)$ *is algebraic. For every* $\lambda \in k \setminus \mathbb{R}$*, the number* $u + \lambda v$ *is transcendental over* F*. Consequently*

$$e,\ u + \lambda v,\ I_0(t_j), C(t_j), \beta(t_j) \quad (1 \le j \le s)$$

are algebraically independent over k*.*

Proof. Suppose that $z = u + \lambda v$ is algebraic over F. Conjugating an algebraic equation shows that $\bar{z} = u + \bar{\lambda} v$ is also algebraic over F, since $\overline{F} = F$. The invertible linear system gives

$$v = \frac{z - \bar{z}}{\lambda - \bar{\lambda}}, \qquad u = \frac{\lambda \bar{z} - \bar{\lambda} z}{\lambda - \bar{\lambda}}.$$

Thus u, v are algebraic over F. Finite reconstruction then makes γ, δ algebraic over F, and $e \in F$ makes $\eta = \gamma + \delta/e$ algebraic over F. This contradicts Theorem 3.3. The other $3s+1$ displayed numbers are already algebraically independent, so adjoining a number transcendental over their field gives the final assertion. □

Corollary 9.2 (Every nonreal algebraic slope works)**.** *For every $\lambda \in \overline{\mathbb{Q}} \setminus \mathbb{R}$, the numbers*

$$e,\ \gamma + \lambda\delta,\ I_0(t_j), C(t_j), \beta(t_j) \quad (1 \le j \le s)$$

are algebraically independent. In particular,

$$e,\ \gamma + i\delta,\ I_0(2),\ C,\ \gamma + \kappa$$

are algebraically independent.

Proof. Apply Lemma 9.1 with $(u, v) = (\gamma, \delta)$. □

This strengthens an "at most one exceptional slope" formulation: any possible exceptional algebraic slope is real. It does not prove that either real component γ or δ is individually transcendental.

9.1 Two positive mixed Hankel integrals

The mixed-Hankel construction of [8] defines

$$G_m = \int_0^\infty x^m e^{-x}(x - \log x)\,dx = m!(\gamma + m + 1 - H_m), \qquad \mathcal{I}_{r+1} = \int_0^\infty \frac{e^{-x}}{(1+x)^{r+1}}\,dx.$$

Here $H_0 = 0$, $\mathcal{I}_1 = \delta$, $\mathcal{I}_2 = 1 - \delta$, and $\mathcal{I}_3 = \delta/2$. Define the first two mixed determinants by

$$\mathcal{H}_1 = G_0\mathcal{I}_1 = (\gamma + 1)\delta,$$
$$\mathcal{H}_2 = \det \begin{pmatrix} G_1\mathcal{I}_1 & G_2\mathcal{I}_2 \\ G_2\mathcal{I}_2 & G_3\mathcal{I}_3 \end{pmatrix}.$$

Both are strictly positive. For $\mathcal{H}_2$, the entries are moments M_r, $r = 0, 1, 2$, of the product variable $y = xt$ under the positive measure

$$xe^{-x}(x - \log x)\,\frac{e^{1-1/t}}{t}\,dx\,dt \quad \text{on } (0,\infty) \times (0,1).$$

Indeed, $\mathcal{I}_{r+1} = \int_0^1 t^r e^{1-1/t}\,dt/t$. Since $x - \log x \ge 1$ and y is not almost everywhere constant, strict Cauchy–Schwarz gives $\mathcal{H}_2 = M_0M_2 - M_1^2 > 0$.

Proposition 9.3 (Explicit degree-four reconstruction)**.** *One has*

$$2\mathcal{H}_2 = -18 + 36\delta - 5\delta^2 - 24\gamma + 48\gamma\delta - 5\gamma\delta^2 - 8\gamma^2 + 16\gamma^2\delta - 2\gamma^2\delta^2.$$

The number δ satisfies

$$2T^4 + (\mathcal{H}_1 - 4)T^3 + (2\mathcal{H}_1^2 - 16\mathcal{H}_1 + 2\mathcal{H}_2 + 2)T^2 + (-16\mathcal{H}_1^2 + 8\mathcal{H}_1)T + 8\mathcal{H}_1^2 = 0, \tag{22}$$

and $\gamma = \mathcal{H}_1/\delta - 1$. Hence

$$[k(\gamma, \delta) : k(\mathcal{H}_1, \mathcal{H}_2)] \le 4.$$

Proof. Substitute $G_1 = \gamma + 1$, $G_2 = 2\gamma + 3$, and $G_3 = 6\gamma + 13$ into the determinant, using the three stated I-values, and expand. Since $\delta > 0$, the relation $\mathcal{H}_1 = (\gamma + 1)\delta$ gives $\gamma = \mathcal{H}_1/\delta - 1$. Substitute it in the expanded expression for $2\mathcal{H}_2$ and clear δ^2 to obtain (22). Its leading coefficient is 2, so it is a nonzero polynomial, proving the degree bound. □

Theorem 9.4 (Joint independence of a complex positive-integral value)**.** *Put $Z = \mathcal{H}_1 + i\mathcal{H}_2$. For every finite collection of distinct positive algebraic $t_1, \dots, t_s$, the numbers*

$$e,\ Z,\ I_0(t_j), C(t_j), \beta(t_j) \quad (1 \le j \le s)$$

are algebraically independent over $\overline{\mathbb{Q}}$. In particular, $C, \gamma + \kappa, Z$ are algebraically independent.

Proof. Use Proposition 9.3 and Lemma 9.1 with $(u, v, \lambda) = (\mathcal{H}_1, \mathcal{H}_2, i)$. □

The transcendence of Z alone was already proved in the source mixed-Hankel construction. The new conclusion is its joint independence with the Bessel coordinates and the explicit Euler–Bessel combination. Positivity alone does not imply this: finite arithmetic reconstruction and conjugation are both used.

There is also an exact statement involving the unshifted quotient κ. The established bound $\operatorname{trdeg}_k k(\gamma, \delta, \kappa, C) \ge 3$ implies

$$Z \text{ is transcendental over } k(\kappa, C), \qquad \operatorname{trdeg}_k k(Z, \kappa, C) = 1 + \operatorname{trdeg}_k k(\kappa, C). \tag{23}$$

Indeed, if Z were algebraic over $k(\kappa, C)$, conjugation and Proposition 9.3 would make γ, δ algebraic over that field, forcing the transcendence degree of all four constants to be at most two. No independence of κ, C is assumed or concluded here.

10 Stronger base fields and a criterion for full independence

Set

$$\eta = \operatorname{Ein}_1(1) = \gamma + \delta/e, \qquad \kappa = K_0(2)/I_0(2), \qquad C = I_1(2)/I_0(2), \qquad H = \overline{\mathbb{Q}}(\gamma, \delta, \kappa, C).$$

Let $\mathcal{U}$ denote the countable indexed family

$$\{\operatorname{Ein}_r(\lambda) : \lambda \in \overline{\mathbb{Q}}^{\times},\ r \ge 1, (\lambda, r) \ne (1, 1)\},$$

and define the explicit coefficient field

$$\mathcal{A} = K_{\exp}(I_0(2), \mathcal{U}).$$

Corollary 10.1 (The three-dimensional bound survives base extension)**.** *One has*

$$\operatorname{trdeg}_{\mathcal{A}} \mathcal{A}(\gamma, \delta, \kappa, C) \ge 3.$$

Proof. Theorem 3.3 makes η, $\beta(2) = \gamma + \kappa$, and C algebraically independent over $\mathcal{A}$. All three belong to the field in the assertion, since $e \in \mathcal{A}$ and $\eta = \gamma + \delta/e$. □

This is stronger than merely adjoining independent auxiliary numbers to enlarge the total transcendence degree: the auxiliary values may all occur in coefficients, and three independent dimensions of the original four constants still remain. It does not establish a fourth dimension.

Theorem 10.2 (Rank three forces the entire polyexponential family to stay free)**.** *Suppose* $\operatorname{trdeg}_{\overline{\mathbb{Q}}} H = 3$*. Then the full family*

$$\{\mathrm{Ein}_r(\lambda) : \lambda \in \overline{\mathbb{Q}}^{\times},\ r \geq 1\},$$

including $\eta = \mathrm{Ein}_1(1)$*, is algebraically independent over* H*. More precisely,* e *and* $I_0(2)$ *are algebraically independent over* H*, and* $\mathcal{U}$ *is algebraically independent over* $H(e, I_0(2))$*.*

Proof. For distinct $U_1, \ldots, U_N \in \mathcal{U}$, Theorem 3.3 gives independence of the $N + 5$ numbers

$$e, \eta, I_0(2), C, \gamma + \kappa, U_1, \ldots, U_N.$$

They belong to $H(e, I_0(2), U_1, \ldots, U_N)$. With $N = 0$ and $\operatorname{trdeg} H = 3$, the two newly adjoined generators $e, I_0(2)$ must contribute exactly two dimensions. For general N the same count shows that every U_i contributes one further dimension, proving the more precise assertions.

Finally $\delta \neq 0$, so

$$e = \frac{\delta}{\eta - \gamma}, \qquad H(\eta) = H(e).$$

As e is transcendental over H, so is η. Independence of every finite subfamily of $\mathcal{U}$ over $H(e, I_0(2))$ implies its independence over $H(\eta)$. Adjoining the already transcendental η proves independence of the entire family over H. □

Corollary 10.3 (A relation sufficient for full independence)**.** *If finitely many distinct polyexponential values* $\mathrm{Ein}_r(\lambda)$ *satisfy a nonzero polynomial relation with coefficients in* H*, then*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} H = 4;$$

that is, $\gamma, \delta, \kappa, C$ *are algebraically independent. The same conclusion follows from a nonzero polynomial relation among members of* $\mathcal{U}$ *with coefficients in* $H(e, I_0(2))$*.*

Proof. The known bound gives $3 \leq \operatorname{trdeg} H \leq 4$. Either proposed relation excludes the value three by Theorem 10.2. □

No such relation is proved here. This is a sufficient condition, not a claim that the original four numbers are now independent. Known identities expressing an $\mathrm{Ein}_r(1)$ as a gamma-jet coefficient minus an incomplete tail do not meet this condition unless the additional tail and gamma-jet data are themselves controlled over H. Rapid convergence of auxiliary values to γ is likewise not a polynomial relation over H.

11 Exact relations and rationally shifted independent families

Write $k = \overline{\mathbb{Q}}$, $I = I_0(2)$, and

$$\eta = \gamma + \delta/e, \qquad \beta = \gamma + \kappa, \qquad A = \eta - \beta, \qquad U = 3\eta - 2\beta, \qquad V = \beta.$$

Equivalently, if $\ell = \gamma - 2\kappa$, then $U = \ell + 3\delta/e$ and $V = \ell + 3\kappa$. Theorem 3.3 makes e, I, C, η, β algebraically independent over k.

Proposition 11.1 (The complete relation ideal)**.** *Every two of A, U, V are algebraically independent over $k(e, I, C)$. The evaluation homomorphism*

$$k[X_e, X_I, X_C, X_A, X_U, X_V] \longrightarrow k[e, I, C, A, U, V]$$

has kernel precisely $(X_U - X_V - 3X_A)$. In particular,

$$\operatorname{trdeg}_k k(e, I, C, A, U, V) = 5.$$

Proof. Since $\eta = A + V$ and $\beta = V$, the five numbers e, I, C, A, V are algebraically independent. The only extra coordinate is $U = 3A + V$. Division by the monic polynomial $X_U - X_V - 3X_A$ leaves a remainder in $k[X_e, X_I, X_C, X_A, X_V]$, and independence forces any vanishing remainder to be zero. Finally, any two of $A, 3A + V, V$ recover A, V by an invertible linear transformation. □

The single relation among A, U, V does not persist after arbitrary common nonlinear shifts by C. The following result describes the exception completely, even when the three rational functions differ.

Theorem 11.2 (Rational shifts and their exact exception)**.** *Let $f, g, h \in k(T)$ be nonconstant, and put*

$$X = C + f(A), \qquad Y = C + g(U), \qquad Z = C + h(V).$$

Then e, I, X, Y, Z are algebraically independent over k, except when

$$\begin{gathered} f(T) = \alpha T + a_0, \quad g(T) = \rho T + b_0, \quad h(T) = \chi T + c_0, \\ \alpha\rho\chi \neq 0, \qquad \rho(3\chi + \alpha) = \alpha\chi. \end{gathered} \tag{24}$$

In this exceptional case, e, I, X, Z are algebraically independent, and every polynomial relation among e, I, X, Y, Z follows from

$$Y - b_0 = \frac{3\rho}{\alpha}(X - a_0) + \frac{\rho}{\chi}(Z - c_0). \tag{25}$$

Proof. Put $\mathcal{F} = k(e, I)$. Proposition 11.1 makes C, A, V algebraically independent over $\mathcal{F}$. All rational evaluations in the statement are therefore defined, and it suffices to study the rational functions

$$c + f(x), \qquad c + g(3x + y), \qquad c + h(y)$$

in the independent variables c, x, y over $\mathcal{F}$. In characteristic zero, their algebraic independence is equivalent to their Jacobian determinant being nonzero. This determinant is

$$D = 3g'(3x + y)h'(y) - f'(x)h'(y) + f'(x)g'(3x + y).$$

The three derivatives are nonzero rational functions. On defining $F = 1/f'$, $G = 1/g'$, and $J = 1/h'$, the identity $D = 0$ becomes

$$G(3x + y) = 3F(x) + J(y). \tag{26}$$

Differentiate with respect to x and y. The result is

$$G'(3x + y) = F'(x) = J'(y).$$

Since x and y are independent variables, these rational functions are a common constant $s \in k$. Hence, for constants $b, d \in k$,

$$F(T) = sT + b, \qquad J(T) = sT + d, \qquad G(T) = sT + 3b + d.$$

If $s \neq 0$, the equality $f'(T) = 1/(sT + b)$ is impossible: the derivative of a rational function has zero residue at every finite pole, whereas the right side has residue $1/s$ at $T = -b/s$. Consequently $s = 0$. The nonzero constant reciprocals of the derivatives imply that f, g, h are affine with nonzero slopes α, ρ, χ, and (26) reduces to

$$\frac{1}{\rho} = \frac{3}{\alpha} + \frac{1}{\chi}.$$

This is exactly (24); conversely that condition makes $D = 0$ and gives (25).

In the exceptional case, $c + \alpha x + a_0$ and $c + \chi y + c_0$ are algebraically independent over $\mathcal{F}$, for example by their rank-two Jacobian. Thus e, I, X, Z are independent, and division by the monic linear relation in Y proves the assertion about all relations, exactly as in Proposition 11.1. □

Corollary 11.3 (Every common rational shift works)**.** *For every nonconstant $f \in k(T)$, the five numbers*

$$e,\ I,\ C + f(A),\ C + f(U),\ C + f(V)$$

are algebraically independent over k. This includes $f(T) = T$, $f(T) = T^m$ for every integer $m \geq 1$, and $f(T) = 1/T$.

Proof. The only possible exception would require f to be affine and its common nonzero slope α to satisfy $4\alpha^2 = \alpha^2$. This cannot occur in characteristic zero. □

12 Arithmetic consequences of the Bessel primitive

Write

$$B = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n(n!)^2} = 2H_{2i}(1), \qquad \mathcal{L} = B - \gamma.$$

The analytic interpretation of $\mathcal{L}$ as a critical Newton–Laguerre limit is established in the approximation section. Here only its arithmetic consequences are needed.

Corollary 12.1 (The new limit joins the one-defect family)**.** *With $\nu = -\pi Y_0(2)/(2J_0(2))$ and $C_J = J_1(2)/J_0(2)$,*

$$\operatorname{trdeg}_k k(\gamma, \delta, \kappa, C, \nu, C_J, \mathcal{L}) \geq 6.$$

Moreover, at most one of

$$\gamma,\ \delta,\ \kappa,\ \gamma - 2\kappa,\ \nu,\ \gamma - 2\nu,\ \mathcal{L}$$

is algebraic. If $\mathcal{L}$ is algebraic, the six original numbers $\gamma, \delta, \kappa, C, \nu, C_J$ are all algebraically independent; indeed they are jointly independent with $e, I_0(2), J_0(2)$.

Proof. Theorem 4.3 gives independence of

$$e, \eta, I_0(2), C, \gamma + \kappa, J_0(2), C_J, \gamma + \nu, B.$$

The seven coordinates $e, \eta, C, \gamma + \kappa, C_J, \gamma + \nu, B$ belong to the field in the first assertion after adjoining e; removing e lowers transcendence degree by at most one. Similarly $e, \eta, \gamma + \kappa, \gamma + \nu, B$ belong to $k(\gamma, \delta, \kappa, \nu, \mathcal{L}, e)$, so $\operatorname{trdeg} k(\gamma, \delta, \kappa, \nu, \mathcal{L}) \geq 4$. Any two different forms in the displayed list have linearly independent coefficient vectors in these five variables. If both were algebraic, the latter transcendence degree would be at most three, a contradiction.

Finally, if $\mathcal{L} = c \in k$, then $\gamma = B - c$. The identities $\delta = e(\eta - \gamma)$, $\kappa = (\gamma + \kappa) - \gamma$, and $\nu = (\gamma + \nu) - \gamma$ show that the nine independent input coordinates generate exactly $k(e, \gamma, \delta, \kappa, C, \nu, C_J, I_0(2), J_0(2))$. Its nine displayed generators are therefore independent. □

Corollary 12.2 (Explicit new independent combinations)**.** *The five numbers*

$$\mathcal{L}-\kappa,\quad \mathcal{L}-\nu,\quad \mathcal{L}-\delta/e,\quad \mathcal{L}+\mathrm{Ei}(1),\quad \mathcal{L}+\mathrm{Ci}(1)$$

are algebraically independent. They remain jointly independent after adjoining $e, I_0(2), J_0(2), C, C_J$*.*

Proof. Put $M=\mathrm{Ein}(-1)$ and $R=(\mathrm{Ein}(i)+\mathrm{Ein}(-i))/2$. Then $\mathrm{Ei}(1)=\gamma-M$ and $\mathrm{Ci}(1)=\gamma-R$. The five displayed numbers equal

$$B-(\gamma+\kappa),\quad B-(\gamma+\nu),\quad B-\eta,\quad B-M,\quad B-R.$$

The six coordinates $B,\gamma+\kappa,\gamma+\nu,\eta,M,R$ are independent over $k(e,I_0(2),J_0(2),C,C_J)$ by Theorem 4.3 and the invertible change from $\mathrm{Ein}(\pm i)$ to their real and imaginary combinations. The five differences are a full-rank linear transformation of six independent variables, which proves the assertion. □

Corollary 12.3 (A criterion over the original four-constant field)**.** *Let* $H_4=k(\gamma,\delta,\kappa,C)$*. If* $\mathcal{L}$ *is algebraic over* H_4*, then* $\mathrm{trdeg}_k H_4=4$*. Moreover*

$$C,\ C_J,\ \mathcal{L}-\delta/e,\ \mathcal{L}-\kappa,\ \mathcal{L}-\nu$$

are jointly independent over $K_{\exp}$*, and remain so after adjoining* $B=\gamma+\mathcal{L}$*. If* $\mathcal{L}\in k$*, then* $\gamma,\delta,\kappa,C,\nu,C_J$ *are jointly independent over* $K_{\exp}$*.*

Proof. The six independent numbers $e, I_0(2), \eta, \gamma+\kappa, C, B$ lie in $H_4(e,I_0(2),\mathcal{L})$. Under the algebraicity assumption, its transcendence degree is at most $\mathrm{trdeg}\,H_4+2$, so $\mathrm{trdeg}\,H_4\geq 4$. For the second assertion, use the invertible change of coordinates

$$(C,C_J,\eta,\gamma+\kappa,\gamma+\nu,B)\longmapsto(C,C_J,B-\eta,B-(\gamma+\kappa),B-(\gamma+\nu),B).$$

For the last assertion, $\gamma=B-\mathcal{L}$ and $\delta=e(\eta-\gamma)$ give the same invertible rational change as above, now over $K_{\exp}$, where e is already a coefficient. □

13 Inverse-function points and arithmetic consequences

Let $q_*, x_*>0$ be defined by

$$\mathrm{Ein}(q_*)=\gamma,\qquad \beta(x_*)=\gamma.$$

The second equation is equivalent to $K_0(x_*)/I_0(x_*)=-\log(x_*/2)$. Both roots exist uniquely. The first defining function is strictly increasing, starts at zero, and is unbounded. For the second,

$$\beta'(x)=\frac{1-I_0(x)^{-2}}{x}>0\quad(x>0),$$

while $\beta(0)=0$ by its series and $\beta(x)\to\infty$. Moreover $0<q_*<1$ and $0<x_*<2$, since $\eta>\gamma$ and $\beta(2)=\gamma+\kappa>\gamma$. Numerically,

$$q_*=0.67635507786536\ldots,\qquad x_*=1.87744825105766\ldots.$$

Theorem 13.1 (Conditional full independence and E-integrality)**.** *If either* q_* *or* x_* *is algebraic, the nine numbers*

$$e, I_0(2), J_0(2), \gamma, \delta, \kappa, C, \nu, C_J$$

are algebraically independent over k*. The six numbers* $\gamma,\delta,\kappa,C,\nu,C_J$ *are then jointly independent over* $K_{\exp}$ *as well. The two roots cannot both be algebraic. Furthermore,*

$$q_*\in k\implies \gamma,\delta\in\mathbf{E},\qquad x_*\in k\implies \gamma,\delta\notin\overline{\mathbf{E}}.$$

Proof. If $q_* \in k$, add $\mathrm{Ein}(q_*)$ to the joint input at the Ein point 1 and Bessel scales $2, 2i$. Its point is different from 1. Substituting $\gamma = \mathrm{Ein}(q_*)$, $\delta = e(\eta - \gamma)$, $\kappa = \beta(2) - \gamma$, and $\nu = \beta(2i) - \gamma$ is an invertible rational change of coordinates. This proves both independence assertions. It also shows $\gamma, \delta \in \mathbf{E}$.

If $x_* \in k$, use the additional Bessel scale x_*, which is neither ± 2 nor $\pm 2i$, and retain the extra coordinate $\beta(x_*) = \gamma$. The same substitutions prove independence. The image of $\beta(x_*)$ under Φ_- has genuine poles, by the positive-scale phase formula; its numerator at a zero of the denominator does not vanish. The monic-relation pole criterion gives $\gamma \notin \overline{\mathbf{E}}$. If δ were integral, then $\gamma = \eta - e^{-1}\delta$ would be integral, since $e^{-1} \in \mathbf{E}$ and integral elements form a ring. Hence $\delta \notin \overline{\mathbf{E}}$ too.

If both roots were algebraic, the joint independence of $\mathrm{Ein}(q_*)$ and $\beta(x_*)$ would be contradicted by their equality. The incompatible integrality conclusions provide another proof. □

Let $H_4 = k(\gamma, \delta, \kappa, C)$. Theorem 13.1 and Corollary 4.6 imply

$$\mathrm{trdeg}_k\, H_4 = 3 \implies q_*, x_* \text{ transcendental and } \mathcal{L} \text{ transcendental over } H_4.$$

Thus any polynomial relation among the original four constants would force both roots to be transcendental. Conversely, algebraicity of either root supplies a sufficient condition for rank four; no algebraicity assertion about either root has been proved. In either algebraic-root case,

$$\dim_k \mathrm{span}_k\{1, e, \delta, e\eta\} = 4.$$

Indeed $e\eta = e\gamma + \delta$, and a linear relation would be a polynomial relation among the independent numbers e, γ, δ.

14 Independent-block alternatives and the stronger conditional family

Put $k = \overline{\mathbb{Q}}$ and abbreviate

$$I = I_0(2), \quad I' = I_1(2), \quad K = K_0(2), \quad J = J_0(2), \quad J' = J_1(2), \quad Y_* = \pi Y_0(2),$$

$$\kappa = K/I, \quad C = I'/I, \quad C_J = J'/J, \quad \eta = \mathrm{Ein}(1) = \gamma + \delta/e, \quad \beta_I = \gamma + \kappa, \quad \beta_J = \gamma - \frac{Y_*}{2J}.$$

In this section, Y_0 denotes the classical Bessel function of the second kind. It is not the harmonic-coefficient entire function used in the construction of the joint E-system. Also set

$$B = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n(n!)^2}, \qquad \mathcal{L} = B - \gamma.$$

Theorem 4.3 gives the independence in

$$e, \eta, I, C, \beta_I, J, C_J, \beta_J, B \quad \text{are algebraically independent over } k. \tag{27}$$

The assertions below that involve B or $\mathcal{L}$ use this primitive extension; the two- and three-block assertions need only the corresponding subtuples. None of these assertions presupposes the transcendence of γ.

14.1 A finite-block lemma

Lemma 14.1. *Let $X_1, \dots, X_r$ be finite tuples whose concatenation is algebraically independent over an algebraically closed field $k \subset \mathbb{C}$. If $g \in \mathbb{C}$ is transcendental over k, then g is algebraic over at most one of the fields $k(X_j)$. Consequently, at least $r-1$ of the tuples (X_j, g) are algebraically independent over k. For $r \geq 2$, transcendence of g is equivalent to this last assertion.*

Proof. Suppose that g is algebraic over both $k(X_i)$ and $k(X_j)$, where $i \neq j$. Write $d_i = |X_i|$ and $d_j = |X_j|$. The two finitely generated fields $F_i = k(X_i, g)$ and $F_j = k(X_j, g)$ have transcendence degrees d_i and d_j, respectively. Their compositum contains the independent concatenation (X_i, X_j), and their intersection contains $k(g)$. Submodularity of transcendence degree gives

$$d_i + d_j \geq \operatorname{trdeg}_k(F_i F_j) + \operatorname{trdeg}_k(F_i \cap F_j) \geq d_i + d_j + 1,$$

a contradiction. Thus g is transcendental over all but at most one $k(X_j)$, which is equivalent to the asserted independence of (X_j, g) since X_j is already independent. Conversely, when $r \geq 2$, at least one independent tuple containing g implies that g is transcendental. □

The lemma is the finite-block version of the algebraic-tail intersection argument in manuscript [021] [9], Lemma `em:tail-intersection`. It is an elementary field-theoretic input, rather than a new transcendence theorem.

Theorem 14.2 (Two, three, and four independent worlds)**.** *The following equivalences hold.*

1. *γ is transcendental if and only if at least one of*

$$(e, \gamma, \delta), \qquad (I, I', K, \gamma)$$

is algebraically independent over k.

2. *γ is transcendental if and only if at least two of*

$$(e, \gamma, \delta), \qquad (I, I', K, \gamma), \qquad (J, J', Y_*, \gamma)$$

are each algebraically independent over k.

3. *With the primitive extension, γ is transcendental if and only if at least three of the four tuples*

$$(e, \gamma, \delta), \qquad (I, I', K, \gamma), \qquad (J, J', Y_*, \gamma), \qquad (\gamma, \mathcal{L}) \tag{28}$$

are each algebraically independent over k.

These are statements about the individual tuples, not about the independence of their concatenation.

Proof. Use the respective first two, first three, or all four blocks

$$X_1 = (e, \eta), \quad X_2 = (I, I', \beta_I), \quad X_3 = (J, J', \beta_J), \quad X_4 = (B).$$

Their concatenation is independent by (27), since replacing C by $I' = IC$ and C_J by $J' = JC_J$ gives invertible rational changes of coordinates. The nonvanishing of I, J follows also from their transcendence (and for I from positivity). For each block, adjoining γ and making another invertible rational change of coordinates gives

$$\delta = e(\eta - \gamma), \qquad K = I(\beta_I - \gamma), \qquad Y_* = 2J(\gamma - \beta_J), \qquad \mathcal{L} = B - \gamma.$$

The displayed tuples and (X_j, γ) therefore generate the same fields and have the same numbers of entries. Apply Lemma 14.1. If γ were algebraic, every tuple in question would be dependent, proving the reverse implications. □

The first assertion does not imply that both of its tuples are independent whenever γ is transcendental. At the level of these algebraic-independence inputs, for instance, the assignment $g = \eta - d/e$ with $0 \neq d \in k$ makes the corresponding $\delta = d$ algebraic while leaving g transcendental and independent of the Bessel block. Such a formal example does not assign a new value to the actual Euler constant; it shows the limitation of this abstract implication. The alternatives in Theorem 14.2 do not prove independence of $\gamma, \delta, \kappa, C$.

14.2 A stronger consequence if the original four are dependent

Corollary 14.3. *Let $H_4 = k(\gamma, \delta, \kappa, C)$. If the four generators of H_4 are algebraically dependent, then*

$$e,\ K_0(2),\ J_0(2),\ J_1(2),\ \pi Y_0(2),\ \mathcal{L} \quad \text{are algebraically independent over } H_4. \tag{29}$$

In particular, $K_0(2)$ and $\mathcal{L}$ are algebraically independent over $H_4(e)$. Even without the J-block, the shorter conclusion that $e, K_0(2), \mathcal{L}$ are independent over H_4 follows.

Proof. The five independent values e, η, I, C, β_I lie in $H_4(e, I)$, so $\operatorname{trdeg}_k H_4 \geq 3$. Under the stated hypothesis equality holds. By (27), the field

$$F = H_4(e, I, J, C_J, \beta_J, B)$$

has transcendence degree at least nine over k. It is obtained from the rank-three field H_4 by adjoining six elements, so its degree is exactly nine and these six elements are jointly independent over H_4. Over H_4 one has the invertible transformations

$$I \longleftrightarrow K = \kappa I, \qquad (J, C_J, \beta_J) \longleftrightarrow (J, J' = JC_J, Y_* = 2J(\gamma - \beta_J)), \qquad B \longleftrightarrow \mathcal{L} = B - \gamma.$$

Here $\kappa > 0$ and $J \neq 0$. This proves (29) and its subtuple conclusions. For the shorter proof without the J-block, the six independent entries $e, \eta, I, C, \beta_I, B$ lie in $H_4(e, I, B)$; the identical rank count proves independence of e, I, B over H_4. □

Thus, for example, algebraicity of either $K_0(2)$ or $\mathcal{L}$ over $H_4(e)$ would force $\operatorname{trdeg}_k H_4 = 4$. This is a conditional criterion, not a proof of that algebraicity or of rank four.

15 Arithmetic of the two mirror limits

The approximation results identify

$$L_- = \ell = \gamma - 2\kappa, \qquad L_+ = \gamma + 2\kappa.$$

Both differ from the Newton–Laguerre primitive limit $\mathcal{L} = B - \gamma$. Their algebraic consequences are as follows.

Corollary 15.1. *Among $\gamma, \delta, \kappa, \ell, L_+$, at most one is algebraic. More generally, at most one of δ, κ, and all the numbers $\gamma + c\kappa$ with $c \in k$ can be algebraic.*

Proof. Two distinct forms among κ and $\gamma + c\kappa$ would, if both algebraic, force both γ and κ algebraic, contradicting transcendence of $\beta_I = \gamma + \kappa$. If δ and κ were algebraic, the identity $e(\eta - \beta_I) = \delta - e\kappa$ would contradict independence of e, η, β_I. If δ and $s = \gamma + c\kappa$ were algebraic, the case $c = 1$ is already excluded by $s = \beta_I$. For $c \neq 1$, write $\gamma = (s - c\beta_I)/(1 - c)$ and use $\eta = \gamma + \delta/e$ to obtain the same forbidden algebraic relation among e, η, β_I. The five-number assertion follows by taking $c = 0, -2, 2$. □

Corollary 15.2. *One has*

$$\operatorname{trdeg}_k k(\delta, C, L_+, L_-) \geq 3.$$

If either L_+ or L_- is algebraic, then γ, δ, C are algebraically independent.

Proof. The identities

$$\gamma = (L_+ + L_-)/2, \qquad \kappa = (L_+ - L_-)/4$$

give $k(\delta, C, L_+, L_-) = H_4$, proving the first assertion. If either mirror limit is algebraic, then κ is a k-affine function of γ, so $H_4 = k(\gamma, \delta, C)$. The established lower bound three forces these three generators to be independent. □

The mirror pair is an invertible linear reparametrization of γ, κ. It resolves an asymptotic identification without adding a new independent coordinate to H_4.

16 Upper-half-plane values over the full Bessel-anchor field

The multiscale primitive extension and the complex-scale joint theorem can be combined with conjugation without spending any additional real adjunction beyond γ. For each nonzero algebraic scale t, put

$$H_t(1) = -\sum_{n\geq 1} \frac{(t/2)^{2n}}{2n(n!)^2}.$$

Let $\mathcal{T}$ be any choice of one representative from each sign class in $\overline{\mathbb{Q}}^\times/\{\pm 1\}$. The field below is independent of that choice: $I_0, \beta, H_t(1)$ are even in t, while $C(-t) = -C(t)$. Define

$$\begin{aligned} F_0 = K_{\exp}\big(&\operatorname{Ein}_r(a) : a \in \overline{\mathbb{Q}} \cap \mathbb{R}^\times,\ r \geq 1; \\ &I_0(t), C(t), \beta(t), H_t(1) : t \in \mathcal{T}\big). \end{aligned} \tag{30}$$

Theorem 4.3 gives joint independence of these coordinates and any finite family of polyexponential values at additional distinct nonreal arguments. All assertions about the infinite fields are interpreted through finite subsets of their generators.

Theorem 16.1 (Upper-half-plane independence with all Bessel anchors)**.** *The entire family*

$$\{\operatorname{Ein}_r(\lambda) : \lambda \in \overline{\mathbb{Q}},\ \operatorname{Im}\lambda > 0,\ r \geq 1\}$$

is algebraically independent over $F_0(\gamma)$.

Proof. The field F_0 is stable under complex conjugation. The field $K_{\exp}$ is stable because conjugation sends e^α to $e^{\overline{\alpha}}$, though $K_{\exp}$ is not a real field. The real-point polyexponential values are real. For each algebraic t, conjugation sends

$$(I_0(t), C(t), \beta(t), H_t(1)) \quad \text{to} \quad (I_0(\bar{t}), C(\bar{t}), \beta(\bar{t}), H_{\bar{t}}(1)).$$

The conjugate scale is represented in $\mathcal{T}$ up to sign, and the even/odd identities above put all these values back in F_0. Thus arbitrary complex algebraic Bessel scales, not only real and imaginary scales, are allowed as anchors.

Choose a finite tuple $u_1, \ldots, u_m$ from the upper-half-plane family. Its conjugates correspond to the same orders at lower-half- plane arguments. These indexed values are all distinct from one

another and from the real-point polyexponential values generating F_0. The joint theorem therefore makes

$$u_1, \dots, u_m, \overline{u_1}, \dots, \overline{u_m}$$

algebraically independent over F_0. If the first m values were dependent over $F_0(\gamma)$, their rank there would be at most $m-1$. Since γ is real and F_0 is conjugation-stable, the conjugate tuple would satisfy the same bound. Their union would then have rank at most $2m-2$ over $F_0(\gamma)$, and at most $2m-1$ over F_0, since adjoining one element increases transcendence degree by at most one. This contradicts its rank $2m$ over F_0. Every finite tuple is therefore independent. □

Corollary 16.2 (The primitive and classical Bessel anchors are allowed)**.** *Put*

$$H_4 = \overline{\mathbb{Q}}(\gamma, \delta, \kappa, C), \qquad \nu = -\frac{\pi Y_0(2)}{2J_0(2)}, \qquad C_J = \frac{J_1(2)}{J_0(2)},$$

$$B = 2H_{2i}(1), \qquad \mathcal{L} = B - \gamma.$$

The upper-half-plane family of Theorem 16.1 is algebraically independent over

$$H_4\big(\nu, C_J, \mathcal{L}, e, I_0(2), J_0(2)\big). \tag{31}$$

This coefficient field also contains $K_0(2)$, $J_1(2)$, and $\pi Y_0(2)$. The conclusion is unconditional with respect to whether H_4 has transcendence degree three or four.

Proof. All the generators in (31) lie in $F_0(\gamma)$. Indeed,

$$\delta = e\big(\mathrm{Ein}(1) - \gamma\big), \quad \kappa = \beta(2) - \gamma, \quad \nu = \beta(2i) - \gamma, \quad C_J = -iC(2i),$$

and $J_0(2) = I_0(2i)$ and $\mathcal{L} = 2H_{2i}(1) - \gamma$. The three further values are recovered by

$$K_0(2) = \kappa I_0(2), \qquad J_1(2) = C_J J_0(2), \qquad \pi Y_0(2) = -2\nu J_0(2).$$

Apply Theorem 16.1 and restrict its coefficient field. □

The proof does not permit an arbitrary extra real constant to be adjoined. In particular it provides no permission to adjoin $\log a$ or $\sqrt{\pi}$ unless that particular constant is already shown to belong to the defined field. The new anchors above are permitted because their required real information is generated by the joint E-coordinates and the same single real constant γ; they are not separate real adjunctions.

17 The exact cost of finitely many real adjunctions

The conjugation argument has a general finite-defect form. The result below needs finite transcendence degree of the coefficient extension, not a finite degree or a specified list of generators.

Theorem 17.1 (Conjugation and the half-defect bound)**.** *Let $F \subseteq L \subseteq \mathbb{C}$ be fields stable under complex conjugation, with $d = \mathrm{trdeg}_F L < \infty$. Let $(u_i)_{i \geq 1}$ be a countable indexed family such that, for every finite set of indices T, the concatenation*

$$(u_i, \overline{u_i} : i \in T)$$

is algebraically independent over F. Put $h = \lfloor d/2 \rfloor$. Then for every finite T,

$$\mathrm{trdeg}_L\, L(u_i : i \in T) \geq |T| - h. \tag{32}$$

At most h of the u_i are algebraic over L. One can delete at most h members of the entire family so that the rest are jointly algebraically independent over L.

Moreover, there is a finite set of indices T_0 such that all remaining values are jointly algebraically independent over $L(u_i : i \in T_0)$. If $\mathfrak{I}$ is the ideal of all polynomial relations in $L[X_1, X_2, \ldots]$, then

$$\mathfrak{I} = (\mathfrak{I} \cap L[X_i : i \in T_0])\, L[X_1, X_2, \ldots]. \tag{33}$$

The finite-variable prime ideal on the right has height at most h. In particular, all relations admit finitely many polynomial generators. If $d \leq 3$, the global relation ideal is either zero or generated by one irreducible polynomial involving only finitely many variables.

Proof. For a finite tuple of length m, write $r = \operatorname{trdeg}_L L(u_i : i \in T)$. Conjugation stability makes the conjugate tuple have the same rank r over L. The independent concatenation lies in $L(u_i, \overline{u_i} : i \in T)$, whose rank over F is at most $d + 2r$. Thus

$$2m \leq d + 2r.$$

Since $m - r$ is an integer, this proves (32). Applying it to any finite tuple of members algebraic over L gives their number at most h.

Enumerate the family and greedily retain u_i exactly when it is transcendental over L and the previously retained values. At each finite stage the retained tuple is a transcendence basis for the field generated by all values processed so far. Inequality (32) therefore permits at most h rejections in total. All retained values are jointly independent.

If there is no rejection, take T_0 empty. Otherwise, take T_0 to be the finite initial segment ending with the last rejection. Every rejected value is algebraic over the previously retained ones, so $L(u_i : i \in T_0)$ is algebraic over the field generated by the retained values in that segment. All later values are jointly independent over the latter field, and remain so after this algebraic extension. This proves the asserted independence over the finite core.

To obtain (33), group a polynomial relation by monomials in variables outside T_0. Independence of their values over $L(u_i : i \in T_0)$ says that every coefficient vanishes at the core tuple. Thus every coefficient lies in the finite-core relation ideal, which proves the displayed equality. Its height is

$$|T_0| - \operatorname{trdeg}_L L(u_i : i \in T_0) \leq h.$$

The polynomial ring in the finitely many core variables is noetherian, so that ideal is finitely generated. If $h \leq 1$, a nonzero prime ideal of that height in this UFD is principal and generated by an irreducible polynomial. Its extension gives the final assertion. □

Remark 17.2 (Sharpness as a field-theoretic statement). The factor $1/2$ cannot be improved under these hypotheses alone. Take real numbers $x_1, y_1, \ldots, x_h, y_h$ algebraically independent over k, put $u_j = x_j + iy_j$, and adjoin all $2h$ real coordinates to $F = k$. The values and their conjugates were independent over F, but all h values now lie in the enlarged field. Additional independent real generators give the corresponding examples for odd d.

Corollary 17.3 (A finite relation core after adjoining Euler's constant, π, and $\log 2$)**.** *Let F_0 be the field in (30), or its enlargement obtained by replacing the depth-one primitives $H_t(1)$ by all $H_{t,r}(1)$ with $r \geq 1$. Put*

$$L = F_0(\gamma, \pi, \log 2).$$

For the family

$$\mathcal{U} = \{\mathrm{Ein}_r(\lambda) : \lambda \in \overline{\mathbb{Q}},\ \operatorname{Im}\lambda > 0,\ r \geq 1\},$$

every m-element subfamily has rank at least $m-1$ over L. At most one member is algebraic over L, and deleting at most one member makes the entire family algebraically independent over L. All its polynomial relations over L are generated by a single finite-variable irreducible polynomial, if there is any relation at all. The same statements hold over the algebraic closure of L inside $\mathbb{C}$.

Proof. The conjugation stability of F_0 was proved in Theorem 16.1. The same proof applies to all primitive depths, because conjugation sends $H_{t,r}(1)$ to $H_{\bar{t},r}(1)$ and these functions are even in the scale. The joint theorem, with Theorem 5.3 for the enlarged anchor, makes every finite upper-half-plane tuple and its conjugates jointly independent over F_0. The three adjoined constants are real, so L is conjugation-stable and $\operatorname{trdeg}_{F_0} L \le 3$. Apply Theorem 17.1. Its algebraic closure is again conjugation-stable and has the same transcendence degree over F_0. □

For example, for every $N \ge 1$,

$$\operatorname{trdeg}_L L(\mathrm{Ein}(i), \mathrm{Ein}(2i), \dots, \mathrm{Ein}(Ni)) \ge N-1.$$

The corollary neither identifies a dependent member nor asserts that any dependence exists. In particular, it gives no arithmetic classification of γ or of δ.

Part III
Integral intersections of rational fields

18 Continued phase embeddings and their Bessel images

Put $k = \overline{\mathbb{Q}}$. Let $\mathbf{E}$ be the ring of E-function values at algebraic points and let $\overline{\mathbf{E}}$ be its integral closure inside $\mathbb{C}$. Thus "integral" means satisfying a monic polynomial with coefficients in $\mathbf{E}$; it does not mean algebraic over its fraction field. The phase-embedding construction from manuscript [033] [19] is reproduced here. The arithmetic analytic input in this section is the Fischler–Rivoal phase embedding and the properties of its component functions [3, §2 and Proposition 3.1]. In particular, the conclusions here do not require a new algebraic-independence input.

18.1 Continuation of the phase embedding

Write

$$\mathcal{R} = \mathbb{C}[\overline{\mathbb{Q}}_{\mathrm{add}}] = \left\{ \sum_{\rho \in k} c_\rho e^{\rho X} : c_\rho \in \mathbb{C},\ \text{finite support} \right\}.$$

The equality with the group algebra uses linear independence of distinct exponential functions. Fix $-\pi/2 < \theta < \pi/2$ and use the right-lateral convention at an exceptional direction. If f is an E-function, let

$$f(z) = \sum_{\rho} f_\rho(z) e^{\rho z}$$

be its canonical component decomposition. Fischler–Rivoal state immediately before their Theorem 2.3 that each $e^{\rho z} f_\rho(z)$ solves an E-operator whose only possible finite singularity is zero. Consequently every component admits holomorphic continuation along every path in $\mathbb{C}^\times$. Let M^j mean continuation around a based loop at 1 of counterclockwise winding number $j \in \mathbb{Z}$.

Lemma 18.1 (Continued embeddings)**.** *For every $j \in \mathbb{Z}$, the prescription*

$$\Phi_\theta^{[j]}(f(1))(X) = \sum_\rho (M^j f_\rho)(1) e^{\rho X}$$

defines an injective k-algebra homomorphism $\mathbf{E} \longrightarrow \mathcal{R}$. It satisfies

$$\Phi_\theta^{[j]}(\xi)(1) = \xi, \qquad \Phi_\theta^{[j]}(e^a) = e^{aX} \quad (a \in k).$$

Proof. If $f(1) = g(1)$, Beukers' zero-removal theorem, as used in the proof of Fischler–Rivoal's Proposition 3.1, gives an E-function h with $f - g = (z-1)h$. Choose one sufficiently small positive perturbation of θ for all three functions. Uniqueness of the canonical decompositions gives

$$f_\rho - g_\rho = (z-1)h_\rho.$$

Continue this identity around the same loop. Since the continued h_ρ is holomorphic at the endpoint 1, evaluation there shows that the proposed map is independent of the representative.

The component identities for sums and products are

$$(f+g)_\rho = f_\rho + g_\rho, \qquad (fg)_\rho = \sum_{\alpha+\beta=\rho} f_\alpha g_\beta.$$

These identities hold initially by uniqueness and remain true after continuation. They prove additivity and multiplicativity. Constant functions and pure exponentials give the stated normalizations. Finally the entire function f is single valued, so continuation of its total component sum yields

$$\sum_\rho (M^j f_\rho)(1) e^\rho = f(1).$$

This is evaluation compatibility and proves injectivity. □

The summation direction is held fixed in this construction. The continued components are not asserted to be the canonical asymptotic components in a new sector.

Lemma 18.2 (Normality and the pole obstruction)**.** *The ring $\mathcal{R}$ is integrally closed. If $\xi \in \mathrm{Frac}(\mathbf{E}) \cap \overline{\mathbf{E}}$, the extension of every $\Phi_\theta^{[j]}$ to the fraction field sends ξ into $\mathcal{R}$. In particular, an image with a finite pole proves $\xi \notin \overline{\mathbf{E}}$.*

Proof. Suppose $H \in \mathrm{Frac}(\mathcal{R})$ is integral over $\mathcal{R}$. The supports of a numerator and denominator for H, together with the finitely many coefficients of a monic integral equation, generate a finitely generated subgroup $\Gamma \subset k_{\mathrm{add}}$. This subgroup is free abelian, and $\mathbb{C}[\Gamma]$ is a Laurent polynomial ring. We have $H \in \mathrm{Frac}(\mathbb{C}[\Gamma])$, and H is integral over $\mathbb{C}[\Gamma]$. Normality of a Laurent polynomial ring therefore gives $H \in \mathbb{C}[\Gamma] \subset \mathcal{R}$.

Apply a continued embedding to a monic integral equation for ξ. Its image is integral over $\mathcal{R}$ and belongs to $\mathrm{Frac}(\mathcal{R})$, hence lies in $\mathcal{R}$. Every element of $\mathcal{R}$ is entire. Alternatively, at a finite pole of a meromorphic image, the highest power in the monic equation has strictly larger pole order than every other term, which is impossible. □

18.2 Explicit winding formulas

For positive algebraic t define

$$I = I_0(t), \quad J = I_1(t), \quad K = K_0(t), \quad L = K_1(t), \qquad g = \gamma + \log(t/2).$$

All four Bessel values here are positive. Let

$$Y(t) = \sum_{n\geq 0} \frac{H_n (t/2)^{2n}}{(n!)^2}, \qquad \beta(t) = \frac{Y(t)}{I_0(t)} = \gamma + \log(t/2) + \frac{K_0(t)}{I_0(t)}, \qquad C(t) = \frac{I_1(t)}{I_0(t)}.$$

The function Y is the harmonic Bessel E-function, not the ordinary Bessel function Y_0. In particular, $\beta(t)$ and $C(t)$ belong to Frac($\mathbf{E}$). Set

$$\varepsilon = \frac{1}{\pi i}, \qquad \tau = 2\pi i.$$

At direction $\theta = -\pi/4$, the two canonical components of $I_0(tz)$, including their exponential factors, are

$$y_+(z) = I_0(tz) - \varepsilon K_0(tz), \qquad y_-(z) = \varepsilon K_0(tz).$$

These are the standard two-phase Bessel components used in the original pole calculation. These follow by applying the Bessel connection formula to the recessive solution in the adjoining sector [2, Sections 10.27 and 10.34]. The local continuation laws are

$$M^j K_0(tz) = K_0(tz) - j\tau I_0(tz), \qquad M^j K_1(tz) = K_1(tz) + j\tau I_1(tz).$$

Define

$$\begin{aligned} A_j &= e^{-t}((1+2j)I - \varepsilon K), & B_j &= e^{t}(\varepsilon K - 2jI), \\ P_j &= e^{-t}((1+2j)J + \varepsilon L), & Q_j &= e^{t}(-2jJ - \varepsilon L). \end{aligned}$$

The imaginary parts of εK show that $A_j B_j \neq 0$ for every integer j. Differentiating the two component identities gives

$$\Phi^{[j]}_{-\pi/4}(I_0(t)) = A_j e^{tX} + B_j e^{-tX}, \tag{34}$$

$$\Phi^{[j]}_{-\pi/4}(I_1(t)) = P_j e^{tX} + Q_j e^{-tX}. \tag{35}$$

The Wronskian identity $IL + JK = 1/t$ implies

$$A_j Q_j - B_j P_j = -\frac{\varepsilon}{t} \neq 0. \tag{36}$$

For completeness, the canonical decomposition of $Y(tz)$ is

$$Y(tz) = (g + \log z) y_+(z) + \big(1 + \varepsilon(g + \log z)\big) K_0(tz).$$

Continue both brackets and K_0 around the same loop, using $M^j \log z = \log z + j\tau$. At $z = 1$ this gives

$$\Phi^{[j]}_{-\pi/4}(Y(t)) = (g + j\tau) A_j e^{tX} + (g + j\tau + \pi i) B_j e^{-tX}. \tag{37}$$

These formulas also check evaluation compatibility at $X = 1$.

19 The full rational two-variable Bessel field

The winding argument closes a larger field than the two one-variable fields separately. Throughout this section, $k = \overline{\mathbb{Q}}$, $t > 0$ is algebraic, and

$$I = I_0(t), \quad J = I_1(t), \quad K = K_0(t), \quad L = K_1(t), \qquad \beta = \gamma + \log(t/2) + K/I, \quad C = J/I.$$

The independence needed below will follow directly from the winding embeddings; no additional joint-specialization input is required in this section.

Theorem 19.1 (Full rational Bessel-field exclusion)**.** *Let $\overline{\mathbf{E}}$ denote the elements of $\mathbb{C}$ integral over the ring $\mathbf{E}$ of E-function values. Then*

$$\boxed{\overline{\mathbf{E}} \cap k(\beta, C) = k.}$$

Equivalently, every nonconstant rational function in β, C with algebraic coefficients lies outside $\overline{\mathbf{E}}$.

Proof. Put $g = \gamma + \log(t/2)$ and $\tau = 2\pi i$. For each integer j, write the continued phase images using

$$\begin{aligned} A_j &= e^{-t}((1+2j)I + iK/\pi), & B_j &= e^{t}(-2jI - iK/\pi), \\ P_j &= e^{-t}((1+2j)J - iL/\pi), & Q_j &= e^{t}(-2jJ + iL/\pi). \end{aligned}$$

All A_j, B_j are nonzero, since I, K are positive. The Bessel Wronskian $IL + JK = 1/t$ gives

$$A_jQ_j - B_jP_j = \frac{i}{\pi t}.$$

With $q = e^{2tX}$, the two images are

$$x_j(X) = g + j\tau + \frac{i\pi B_j}{A_jq + B_j}, \qquad y_j(X) = \frac{P_jq + Q_j}{A_jq + B_j}. \tag{38}$$

They satisfy

$$y_j = C + r_j(x_j - \beta), \qquad r_j = \frac{1}{\pi^2 t A_j B_j}. \tag{39}$$

Indeed the slope follows by eliminating q, and the intercept follows from evaluation of both continued phase images at $X = 1$.

These slopes are nonzero and take infinitely many distinct values. More precisely,

$$A_jB_j = -2j(2j+1)I^2 - \frac{i(4j+1)IK}{\pi} + \frac{K^2}{\pi^2}, \qquad r_j \sim -\frac{1}{4\pi^2 t I^2 j^2} \quad (j \longrightarrow +\infty).$$

For every fixed nonzero r, the equation $r_j = r$ has at most two integer solutions, since it is a nonconstant quadratic equation in j.

This also proves that β, C are algebraically independent over k. Indeed, if a nonzero $H \in k[U, V]$ satisfied $H(\beta, C) = 0$, applying each continued phase embedding would make H vanish identically on the line (39). There are infinitely many distinct such lines, whereas a nonzero polynomial has only finitely many distinct linear factors. This is impossible. In particular, both β and C are transcendental.

As X ranges over $\mathbb{C}$, q ranges over $\mathbb{C}^\times$. Consequently (38) covers the projective line (39) except for the two finite points

$$\begin{aligned} U_j &= (g + j\tau,\ P_j/A_j), \\ V_j &= (g + j\tau + i\pi,\ Q_j/B_j). \end{aligned}$$

In particular, its point at infinity is attained at $q = -B_j/A_j \neq 0$. The omitted points satisfy

$$(U_j)_1 = j\tau + O(1), \quad (V_j)_1 = j\tau + O(1), \qquad (U_j)_2 = C + O(j^{-1}), \quad (V_j)_2 = C + O(j^{-1}). \tag{40}$$

Take a nonconstant $F \in k(U,V)$ and write $F = P/Q$ with coprime $P, Q \in k[U,V]$, $Q \neq 0$. Suppose first that Q is nonconstant. We choose a sufficiently large positive integer j with the following three properties.

First, the polynomial

$$Q_j^{\mathrm{line}}(U) = Q\big(U, C + r_j(U - \beta)\big)$$

is nonconstant. To see this, let $d > 0$ be the total degree of Q and let Q_d be its top homogeneous part. The coefficient of U^d in this restriction is $Q_d(1, r_j)$. The polynomial $Q_d(1,T)$ is nonzero, so it vanishes for only finitely many slopes, and thus for only finitely many integers j.

Second, neither omitted point lies on $Q = 0$. Write

$$Q(U,V) = \sum_{a=0}^{D} q_a(V) U^a, \qquad q_D \neq 0.$$

Transcendence of C implies $q_D(C) \neq 0$. The asymptotics (40) show that, for $D > 0$,

$$Q(U_j) = q_D(C)(j\tau)^D + O(j^{D-1}), \qquad Q(V_j) = q_D(C)(j\tau)^D + O(j^{D-1}).$$

For $D = 0$ the two values tend to $q_0(C) \neq 0$. Therefore both values are nonzero for all sufficiently large j.

Third, the line (39) contains no common zero of P, Q. Since P, Q are coprime in $k[U,V]$, their common affine zero set is finite, with all its points in k^2. A point (a,b) in this set can lie on the line only if

$$r_j = \frac{b - C}{a - \beta}.$$

Here $a - \beta \neq 0$, and the ratio is nonzero because $b - C \neq 0$. Each common zero therefore excludes at most two integers j.

Choose j satisfying all three properties. The nonconstant polynomial Q_j^{line} has a finite complex root. At this root the restricted numerator is nonzero, and the corresponding point is not one of U_j, V_j. Hence it is attained by (38) at a finite X, and $F(x_j(X), y_j(X))$ has a pole there.

If Q is constant, P is nonconstant. The same leading-homogeneous- part argument chooses j for which $P(U, C + r_j(U - \beta))$ is a nonconstant polynomial in U. It has a pole at the point at infinity of the line, which is attained at $q = -B_j/A_j \neq 0$, again giving a pole at a finite X.

Finally, $\beta, C \in \mathrm{Frac}(\mathbf{E})$, so every continued phase embedding extends to their rational field. If $F(\beta, C)$ were integral over $\mathbf{E}$, applying this embedding to a monic integral equation would give a monic equation with entire coefficients for $F(x_j(X), y_j(X))$. At a pole, the highest power has a uniquely largest pole order, which is impossible. Thus no such nonconstant $F(\beta, C)$ is integral. Constants in k are already in $\mathbf{E}$, proving the equality. □

Corollary 19.2 (Earlier reciprocal exceptions are removed)**.** *For every positive algebraic t and every $\alpha \in k$,*

$$1/(\beta(t) - \alpha), \quad 1/(C(t) - \alpha) \notin \overline{\mathbf{E}}.$$

Both denominators are nonzero. At $t = 2$, writing $V = \gamma + \kappa$,

$$\overline{\mathbf{E}} \cap k(V, C) = k.$$

In particular the original Theorem B needs no assumption $\gamma + \log(t/2) \neq 0$ for its reciprocal conclusion.

Proof. These are nonconstant rational functions of the two algebraically independent coordinates in Theorem 19.1. □

Remark 19.3 (Scope). The theorem is a two-variable rational-field statement, not just a statement about two separate one-variable fields. For example, it excludes every nonconstant quotient $P(\beta, C)/Q(\beta, C)$ with $P, Q \in k[U, V]$, including denominators that mix both coordinates. The independent-scale extension is proved in Theorem 21.4; arbitrary rationally dependent scales are not covered by that rational-field conclusion.

20 Mixed sums outside the integral closure of E-values

Let $\mathbf{E}$ be the ring of values of E-functions at algebraic points, and let $\overline{\mathbf{E}}$ be its integral closure in $\mathbb{C}$. This is a ring containing $\overline{\mathbb{Q}}$. In particular it is a $\overline{\mathbb{Q}}$-vector subspace of $\mathbb{C}$. We use the zero-winding morphisms $\Phi_\pm = \Phi_{\pm\pi/4}$ from Lemma 18.1. Their explicit images are the $j = 0$ case of the preceding winding formulas and its conjugate. To avoid conflicting with coefficients of a linear combination, denote the image coefficients by A_t, B_t, C_t, D_t. With $I = I_0(t)$, $K = K_0(t)$ and $g_t = \gamma + \log(t/2)$, they are

$$\begin{aligned} A_t &= e^{-t}(I + iK/\pi), & B_t &= -ie^t K/\pi, \\ C_t &= e^{-t}(I_1(t) - iK_1(t)/\pi), & D_t &= ie^t K_1(t)/\pi, \\ \Phi_-(I_0(t))(x) &= A_t e^{tx} + B_t e^{-tx} =: D_t^*(x), \\ \Phi_-(I_1(t))(x) &= C_t e^{tx} + D_t e^{-tx}, \\ \Phi_-(Y(t))(x) &= g_t D_t^*(x) + e^t K e^{-tx}. \end{aligned} \tag{41}$$

Here $\Phi_\pm = \Phi_{\pm\pi/4}$. The plus images have conjugate coefficients. Equation (4) gives

$$A_t D_t - B_t C_t = \frac{i}{\pi t}. \tag{42}$$

The complex constants g_t in these formulas are coefficients of the images; their appearance does not assert $\gamma \in \mathbf{E}$.

Lemma 20.1 (Pole test)**.** *If $u, v \in \mathbf{E}$, $u \neq 0$, and some Φ_θ and $x_0 \in \mathbb{C}$ satisfy $\Phi_\theta(u)(x_0) = 0 \neq \Phi_\theta(v)(x_0)$, then $v/u \notin \overline{\mathbf{E}}$.*

Proof. A monic equation for v/u over $\mathbf{E}$, multiplied by a suitable power of u, would give an identity whose image at x_0 is $\Phi_\theta(v)(x_0)^d = 0$. No extension of the morphism to $\overline{\mathbf{E}}$ is used. □

Lemma 20.2 (Ordered nearest zeros)**.** *The smallest positive imaginary part of a zero of D_t^* is*

$$y(t) = \frac{1}{2t} \arctan \frac{\pi I_0(t)}{K_0(t)}.$$

The function y is strictly decreasing on $(0, \infty)$.

Proof. Writing $q(t) = \arctan(\pi I/K) \in (0, \pi/2)$, the equation $e^{2tx} = -B_t/A_t$ shows that the zero imaginary parts are $q(t)/(2t) + \pi n/t$, $n \in \mathbb{Z}$. The Wronskian gives

$$q'(t) = \frac{\pi}{t(K^2 + \pi^2 I^2)}.$$

It remains to show $tq' < q$. Put $r = \pi I/K$. If $IK \geq 1$, then

$$q = \arctan r > \frac{r}{1+r^2} = \frac{\pi IK}{K^2+\pi^2 I^2} \geq tq'.$$

If $IK < 1$, use $I \geq 1$ to obtain $r > \pi I^2 \geq \pi$ and

$$q > \arctan \pi > \frac{1}{\pi} \geq \frac{\pi}{K^2+\pi^2 I^2} = tq'.$$

Thus $y' = (tq' - q)/(2t^2) < 0$. □

Theorem 20.3 (Arbitrary finite mixed sums)**.** *For distinct positive algebraic $t_1, \ldots, t_k$ and $a_j, b_j \in \overline{\mathbb{Q}}$, not all zero,*

$$\sum_{j=1}^{k} (a_j C(t_j) + b_j \beta(t_j)) \notin \overline{\mathbf{E}}. \tag{43}$$

The conclusion is unchanged on adding any element of $\overline{\mathbf{E}}$.

Proof. Keep only active indices, for which $(a_j, b_j) \neq (0,0)$, and choose the largest active argument T, writing a_T, b_T for its coefficients. At the nearest upper zero of D_T^*, Lemma 20.2 shows that no other active denominator vanishes. At this zero, the numerator belonging to scale T has image equal to a nonzero exponential factor times

$$a_T \frac{i}{\pi T A_T} + b_T e^T K_0(T). \tag{44}$$

For the plus morphism, use the conjugate zero in the lower half-plane. The analogous expression is $-a_T i/(\pi T \overline{A_T}) + b_T e^T K_0(T)$. If both expressions vanished, subtraction would give $a_T(1/A_T + 1/\overline{A_T}) = 0$. Since $\operatorname{Re} A_T = e^{-T} I_0(T) > 0$, it follows that $a_T = 0$ and then $b_T = 0$, a contradiction. This argument does not assume that a_T, b_T are real.

Write (43) over the common denominator $u = \prod I_0(t_j)$, which is a nonzero E-value. Its numerator v is also an E-value. For at least one of the two morphisms the chosen zero of its image denominator leaves precisely the nonzero active numerator above, multiplied by nonzero images of the other denominators. Lemma 20.1 applies. The assertion concerning addition follows because $\overline{\mathbf{E}}$ is a ring. □

Corollary 20.4. *The $2k$ classes of $C(t_j), \beta(t_j)$ in $\mathbb{C}/\overline{\mathbf{E}}$ are linearly independent over $\overline{\mathbb{Q}}$. The space of $\overline{\mathbb{Q}}$-linear forms in $\gamma, C(t_j), \mu(t_j)$ whose value belongs to $\overline{\mathbf{E}}$ has dimension at most one.*

Proof. The first assertion is Theorem 20.3. In a linear form, replace $\mu(t_j)$ by $\beta(t_j) - \gamma$. Two independent forms with values in $\overline{\mathbf{E}}$ would have a nonzero linear combination without γ, contradicting the first assertion. □

Remark 20.5. This is non-integrality over the E-value ring, not transcendence over its fraction field. All the $C(t_j)$ and $\beta(t_j)$ already belong to $\operatorname{Frac}(\mathbf{E})$. The distinctness hypothesis is weaker than the rational independence needed for the multiscale rational-field classifications below.

21 The full rational field at independent Bessel scales

The independent-scale hypothesis permits one to freeze all but one coordinate in the exponential torus. This avoids any generic-position assumption about the synchronized family of product lines.

Lemma 21.1 (Intersection with an exponential sublattice)**.** *Let G be a torsion-free abelian group and $\Lambda \subseteq G$ a subgroup. Inside the fraction field of $\mathbb{C}[G]$ one has*

$$\mathbb{C}[G] \cap \mathrm{Frac}(\mathbb{C}[\Lambda]) = \mathbb{C}[\Lambda].$$

Proof. Decompose the group algebra as the direct sum of its G/Λ coset components. If $H \in \mathbb{C}[G]$ and $DH = N$ with nonzero $D \in \mathbb{C}[\Lambda]$ and $N \in \mathbb{C}[\Lambda]$, then every nonidentity coset component H_c satisfies $DH_c = 0$. The group algebra is a domain, so all these components vanish. Thus H is supported on Λ. □

Lemma 21.2 (The one-pair pole argument over a larger coefficient field)**.** *Let $F \subseteq \mathbb{C}$ be a field. Fix a positive algebraic t, and let $\beta = \beta(t)$ and $C = C(t)$ be algebraically independent over F. With the constants and winding formulas of the one-point Bessel theorem, put*

$$x_j(q) = g + j\tau + \frac{i\pi B_j}{A_j q + B_j}, \qquad y_j(q) = \frac{P_j q + Q_j}{A_j q + B_j}.$$

For every nonconstant $f \in F(U, V)$ there is an integer j such that $f(x_j(q), y_j(q))$ has a pole at a point of $\mathbb{C}^\times$. This is a statement about rational functions with complex coefficients; no phase embedding is asserted to fix F.

Proof. The winding formulas give

$$y_j = C + r_j(x_j - \beta), \qquad r_j = \frac{1}{\pi^2 t A_j B_j}, \qquad r_j \sim -\frac{1}{4\pi^2 t I_0(t)^2 j^2}.$$

Each fixed nonzero slope occurs for at most two integer windings. The image of $q \in \mathbb{C}^\times$ is the projective line displayed above with precisely two finite points omitted. Their first coordinates are $j\tau + O(1)$, and their second coordinates are $C + O(j^{-1})$. The point at infinity of the line is attained at $q = -B_j/A_j \in \mathbb{C}^\times$.

Write $f = P/Q$ in lowest terms in $F[U, V]$. If Q is nonconstant, the restriction $Q(U, C + r_j(U - \beta))$ is nonconstant for all but finitely many j: its top coefficient is $Q_d(1, r_j)$, where Q_d is the nonzero highest homogeneous part. Write $Q(U, V) = \sum_{a=0}^{D} q_a(V) U^a$, $q_D \neq 0$. Since C is transcendental over F, $q_D(C) \neq 0$. At each omitted point the value of Q is therefore $q_D(C)(j\tau)^D + O(j^{D-1})$ when $D > 0$, and tends to $q_0(C) \neq 0$ when $D = 0$. Neither omitted point lies on $Q = 0$ for sufficiently large j.

The common affine zeros of the coprime polynomials P, Q form a finite set, all of whose coordinates are algebraic over F. For such a zero (a, b), transcendence of β, C over F implies $a \neq \beta$, $b \neq C$. It lies on the selected line only when

$$r_j = \frac{b - C}{a - \beta},$$

which excludes at most two windings per common zero. Choose a sufficiently large j avoiding these finitely many exceptions. A root of the nonconstant restricted denominator is then an attained point where the numerator is nonzero, and yields the required pole.

If Q is constant, the same highest-homogeneous-part argument makes the restriction of P nonconstant for some j. Its pole at infinity is attained at $q = -B_j/A_j$, proving this case as well. □

Lemma 21.3 (Independence from the frozen line pencils)**.** *For positive algebraic $t_1, \dots, t_m$ linearly independent over $\mathbb{Q}$, the $2m$ numbers $\beta(t_i), C(t_i)$ are algebraically independent over $\overline{\mathbb{Q}}$.*

Proof. Fix an index i and put $F_i = \overline{\mathbb{Q}}(\beta(t_l), C(t_l) : l \neq i)$. Suppose a nonzero polynomial $H \in F_i[U, V]$ vanishes at $(\beta(t_i), C(t_i))$. Express its coefficients as rational functions of the other actual generators, choosing denominators nonzero at those values. Apply each continued embedding to this relation. The winding formulas involve only $q_l = e^{2t_l X}$, which are algebraically independent as exponential functions because the t_l are rationally independent. Hence the embedded relation is a rational identity in these independent variables. Set $q_l = e^{2t_l}$ for every $l \neq i$. Evaluation compatibility recovers every original coefficient of H; its coefficient denominators remain nonzero. Thus

$$H\big(x_{i,j}(q_i), y_{i,j}(q_i)\big) = 0 \qquad \text{for every } j \in \mathbb{Z}.$$

The polynomial H therefore vanishes identically on infinitely many distinct affine lines, with slopes $r_{i,j}$. A nonzero polynomial in two variables has only finitely many distinct linear factors, a contradiction. This proves independence of each pair over the field of all other pairs. Successively deleting pairs gives total transcendence degree $2m$. □

Theorem 21.4 (Full rational exclusion at independent scales)**.** *Let $t_1, \dots, t_m$ be positive algebraic numbers linearly independent over $\mathbb{Q}$. Then*

$$\boxed{\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(\beta(t_1), C(t_1), \dots, \beta(t_m), C(t_m)) = \overline{\mathbb{Q}}.}$$

Proof. Abbreviate $\beta_i = \beta(t_i)$ and $C_i = C(t_i)$; these are jointly independent by Lemma 21.3. Suppose $\xi = R(\beta_1, C_1, \dots, \beta_m, C_m)$ is integral over $\mathbf{E}$, where R is a rational function with algebraic coefficients. The continued embedding $\Phi^{[j]}_{-\pi/4}$ sends it to an element H_j of the integrally closed exponential-polynomial ring

$$\mathcal{R} = \mathbb{C}[\overline{\mathbb{Q}}_{\mathrm{add}}].$$

The explicit winding formulas also put H_j in the fraction field of $\mathbb{C}[\Lambda]$, where $\Lambda = \sum_i \mathbb{Z}(2t_i)$. Lemma 21.1 therefore gives

$$H_j \in \mathbb{C}[q_1^{\pm 1}, \dots, q_m^{\pm 1}], \qquad q_i = e^{2t_i X}.$$

Here rational independence of the t_i identifies the displayed ring with a Laurent polynomial ring in m independent variables. This identification is essential: the following argument takes place on the full exponential torus, not just on its analytic one-parameter image.

If R is nonconstant, choose an index i on whose pair of variables it depends. Set

$$F_i = \overline{\mathbb{Q}}(\beta_l, C_l : l \neq i).$$

Joint independence identifies the specialization of R in that pair with a nonconstant rational function $f \in F_i(U, V)$, and makes β_i, C_i algebraically independent over F_i.

For every winding j, specialize $q_l = e^{2t_l}$ for all $l \neq i$. Evaluation compatibility of the continued embeddings says that the two rational images for scale t_l then take the values β_l, C_l, independently of j. Consequently the specialization of H_j is

$$f\big(x_{i,j}(q_i), y_{i,j}(q_i)\big).$$

This identity is valid as a rational identity: after clearing denominators, the specialized denominator cannot vanish identically, since its value at $q_i = e^{2t_i}$ is the nonzero original denominator at the

jointly independent numerical tuple. The specialized H_j is a Laurent polynomial in q_i, and hence has no pole on $\mathbb{C}^\times$.

Lemma 21.2, applied with coefficient field F_i, provides a winding for which the same function has a pole on $\mathbb{C}^\times$. This contradiction shows that R is constant. Conversely, all algebraic constants already belong to $\mathbf{E}$. □

Remark 21.5. The theorem concerns the entire rational field and allows arbitrary mixed numerators and denominators. The windings remain synchronized across all scales. No separate continuation of individual factors, generic product-line intersection, or independent choice of windings is used. Rationally dependent scales are not covered by the freezing argument, because their exponential coordinates satisfy relations.

Corollary 21.6 (Mixed polynomial exclusion)**.** *For positive algebraic $t_1, \dots, t_m$ linearly independent over $\mathbb{Q}$,*

$$\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}[\beta(t_1), C(t_1), \dots, \beta(t_m), C(t_m)] = \overline{\mathbb{Q}}.$$

The same equality holds after omitting any of the displayed variables, in particular for the polynomial ring in the $\beta(t_i)$ alone.

Proof. A polynomial is a rational function, so Theorem 21.4 applies. This includes the earlier polynomial conclusions without a separate torus-density proof. □

22 Complex scales and rational fields without Euler's constant

The positivity assumption in the preceding Bessel integral-intersection theorems can be removed. It is convenient to use a component description that is independent of a particular sector-dependent sign. Throughout this section $k = \overline{\mathbb{Q}}$. The functions $\beta(t) = Y(t)/I_0(t)$ and $C(t) = I_1(t)/I_0(t)$ retain their branch-free definitions. Auxiliary occurrences of $K_0(t)$ and $\log(t/2)$ use compatible branches.

Lemma 22.1 (Complex Bessel components and their winding pencils)**.** *Let $t \in k^\times$. Choose a nonexceptional direction $\theta \in (-\pi/2, \pi/2)$ for the canonical component decompositions of $I_0(tz)$ and $Y(tz)$. Write*

$$I = I_0(t), \quad J = I_1(t), \quad K = K_0(t), \quad L = K_1(t), \qquad g = \gamma + \log(t/2), \quad \tau = 2\pi i.$$

There are constants $c, d \in \mathbb{C}$, with $d \neq 0$, such that the two components of $I_0(tz)$, including their exponential factors, are

$$f_+(z) = cI_0(tz) + dK_0(tz), \qquad f_-(z) = (1-c)I_0(tz) - dK_0(tz). \tag{45}$$

Their exponential phases are t and $-t$, respectively. Put $a = (1-c)/d$ and $b = -c/d$. The corresponding components of $Y(tz)$ are

$$(g + \log z + a)f_+(z), \qquad (g + \log z + b)f_-(z). \tag{46}$$

For integer j, define

$$A_j = e^{-t}((c - d\tau j)I + dK), \qquad B_j = e^{t}((1 - c + d\tau j)I - dK),$$
$$P_j = e^{-t}((c - d\tau j)J - dL), \qquad Q_j = e^{t}((1 - c + d\tau j)J + dL).$$

Then $A_jB_j \neq 0$ for all but finitely many j. For such j, the continued images of $\beta(t), C(t)$, with $q = e^{2tX}$, are

$$x_j(q) = g + \tau j + a - \frac{B_j}{d(A_jq + B_j)}, \qquad y_j(q) = \frac{P_jq + Q_j}{A_jq + B_j}. \tag{47}$$

They parametrize the line

$$y = C(t) + r_j(x - \beta(t)), \qquad r_j = -\frac{d^2}{tA_jB_j} \sim -\frac{1}{4\pi^2 t I_0(t)^2 j^2} \quad (j \longrightarrow +\infty). \tag{48}$$

Each fixed nonzero slope occurs at most twice. The image of $q \in \mathbb{C}^\times$ omits precisely two finite points of this projective line. Their first coordinates are $j\tau + O(1)$ and their second coordinates are $C(t) + O(j^{-1})$. The point at infinity is attained at $q = -B_j/A_j \in \mathbb{C}^\times$.

Proof. The classical two-exponential expansion of the order-zero Bessel equation gives nonzero formal components with phases t and $-t$. Their directional sums are independent solutions of the same second-order equation; equivalently they are the appropriate rotated Hankel solutions. This description follows from the Bessel connection and large-argument formulas in [2, Sections 10.17, 10.27, and 10.40]. Both phases occur: a single phase with a power-series amplitude on a sector of opening greater than π would contradict the two different dominant exponentials on rays with positive and negative $\mathrm{Re}(tz)$. Directional summation commutes with the differential equation. Consequently the two sums are independent solutions, and their sum is $I_0(tz)$.

Since $I_0(tz), K_0(tz)$ form a fundamental pair, the two components therefore have the form (45). If $d = 0$, both would be constant multiples of $I_0(tz)$, contrary to their independence. The constants a, b satisfy $af_+ + bf_- = K_0(tz)$. Hence the sum of the expressions in (46) is

$$(g + \log z)I_0(tz) + K_0(tz) = Y(tz).$$

Multiplication by $\log z$ preserves the corresponding exponential phase and the logarithmic Nilsson–Gevrey class. Uniqueness of the canonical decomposition [3, Theorem 2.3] proves (46).

The classical continuation formulas are

$$M^jK_0(tz) = K_0(tz) - j\tau I_0(tz), \qquad M^jK_1(tz) = K_1(tz) + j\tau I_1(tz), \qquad M^j \log z = \log z + j\tau;$$

see [2, Section 10.34]. Applying Lemma 18.1 to (45), its derivative divided by t, and (46) gives the coefficients A_j, B_j, P_j, Q_j and the two rational images (47). The Wronskian gives

$$A_jQ_j - B_jP_j = d(IL + JK) = d/t \neq 0.$$

Moreover $I \neq 0$ by Corollary 3.2, also included in the coordinate system of Theorem 3.3. Thus A_jB_j is a quadratic polynomial in j with leading coefficient $-d^2\tau^2I^2 \neq 0$. Only finitely many integer windings need be discarded; no assertion that every winding is nondegenerate is required.

Eliminating q now yields the slope in (48); evaluation at $X = 1$ gives its stated intercept. The leading coefficient of A_jB_j gives the slope asymptotic, and its degree gives the bound of two on repeated nonzero slopes. At $q = \infty$ and $q = 0$ the respective omitted points are

$$(g + \tau j + a, P_j/A_j), \qquad (g + \tau j + b, Q_j/B_j).$$

Their second coordinates tend to J/I at rate $O(j^{-1})$. Finally the common denominator vanishes at the displayed nonzero value of q, while the nonzero Wronskian prevents degeneration of the projective parametrization. □

Theorem 22.2 (Exact rational intersection at complex independent scales)**.** *Let $t_1, \dots, t_m \in k^\times$ be linearly independent over $\mathbb{Q}$. Then*

$$\boxed{\overline{\mathbf{E}} \cap k(\beta(t_j), C(t_j) : 1 \le j \le m) = k.}$$

The $2m$ displayed coordinates are algebraically independent over k. In particular, the one-point conclusion holds at every nonzero algebraic complex t.

Proof. Joint independence follows from Theorem 3.3: rational independence excludes repeated scales modulo sign. Choose one common direction that is nonexceptional for the finitely many Bessel functions under consideration. Such directions exist because each function excludes only finitely many directions modulo 2π. Use this one direction and a common integer winding j for every scale.

We first explain precisely why the coefficient-field pole argument survives the complex extension. Suppose $F \subseteq \mathbb{C}$ and $\beta(t), C(t)$ are independent over F. For a nonconstant $f = P/Q \in F(U, V)$ in lowest terms, the proof of Lemma 21.2 uses only: infinitely many distinct nonzero line slopes with finitely many repetitions; the stated asymptotics of the two omitted points; and attainment of the point at infinity. All three properties hold by Lemma 22.1 after finitely many windings have been discarded. Indeed, if Q is nonconstant, its restriction to the line is nonconstant for all but finitely many slopes. Writing $Q(U, V) = \sum_{a=0}^{D} q_a(V) U^a$ with $q_D \neq 0$, transcendence of $C(t)$ over F gives $q_D(C(t)) \neq 0$; hence the denominator does not vanish at either omitted point for sufficiently large j. The finitely many common affine zeros of P, Q have coordinates algebraic over F and exclude only finitely many slopes. A root of the restricted denominator therefore gives an attained pole. If Q is constant, a nonconstant restriction of P has a pole at the attained point at infinity. Thus some sufficiently large j produces a pole in $\mathbb{C}^\times$ in every case.

Now let ξ belong to the intersection in the theorem. Normality and Lemma 18.2 put every continued image of ξ in $\mathbb{C}[k_{\mathrm{add}}]$. The explicit images put it also in

$$\operatorname{Frac} \mathbb{C}[q_1^{\pm 1}, \dots, q_m^{\pm 1}], \qquad q_j = e^{2t_j X}.$$

The q_j are algebraically independent as exponential functions. Lemma 21.1 consequently puts the image in the displayed Laurent polynomial ring. If ξ is nonconstant, choose a coordinate pair on which it depends, and freeze every other q_l at e^{2t_l}. Evaluation compatibility recovers the original coefficient field generated by the other Bessel pairs. This is the same rational specialization used in Theorem 21.4; its denominator is nonzero because it is nonzero at the original numerical tuple. The resulting Laurent polynomial has no pole on $\mathbb{C}^\times$, whereas the preceding one-pair argument supplies a winding for which it must have such a pole. This contradiction proves that $\xi \in k$. The opposite inclusion is immediate. □

Corollary 22.3 (The Mahler coordinate and its derivative ratio)**.** *For every positive algebraic α, put*

$$M_\alpha = \frac{\pi Y_0(\alpha)}{2J_0(\alpha)} - \log(\alpha/2) - \gamma, \qquad R_\alpha = \frac{J_1(\alpha)}{J_0(\alpha)}.$$

Then M_α, R_α are algebraically independent and

$$\overline{\mathbf{E}} \cap k(M_\alpha, R_\alpha) = k.$$

In particular M_α is not integral over $\mathbf{E}$.

Proof. The Bessel connection formulas give

$$\beta(i\alpha) = \gamma + \log(\alpha/2) - \frac{\pi Y_0(\alpha)}{2J_0(\alpha)}, \qquad C(i\alpha) = i\frac{J_1(\alpha)}{J_0(\alpha)}.$$

The denominators are nonzero by Corollary 3.2, since $J_0(\alpha) = I_0(i\alpha)$. Apply Theorem 22.2 at $t = i\alpha$ and make the invertible algebraic changes of coordinates $M_\alpha = -\beta(i\alpha)$ and $R_\alpha = C(i\alpha)/i$. □

Corollary 22.4 (Exact intersections for three coordinates without γ)**.** *For positive algebraic a, define*

$$D(a) = \frac{K_0(2a)}{I_0(2a)} + \frac{\pi Y_0(2a)}{2J_0(2a)},$$
$$C_I(a) = \frac{I_1(2a)}{I_0(2a)}, \qquad C_J(a) = \frac{J_1(2a)}{J_0(2a)}.$$

If $a_1, \dots, a_m$ are linearly independent over $\mathbb{Q}$, then these $3m$ coordinates are jointly algebraically independent. For

$$F = k(D(a_j), C_I(a_j), C_J(a_j) : 1 \le j \le m)$$

one has $\overline{\mathbf{E}} \cap F = k$. Thus every nonconstant rational expression in these coordinates lies outside $\overline{\mathbf{E}}$.

Proof. The $2m$ complex numbers $2a_j, 2ia_j$ are linearly independent over $\mathbb{Q}$, by taking real and imaginary parts. The identities

$$D(a) = \beta(2a) - \beta(2ia), \qquad C_I(a) = C(2a), \qquad C_J(a) = C(2ia)/i$$

show that F is a subfield of the field in Theorem 22.2. They also identify the displayed $3m$-tuple with part of an invertible algebraic linear change of its $4m$ independent generators. Both conclusions follow. □

Corollary 22.5 (Radical exclusion)**.** *Let F be either a field from Theorem 22.2 or a subfield containing k, including the fields in Corollaries 22.3 and 22.4. Define the set*

$$\mathrm{Rad}(F) = \{y \in \mathbb{C} : y^n \in F \text{ for some integer } n \ge 1\}.$$

Then $\overline{\mathbf{E}} \cap \mathrm{Rad}(F) = k$. The notation denotes a set and does not assert that it is a field.

Proof. If $y \in \overline{\mathbf{E}}$ and $y^n \in F$, the ring property gives $y^n \in \overline{\mathbf{E}} \cap F = k$. Thus y is algebraic over the algebraically closed field k and belongs to k. The converse is immediate. □

Remark 22.6. At $a = 1$, Corollary 22.4 applies to the three explicit coordinates $\kappa + \pi Y_0(2)/(2J_0(2))$, C, and C_J. The argument uses the independent complex scales 2 and $2i$. It does not settle the separate problem of rationally dependent positive scales, such as 2 and 4.

23 Exact multivariate exponential-integral intersections

Put $k = \overline{\mathbb{Q}}$. As before, $\mathbf{E}$ is the ring of E-function values and $\overline{\mathbf{E}}$ is its integral closure in $\mathbb{C}$. The continued phase embeddings give an exact answer for a rational field generated by finitely many exponential-integral values and their exponential units. All independence assumptions on the arguments in this section are over $\mathbb{Q}$, not merely pairwise distinctness.

Theorem 23.1 (Exact exponential-integral rational intersection)**.** *Let $a_1, \dots, a_m, b_1, \dots, b_r$ be nonzero algebraic numbers that are linearly independent over $\mathbb{Q}$, where $m \geq 1$ and $r \geq 0$. Put*

$$E_i = e^{a_i}, \qquad \eta_i = \mathrm{Ein}(a_i), \qquad Z_l = e^{b_l}.$$

Then the $2m + r$ numbers E_i, η_i, Z_l are algebraically independent over k, and

$$\boxed{\overline{\mathbf{E}} \cap k(E_1, \eta_1, \dots, E_m, \eta_m, Z_1, \dots, Z_r) = k[E_1^{\pm 1}, \dots, E_m^{\pm 1}, \eta_1, \dots, \eta_m, Z_1^{\pm 1}, \dots, Z_r^{\pm 1}].} \tag{49}$$

In particular,

$$\overline{\mathbf{E}} \cap k(\eta_1, \dots, \eta_m) = k[\eta_1, \dots, \eta_m], \tag{50}$$

$$\overline{\mathbf{E}} \cap k(Z_1, \dots, Z_r, \eta_1, \dots, \eta_m) = k[Z_1^{\pm 1}, \dots, Z_r^{\pm 1}, \eta_1, \dots, \eta_m]. \tag{51}$$

For a single arbitrary nonzero algebraic a, this gives

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a)) = k[e^a, e^{-a}, \mathrm{Ein}(a)]. \tag{52}$$

23.1 Winding coordinates and independence

Fix one admissible right-lateral summation direction for the finitely many functions involved, and put $\tau = 2\pi i$. With compatible logarithm branches, there are constants

$$c_i = \gamma + \log a_i, \qquad d_i = E_i(\eta_i - c_i)$$

for which the continued phase embeddings satisfy

$$\Phi^{[j]}(E_i) = u_i = e^{a_i X}, \qquad \Phi^{[j]}(\eta_i) = c_i + j\tau + \frac{d_i - j\tau E_i}{u_i}, \qquad \Phi^{[j]}(Z_l) = z_l = e^{b_l X}. \tag{53}$$

These follow by continuing the decomposition $\mathrm{Ein}(a_i z) = c_i + \log z + E_1^{\mathrm{tail}}(a_i z)$: the entire function $\mathrm{Ein}(a_i z)$ is single valued, whereas $\log z$ increases by τ per turn. Different initial compatible branches only change the fixed constants c_i, d_i. Since the arguments a_i, b_l are linearly independent over $\mathbb{Q}$, the phase variables u_i, z_l are algebraically independent over $\mathbb{C}$ as exponential functions.

It is useful to replace η_i by the E-value $V_i = E_i \eta_i$. Formula (53) becomes

$$\Phi^{[j]}(V_i) = (c_i + j\tau)u_i + d_i - j\tau E_i. \tag{54}$$

Thus the image in each (u_i, v_i) coordinate plane is the line through the fixed point (E_i, V_i) with slope $c_i + j\tau$, parameterized by $u_i \in \mathbb{C}^\times$.

Here the claimed joint algebraic independence can be proved directly, without importing the joint Bessel theorem. Suppose that a nonzero $P \in k[U_1, \dots, U_m, V_1, \dots, V_m, Z_1, \dots, Z_r]$ vanishes at these values. Applying every continued embedding gives the polynomial identity

$$P\big(u,\ c_i u_i + d_i + j\tau(u_i - E_i),\ z\big) = 0 \qquad (j \in \mathbb{Z}).$$

The identity is polynomial in the independent phase variables u, z and in $j\tau$. Let D be the largest total degree of P in its V variables, and let P_D be the nonzero part homogeneous of degree D in those variables. The coefficient of $(j\tau)^D$ yields

$$P_D(u, u - E, z) = 0.$$

The numbers $E_1, \dots, E_m$ are algebraically independent over k by Lindemann–Weierstrass: distinct integer combinations of the a_i give distinct algebraic exponential exponents. Consequently the last displayed identity also holds as a polynomial identity with $E_1, \dots, E_m$ replaced by independent indeterminates. The invertible substitution $V_i = U_i - E_i$ would then give $P_D = 0$, a contradiction. This proves the independence of E_i, V_i, Z_l, and hence of E_i, η_i, Z_l.

23.2 A coefficient-field version of the one-point argument

Lemma 23.2 (A line pencil detects every nonunit denominator)**.** *Let $K \subset \mathbb{C}$ be a field, let E be transcendental over K, and fix $c, d \in \mathbb{C}$ and $\tau \neq 0$. For $j \in \mathbb{Z}$, put*

$$v_j(u) = (c + j\tau)u + d - j\tau E.$$

If $F \in K(U, V)$ does not belong to $K[U, U^{-1}, V]$, then for some integer j the rational function $F(u, v_j(u))$ is well defined and has a pole at a point of $\mathbb{C}^\times$.

Proof. Multiplication by a power of U does not affect the assertion. We may write $F = P/Q$ with coprime $P, Q \in K[U, V]$, where Q is nonconstant and is not divisible by U. The omitted point of the parameterized line is $(0, d - j\tau E)$. Since $Q(0, V)$ is a nonzero polynomial,

$$Q(0, d - j\tau E) \neq 0$$

for all but finitely many j; here $\tau E \neq 0$. If q_D is the highest nonzero homogeneous part of Q in total degree, the leading coefficient of its restriction to the line is $q_D(1, c + j\tau)$, nonzero for all but finitely many j. Thus that restriction is nonconstant.

The common zero set of the coprime polynomials P, Q is finite, and its coordinates are algebraic over K. Every line in this pencil passes through $(E, cE + d)$. Since E is not algebraic over K, every common zero belongs to at most one line of the pencil. Avoid the resulting finitely many indices. The restricted Q now has a finite root away from $u = 0$, and the restricted P is nonzero at that root. This is a genuine pole of $F(u, v_j(u))$ in $\mathbb{C}^\times$. □

23.3 Freezing the other coordinates

Let $\mathcal{R}$ again be the exponential-polynomial ring with complex coefficients and algebraic phases, and let Λ be the additive subgroup generated by the a_i, b_l. The associated group ring is

$$\mathcal{R}_\Lambda = \mathbb{C}[u_1^{\pm 1}, \dots, u_m^{\pm 1}, z_1^{\pm 1}, \dots, z_r^{\pm 1}].$$

We recall the subgroup-support fact

$$\mathcal{R} \cap \mathrm{Frac}(\mathcal{R}_\Lambda) = \mathcal{R}_\Lambda. \tag{55}$$

This is Lemma 21.1, applied to the additive group of algebraic phases; no saturation hypothesis is needed.

Proof of the intersection statements. Let

$$\xi \in \overline{\mathbf{E}} \cap k(E_1, V_1, \dots, E_m, V_m, Z_1, \dots, Z_r).$$

All generators are E-values, so $\xi \in \mathrm{Frac}(\mathbf{E})$. Normality of $\mathcal{R}$ and the continued embeddings imply $\Phi^{[j]}(\xi) \in \mathcal{R}$ for every j. The explicit formulas place this image in $\mathrm{Frac}(\mathcal{R}_\Lambda)$, so (55) gives

$$\Phi^{[j]}(\xi) \in \mathbb{C}[u_1^{\pm 1}, \dots, u_m^{\pm 1}, z_1^{\pm 1}, \dots, z_r^{\pm 1}]. \tag{56}$$

Fix an index i and set

$$K_i = k(E_l, V_l \ (l \neq i), Z_1, \dots, Z_r).$$

Regard ξ as a rational function in E_i, V_i over K_i. The established joint independence implies that E_i is transcendental over K_i. In the Laurent-polynomial identity for each winding, substitute $u_l = E_l$ for $l \neq i$ and $z_l = Z_l$ for every l. Formula (54) gives

$$(c_l + j\tau)E_l + d_l - j\tau E_l = V_l,$$

so this substitution fixes every coefficient in K_i for every winding. The right side of (56) becomes a Laurent polynomial in u_i. If the rational function in E_i, V_i did not belong to $K_i[E_i, E_i^{-1}, V_i]$, Lemma 23.2 would give a winding whose specialized image has a pole in $\mathbb{C}^\times$, a contradiction. For precision, the substitutions can always be performed on the product identity "denominator times image equals numerator"; the lemma chooses a winding for which the restricted denominator is a nonzero polynomial. No undefined specialization of a rational expression is required.

Likewise, fix an extra exponential variable Z_l and put all other phase variables equal to their actual values. A rational function in Z_l over the field of the other generators must then be a Laurent polynomial in Z_l: otherwise it has a finite nonzero pole, contradicting (56).

These coordinatewise conclusions imply the global ring membership

$$\xi \in k[E_1^{\pm 1}, \ldots, E_m^{\pm 1}, V_1, \ldots, V_m, Z_1^{\pm 1}, \ldots, Z_r^{\pm 1}].$$

Here is the elementary algebraic justification. This Laurent polynomial ring is a UFD. In a reduced fraction, a nonunit irreducible denominator factor remains a nonunit after localizing all variables outside at least one pair (E_i, V_i) or one variable Z_l. If it contains a V_i, use that pair; otherwise it has at least two Laurent monomials and some exponential variable has different exponents in those monomials. This contradicts the corresponding coordinatewise conclusion.

Conversely, every displayed generator and every allowed exponential inverse is an E-value. The whole displayed ring therefore lies in $\mathbf{E}$. Since $V_i = E_i\eta_i$ and E_i is a unit, this ring is exactly the right side of (49). Finally, algebraic independence allows us to intersect it with $k(\eta_1, \ldots, \eta_m)$, or with $k(Z_1, \ldots, Z_r, \eta_1, \ldots, \eta_m)$, by uniqueness of Laurent expansions in the omitted E_i variables. This gives (50) and (51). □

Remark 23.3 (Scope and the point $a = 1$). For $a = 1$, the new exact ring is

$$\overline{\mathbf{E}} \cap k(e, \gamma + \delta/e) = k[e, e^{-1}, \gamma + \delta/e].$$

Thus adjoining the exponential unit does not create any additional integral rational expressions: precisely the polynomials in the exponential-integral value with Laurent polynomial exponential coefficients survive. The theorem does not assume an arithmetic classification of γ or δ. The multi-argument proof requires the stated $\mathbb{Q}$-linear independence; it does not assert the same classification for arbitrary distinct, linearly dependent arguments, or for arbitrary additional exponential units.

Corollary 23.4 (The one-variable exponential-integral field)**.** *For every nonzero algebraic a,*

$$\overline{\mathbf{E}} \cap k(\mathrm{Ein}(a)) = k[\mathrm{Ein}(a)].$$

Proof. This is the case $m = 1$, $r = 0$ of (50). A singleton containing a nonzero number is linearly independent over $\mathbb{Q}$. □

24 A combined exponential-integral and Bessel intersection

Corollary 24.1 (The exact mixed rational intersection)**.** *Let $a_1, \dots, a_m, b_1, \dots, b_r$ be nonzero algebraic numbers, and let $t_1, \dots, t_s$ be positive algebraic numbers, where $m, s \geq 1$ and $r \geq 0$. Assume that the entire list*

$$a_1, \dots, a_m, b_1, \dots, b_r, 2t_1, \dots, 2t_s$$

is linearly independent over $\mathbb{Q}$. Set

$$E_i = e^{a_i}, \qquad \eta_i = \mathrm{Ein}(a_i), \qquad Z_l = e^{b_l}, \qquad \beta_j = \beta(t_j), \qquad C_j = C(t_j).$$

These $2m + r + 2s$ numbers are algebraically independent over $k = \overline{\mathbb{Q}}$. Write

$$\mathcal{F} = k(E_i, \eta_i, Z_l, \beta_j, C_j : 1 \leq i \leq m,\ 1 \leq l \leq r,\ 1 \leq j \leq s).$$

Then

$$\boxed{\overline{\mathbf{E}} \cap \mathcal{F} = k[E_i^{\pm 1}, \eta_i, Z_l^{\pm 1} : 1 \leq i \leq m,\ 1 \leq l \leq r].} \tag{57}$$

Thus the Bessel coordinates introduce no additional integral rational expressions, even when numerators and denominators mix all the displayed families. Omitting the generators E_i on the left gives the corresponding right side $k[\eta_i, Z_l^{\pm 1}]$; omitting both E_i and Z_l gives $k[\eta_i]$.

Proof. Write the independent phase variables as

$$u_i = e^{a_i X}, \qquad z_l = e^{b_l X}, \qquad q_j = e^{2t_j X}.$$

For winding $h \in \mathbb{Z}$, use the exponential-integral formulas (53) and the Bessel formulas from Lemma 21.2. All images are rational functions of these independent variables. At the original exponential coordinates $u_i = E_i$, $z_l = Z_l$, and $q_j = e^{2t_j}$, evaluation compatibility recovers the original numerical generators for every winding.

First we justify the joint independence needed in the coefficient-field argument. The exponential and exponential-integral generators are independent by Theorem 23.1. For each Bessel pair let F_j be the field of all other displayed generators. If a nonzero polynomial $H \in F_j[U, V]$ vanished at (β_j, C_j), apply every continued embedding and freeze all phase variables except q_j at their original exponential coordinates. As in Lemma 21.3, this recovers the original coefficients of H and forces H to vanish on infinitely many distinct affine Bessel lines. It is therefore the zero polynomial, a contradiction. Coefficient fractions cause no difficulty: choose their denominators nonzero at the original generators, clear them before freezing, and evaluate those denominators at the same original coordinates. Each pair is consequently independent over the field of all the others. Together with the independence of the remaining exponential-integral block, this proves the asserted total independence.

Now let ξ belong to the intersection on the left of (57). Normality of the exponential-polynomial ring and the subgroup-support identity (55), equivalently Lemma 21.1, imply that every continued image of ξ is a Laurent polynomial in u_i, z_l, q_j. Fix a Bessel index j and freeze all the other phase variables at their original exponential coordinates. The result is a Laurent polynomial in q_j and is the one-pair rational image of ξ, viewed as a rational function in β_j, C_j over F_j. If this rational function were nonconstant, Lemma 21.2 would provide a winding for which it has a pole in $\mathbb{C}^\times$. This is impossible for a Laurent polynomial. Thus ξ is independent of that pair of variables.

Repeat over all Bessel pairs. Algebraic independence identifies the numerical field with the corresponding pure rational function field, so being independent of each pair leaves precisely

$$\xi \in k(E_1, \eta_1, \dots, E_m, \eta_m, Z_1, \dots, Z_r).$$

Theorem 23.1 now gives the right side of (57). Conversely, all its generators, including the indicated exponential inverses, are E-values. The two statements obtained by omitting generators follow by intersecting with the relevant subfields and using algebraic independence, as in (50)–(51). □

Remark 24.2 (The independence of the phases is essential to this proof). The result uses one synchronized winding for every family; independent windings are never assumed. The free phase variables make the coordinate freezing possible. In particular, it does not cover the resonant field $k(\mathrm{Ein}(1), \beta(2), C(2))$, since its phases 1 and 4 are rationally dependent.

25 Rational-field exclusion at independent positive scales

Throughout this section, $\mathbf{E}$ is the ring of E-function values at algebraic points and $\overline{\mathbf{E}}$ is its integral closure in $\mathbb{C}$. For a positive algebraic number a, put

$$h_a = E_1^{\mathrm{tail}}(a) := \int_a^\infty \frac{e^{-u}}{u}\, du, \qquad g_a = \gamma + \log a, \tag{58}$$

$$\eta_a = \mathrm{Ein}(a) = g_a + h_a, \qquad B_a = \beta(2a) = g_a + \frac{K_0(2a)}{I_0(2a)}, \tag{59}$$

$$D_a = \eta_a - B_a = E_1^{\mathrm{tail}}(a) - \frac{K_0(2a)}{I_0(2a)}, \qquad C_a = \frac{I_1(2a)}{I_0(2a)}. \tag{60}$$

The logarithm in these formulas is real. The values η_a belong to $\mathbf{E}$, while B_a, C_a, D_a belong to $\mathrm{Frac}(\mathbf{E})$.

25.1 The images and their surjectivity

We use the injective $\overline{\mathbb{Q}}$-algebra morphisms of Fischler–Rivoal [3, Proposition 3.1], in the two directions $\Phi_\pm = \Phi_{\pm\pi/4}$. Their target is contained in

$$\mathcal{R} = \left\{ \sum_{\lambda \in S} c_\lambda e^{\lambda x} : S \subset \overline{\mathbb{Q}} \text{ finite},\ c_\lambda \in \mathbb{C} \right\}.$$

They satisfy $\Phi_\pm(\xi)(1) = \xi$ and extend to embeddings of $\mathrm{Frac}(\mathbf{E})$ into $\mathrm{Frac}(\mathcal{R})$. The coefficients of the exponential polynomials are allowed to be arbitrary complex numbers; the phases are algebraic.

Set

$$I_a = I_0(2a), \qquad K_a = K_0(2a), \qquad q_a = e^a h_a, \qquad \rho_a = \frac{\pi I_a}{K_a}, \qquad s_a = e^{-4a}(1 - i\rho_a).$$

All of $I_a, K_a, h_a, q_a, \rho_a$ are positive, and $s_a \neq 0$. With $w = e^{-ax}$, the sectorial Ein and Bessel decompositions give

$$\Phi_-(\eta_a) = g_a + q_a w, \qquad \Phi_-(B_a) = g_a + \frac{i\pi w^4}{w^4 - s_a}. \tag{61}$$

For example, the first formula follows by applying the decomposition $\mathrm{Ein}(az) = g_a + \log z + E_1^{\mathrm{tail}}(az)$; the coefficient of the exponential part at $z = 1$ is $e^a E_1^{\mathrm{tail}}(a)$. For Φ_+, conjugate the coefficients in (61); g_a and q_a remain unchanged.

Lemma 25.1 (Every target value has a nonzero finite preimage)**.** *Let $u, v \in \overline{\mathbb{Q}}^{\times}$. The rational function*

$$R_{a,u,v}(w) = (u+v)g_a + uq_a w + \frac{vi\pi w^4}{w^4 - s_a} \tag{62}$$

maps $\mathbb{C}^{\times}$ onto $\mathbb{P}^1(\mathbb{C})$. Every finite target value has a preimage at which $w^4 - s_a \neq 0$.

Proof. The four roots of $w^4 = s_a$ are nonzero simple poles, since $v \neq 0$. Thus ∞ is attained. For a finite target α, put $c = (u+v)g_a - \alpha$. The numerator of $R_{a,u,v}(w) - \alpha$ is

$$N_\alpha(w) = uq_a w^5 + (c + vi\pi)w^4 - uq_a s_a w - cs_a.$$

This has degree five. If $c \neq 0$, its constant coefficient is nonzero, so it has a nonzero root. If $c = 0$, its root at zero is simple because the coefficient $-uq_a s_a$ of w is nonzero; its remaining four roots, counted with multiplicity, are nonzero. At any root of $w^4 = s_a$ we have $N_\alpha(w) = vi\pi s_a \neq 0$. A nonzero root of N_α is therefore a nonpole preimage of α. □

25.2 Normality and the scale subgroup

Distinct exponential functions with distinct constant phases are linearly independent over $\mathbb{C}$. Thus $\mathcal{R}$ identifies with the group algebra $\mathbb{C}[\overline{\mathbb{Q}}_{\mathrm{add}}]$. For an additive subgroup $\Lambda \subset \overline{\mathbb{Q}}$, write $\mathcal{R}_\Lambda$ for its group algebra, realized by the exponential polynomials whose phases lie in Λ.

Lemma 25.2 (Two group-algebra facts)**.** *The ring $\mathcal{R}$ is integrally closed in its fraction field. Moreover, for every additive subgroup $\Lambda \subset \overline{\mathbb{Q}}$,*

$$\mathcal{R} \cap \mathrm{Frac}(\mathcal{R}_\Lambda) = \mathcal{R}_\Lambda. \tag{63}$$

No primitivity or saturation assumption on Λ is required.

Proof. Normality is Lemma 18.2; the subgroup intersection is Lemma 21.1. Their proofs use respectively finite phase support and the decomposition into additive-subgroup cosets. □

In particular, if $\xi \in \mathrm{Frac}(\mathbf{E})$ is integral over $\mathbf{E}$, applying either morphism to a monic relation and using Lemma 25.2 gives

$$\Phi_\pm(\xi) \in \mathcal{R}. \tag{64}$$

This is stronger than merely asserting that the meromorphic image has no finite pole.

Theorem 25.3 (Full rational-field exclusion)**.** *Let $a_1, \dots, a_m$ be positive algebraic numbers that are linearly independent over $\mathbb{Q}$. Choose $u_j, v_j \in \overline{\mathbb{Q}}^{\times}$, and put*

$$Z_j = u_j \eta_{a_j} + v_j B_{a_j} \qquad (1 \le j \le m).$$

Then the Z_j are algebraically independent over $\overline{\mathbb{Q}}$, and

$$\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(Z_1, \dots, Z_m) = \overline{\mathbb{Q}}. \tag{65}$$

Equivalently, for every $F \in \overline{\mathbb{Q}}(T_1, \dots, T_m) \setminus \overline{\mathbb{Q}}$, the value $F(Z_1, \dots, Z_m)$ is not integral over $\mathbf{E}$.

Proof. Put $w_j = e^{-a_j x}$ and $\Lambda = \sum_{j=1}^m \mathbb{Z} a_j$. Linear independence of the a_j identifies $\mathcal{R}_\Lambda$ with $\mathbb{C}[w_1^{\pm 1}, \dots, w_m^{\pm 1}]$ and its fraction field with $\mathbb{C}(w_1, \dots, w_m)$. Here the w_j may therefore be treated as independent variables. Formula (61) gives

$$\Phi_-(Z_j) = R_j(w_j), \qquad R_j = R_{a_j, u_j, v_j}.$$

The substitution $T_j \mapsto R_j(w_j)$ is an injection of rational function fields over $\mathbb{C}$: each R_j is a nonconstant function of its own independent variable. More explicitly, Lemma 25.1 supplies a nonpole point above every finite tuple of target values, so no nonzero polynomial in the T_j can vanish after this substitution. Since Φ_- fixes $\overline{\mathbb{Q}}$, it follows that the original values $Z_1, \dots, Z_m$ are algebraically independent.

Suppose that $F(Z_1, \dots, Z_m)$ is integral over $\mathbf{E}$. By (64), its image belongs to $\mathcal{R}$. It also belongs to $\mathrm{Frac}(\mathcal{R}_\Lambda)$. Lemma 25.2 therefore gives the rational identity

$$F(R_1(w_1), \dots, R_m(w_m)) = H(w_1, \dots, w_m), \qquad H \in \mathbb{C}[w_1^{\pm 1}, \dots, w_m^{\pm 1}]. \tag{66}$$

Assume F is nonconstant, and choose an index j on which it depends. Fix the other target variables at generic finite complex values α_ℓ ($\ell \neq j$), so that the specialized denominator of F is not the zero polynomial and the specialized rational function

$$f(T_j) = F(\alpha_1, \dots, \alpha_{j-1}, T_j, \alpha_{j+1}, \dots, \alpha_m)$$

is nonconstant. Such choices form a nonempty Zariski open set: if $F = P/Q$ with coprime polynomials, retain a nonzero coefficient of Q as a polynomial in T_j and a nonzero coefficient of $P_{T_j} Q - P Q_{T_j}$. For each $\ell \neq j$, Lemma 25.1 supplies $c_\ell \in \mathbb{C}^\times$ with $R_\ell(c_\ell) = \alpha_\ell$ and with the denominator of R_ℓ nonzero at c_ℓ.

Substitute these c_ℓ in (66), after clearing the rational denominators. This is legitimate because the chosen coordinates are nonzero and are not poles of the R_ℓ, and the remaining denominator $Q(\alpha_1, \dots, R_j(w_j), \dots, \alpha_m)$ is not identically zero. We obtain

$$f(R_j(w_j)) = h(w_j), \qquad h \in \mathbb{C}[w_j^{\pm 1}].$$

The nonconstant rational function f has a pole at some $\alpha \in \mathbb{P}^1(\mathbb{C})$. By Lemma 25.1, R_j takes the value α at a finite nonzero point. A pole pulls back to a pole under a nonconstant rational map, including when $\alpha = \infty$. Thus $f \circ R_j$ has a pole in $\mathbb{C}^\times$, whereas the Laurent polynomial h has none. This contradiction proves that F is constant, necessarily in $\overline{\mathbb{Q}}$. Finally, $\overline{\mathbb{Q}} \subset \mathbf{E}$ proves the reverse inclusion in (65). □

Corollary 25.4 (Tail–Bessel differences)**.** *For linearly independent positive algebraic $a_1, \dots, a_m$,*

$$\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(D_{a_1}, \dots, D_{a_m}) = \overline{\mathbb{Q}}.$$

In particular, every nonconstant rational function over $\overline{\mathbb{Q}}$ of $D_1, D_{\sqrt{2}}, D_{\sqrt{3}}$ is nonintegral over $\mathbf{E}$. For example,

$$\frac{D_1 D_{\sqrt{2}} + 1}{D_{\sqrt{3}} - D_1} \notin \overline{\mathbf{E}};$$

the denominator is nonzero by algebraic independence. For one point, $\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(D_a) = \overline{\mathbb{Q}}$ for every positive algebraic a.

Proof. Take $u_j = 1$ and $v_j = -1$ in Theorem 25.3. The three positive algebraic numbers $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$, as is seen by applying the independent sign-changing automorphisms of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. □

Remark 25.5 (Two different hypotheses on the scales). The joint independence theorem, Theorem 3.3, already makes $D_{a_1}, \dots, D_{a_m}$ algebraically independent whenever the positive algebraic a_j are merely *distinct*. Indeed, it gives joint independence of η_{a_j}, B_{a_j}, and the differences are part of an invertible linear change of coordinates. The stronger rational-field exclusion proved here requires *linear independence over* $\mathbb{Q}$. Distinctness, or even pairwise irrational ratios for three or more scales, does not justify the independent Laurent variables used in its proof. No rational-field exclusion at arbitrary distinct scales is asserted here.

26 The full Ein–Bessel rational intersection at arbitrary positive scales

Write $k = \overline{\mathbb{Q}}$, let $\mathbf{E}$ be the ring of E-function values at algebraic arguments, and let $\overline{\mathbf{E}}$ be the set of complex numbers integral over $\mathbf{E}$. In particular, $\overline{\mathbf{E}}$ is an integral closure of a ring, not the algebraic closure of its fraction field. This section uses two previously established inputs: the continued phase embeddings Φ_j and their finite-pole obstruction, and the joint algebraic independence over k of $\mathrm{Ein}(a), \beta(t), C(t)$ for positive algebraic a, t.

Theorem 26.1 (Full rational intersection, including resonant scales)**.** *For every pair of positive algebraic numbers a, t, put*

$$\eta = \mathrm{Ein}(a), \qquad \beta = \beta(t) = \gamma + \log(t/2) + \frac{K_0(t)}{I_0(t)}, \qquad C = C(t) = \frac{I_1(t)}{I_0(t)}.$$

Then

$$\boxed{\overline{\mathbf{E}} \cap k(\eta, \beta, C) = k[\eta].} \tag{67}$$

There is no restriction on the ratio t/a.

26.1 Normalized phase images and three local charts

All logarithms of positive real numbers in this section are real. Set

$$p = i\pi, \quad \tau = 2\pi i = 2p, \quad r = \frac{2t}{a}, \quad I = I_0(t), \quad h = \frac{1}{tI^2},$$

and, for positive integers j, set

$$c_{a,j} = \gamma + \log a + j\tau, \qquad c_{t,j} = \gamma + \log(t/2) + j\tau,$$
$$d_j = \eta - c_{a,j}, \qquad v_j = \beta - c_{t,j}, \qquad s_j = \frac{h}{v_j(v_j - p)}.$$

Thus $d_j = -\tau j + O(1)$, $v_j = -\tau j + O(1)$, $d_j/v_j \to 1$, and $s_j \neq 0$ for all sufficiently large j, with $s_j = O(j^{-2})$ and $|s_j| \asymp j^{-2}$.

Use the exponential coordinate

$$q = e^{2t(1-X)}.$$

The normalized continued images are

$$E_j(X) := \Phi_j(\eta)(X) = c_{a,j} + d_j e^{a(1-X)}, \tag{68}$$

$$B_j(X) := \Phi_j(\beta)(X) = \beta + \frac{v_j(p - v_j)(q-1)}{p + v_j(q-1)}, \tag{69}$$

$$D_j(X) := \Phi_j(C)(X) = C - \frac{h(q-1)}{p + v_j(q-1)} = C + s_j(B_j(X) - \beta). \tag{70}$$

Here D_j denotes the image of the ratio C, not a Bessel denominator. These identities follow directly from the winding coefficients in the one-point theorem. For example, writing $g_t = \gamma + \log(t/2)$ and $K/I = \beta - g_t$, the two normalized coefficients of the I_0 image are $1 - v_j/p$ and v_j/p. Consequently

$$\Phi_j(I_0(t))(X) = Iq^{-1/2}\frac{p + v_j(q-1)}{p}.$$

The Y image gives (69); the Bessel Wronskian gives (70). The Ein decomposition and its logarithmic continuation give (68).

Whenever q is sufficiently close to 1, use the holomorphic logarithm with $\log 1 = 0$, and put

$$X = 1 - \frac{\log q}{2t}, \qquad E_j = c_{a,j} + d_j q^{1/r}.$$

This is only a local inverse coordinate: the original functions in X remain single valued. In particular, no rationality of r is needed.

Lemma 26.2 (Local degenerations)**.** *Put*

$$u_* = \eta - \frac{p}{r}. \tag{71}$$

The following statements hold as $j \to +\infty$.

1. *If the image B_j is prescribed to be a variable x in a fixed bounded disk, its local inverse near $X = 1$ is given by*

$$q_j(x) = 1 + \frac{p(x-\beta)}{v_j(p - v_j - (x-\beta))}. \tag{72}$$

   On every such disk, uniformly,

$$E_j(x) \longrightarrow \eta, \qquad D_j(x) = C + s_j(x-\beta) \longrightarrow C.$$

2. *If the image D_j is prescribed to be a variable w in a fixed closed disk not containing C, its local inverse near $X = 1$ is given by*

$$q_j(w) = 1 - \frac{p(w-C)}{h + v_j(w-C)}. \tag{73}$$

   Uniformly on that disk,

$$E_j(w) \longrightarrow u_*, \qquad B_j(w) = \beta + \frac{w-C}{s_j}.$$

   In particular, $|B_j(w)| \asymp j^2$ uniformly there.

3. *If $\alpha \in k$, there is a local point near $X = 1$ where $E_j = \alpha$, and at that point*

$$B_j = \frac{r(\alpha-\eta)\tau}{p + r(\alpha-\eta)}j + O(1), \qquad D_j = C + O(j^{-1}). \tag{74}$$

   The displayed coefficient of j is nonzero.

4. The Bessel images have a common simple pole at the local point with

$$q_j^{\mathrm{pole}} = 1 - \frac{p}{v_j}. \tag{75}$$

The Ein image is holomorphic there, and its value e_j^{pole} satisfies $e_j^{\mathrm{pole}} \to u_*$.

All local inverse maps used above are holomorphic and have nonzero derivative for sufficiently large j.

Proof. Solving (69) for q gives (72). Its denominator has size comparable to j^2 on a fixed bounded disk, so $q_j(x) - 1 = O(j^{-2})$ uniformly. The local expansion $q^{1/r} = 1 + (q-1)/r + O((q-1)^2)$, together with $d_j = O(j)$, gives $E_j(x) = \eta + O(j^{-1})$. The formula for $D_j(x)$ is exact. The derivative of the nonconstant Möbius map (72) does not vanish; explicitly, with $b = x - \beta$, it is

$$q_j'(x) = \frac{p(p - v_j)}{v_j(p - v_j - b)^2} \neq 0$$

for all sufficiently large j.

Solving (70) gives (73). On a disk avoiding C, its denominator does not vanish for large j, and

$$q_j(w) - 1 = -\frac{p}{v_j} + O(j^{-2})$$

uniformly. Hence $E_j(w) = \eta - pd_j/(rv_j) + O(j^{-1}) \to u_*$. The exact expression for $B_j(w)$ follows from the line identity in (70). The inverse derivative is

$$q_j'(w) = -\frac{ph}{(h + v_j(w - C))^2} \neq 0.$$

In both charts $q_j \to 1$, so the chosen logarithm defines a finite holomorphic local coordinate X with nonzero derivative.

For the third assertion prescribe

$$z_j = e^{a(1-X_j)} = \frac{\alpha - c_{a,j}}{d_j} = 1 + \frac{\alpha - \eta}{d_j}.$$

This is nonzero and close to 1. Locally $q_j = z_j^r$, and thus

$$q_j - 1 = \frac{r(\alpha - \eta)}{d_j} + O(j^{-2}), \qquad p + v_j(q_j - 1) \longrightarrow p + r(\alpha - \eta).$$

The last limit is nonzero. Otherwise, since η and r are real, one would have $\pi = -r \operatorname{Im} \alpha$, contrary to the transcendence of π. Also $\alpha \neq \eta$, since η is transcendental. Substitution into (69)–(70) gives (74). The local inverse of E_j has derivative $-1/(a(\alpha - c_{a,j}))$, which is nonzero.

Finally, (75) is a nonzero simple zero of $p + v_j(q-1)$ for large j. The numerators of the Bessel polar terms do not vanish there. Moreover

$$e_j^{\mathrm{pole}} = c_{a,j} + d_j(1 - p/v_j)^{1/r} = \eta - \frac{pd_j}{rv_j} + O(j^{-1}) \longrightarrow u_*.$$

The map from X to q has nonzero derivative, so these are also simple poles in the original variable X. □

Lemma 26.3 (Independence at the critical limit)**.** *The numbers u_* and C are algebraically independent over k. In particular, u_* is transcendental.*

Proof. The already established joint theorem makes η, C algebraically independent. If u_* were algebraic over $k(C)$, complex conjugation would make $\overline{u_*}$ algebraic over the same field, because C is real and conjugation preserves k. But $(u_* + \overline{u_*})/2 = \eta$, a contradiction. Together with the transcendence of C, this proves the assertion. □

26.2 Eliminating every denominator

Proof of Theorem 26.1. The reverse inclusion in (67) follows from $\eta \in \mathbf{E}$. For the other inclusion, take

$$\xi = \frac{P(\eta, \beta, C)}{Q(\eta, \beta, C)} \in \overline{\mathbf{E}}, \qquad P, Q \in k[U, V, W], \quad (P, Q) = 1,$$

with $Q \neq 0$. If $P = 0$, the conclusion is immediate. Otherwise, joint independence of η, β, C makes all nonzero polynomial denominators nonzero at the actual tuple. Each continued embedding extends to this rational field. By the pole obstruction, none of its images of ξ can have a finite pole.

Suppose first that Q has an irreducible factor q_0 of positive V-degree. Gauss's lemma makes q_0 irreducible over $k(U, W)$, and $q_0 \nmid P$ makes q_0 and P coprime there. Their resultant with respect to V is a nonzero element of $k[U, W]$. Joint independence of η, C ensures that this resultant, as well as the leading V-coefficient of q_0, remains nonzero at (η, C). Consequently

$$q_0(\eta, V, C), \quad P(\eta, V, C)$$

are coprime polynomials, the first of positive degree. Choose one of its roots x_0 and a sufficiently small closed disk about x_0 on which the second polynomial does not vanish and whose boundary contains no zero of the first.

Use the first chart of Lemma 26.2. On this disk the functions

$$q_0(E_j(x), x, D_j(x)), \qquad P(E_j(x), x, D_j(x))$$

converge uniformly to the two displayed polynomials. Rouché's theorem supplies a zero of the first function in the disk for every sufficiently large j, while the second stays nonzero throughout the disk. At that zero the full denominator Q vanishes and its numerator does not. The local inverse chart transfers the resulting pole to a finite X. This contradicts integrality. Therefore

$$Q \in k[U, W]. \tag{76}$$

Next let $q_0(U, W)$ be an irreducible factor of Q of positive W-degree. Expand the numerator as

$$P(U, V, W) = \sum_{\ell=0}^{M} P_\ell(U, W) V^\ell.$$

Because $q_0 \nmid P$, there is a largest index m for which $q_0 \nmid P_m$. Every P_ℓ with $\ell > m$ is divisible by q_0. Irreducibility and Gauss's lemma give

$$R(U) = \operatorname{Res}_W(q_0, P_m) \in k[U] \setminus \{0\}.$$

Lemma 26.3 implies that $R(u_*) \neq 0$ and that the leading W-coefficient of q_0 is nonzero at u_*. Thus $q_0(u_*, W)$ is nonconstant and coprime to $P_m(u_*, W)$. The same lemma gives $q_0(u_*, C) \neq 0$.

Choose a root w_0 of $q_0(u_*, W)$. It differs from C. Choose a small closed disk about w_0, avoiding C, on whose boundary $q_0(u_*, W)$ does not vanish and throughout which $P_m(u_*, W)$ does not vanish. In the second chart of Lemma 26.2, Rouché's theorem gives a zero w_j of $q_0(E_j(w), w)$ in this disk for all large j. On the disk $P_m(E_j(w), w)$ is uniformly bounded away from zero, all the other coefficient functions are bounded, and $|B_j(w)| \asymp j^2$.

At $w = w_j$, every term with $\ell > m$ vanishes exactly, because q_0 divides its coefficient. Therefore

$$P(E_j(w_j), B_j(w_j), w_j) = B_j(w_j)^m \left(P_m(E_j(w_j), w_j) + O(|B_j(w_j)|^{-1})\right) \neq 0$$

for large j; when $m = 0$, the same assertion holds with no lower terms. The denominator vanishes there. This is again a finite pole, contradicting integrality. Combining this with (76), we obtain $Q \in k[U]$.

If Q were nonconstant, choose a root $\alpha \in k$. Coprimality gives $P(\alpha, V, W) \neq 0$. Let m be its degree in V, and write its leading coefficient as $b_m(W) \in k[W] \setminus \{0\}$. Transcendence of C gives $b_m(C) \neq 0$. The third chart of Lemma 26.2 sets $E_j = \alpha$, makes B_j diverge linearly with nonzero coefficient, and makes $D_j \to C$. Consequently $P(\alpha, B_j, D_j) \neq 0$ for all large j. The denominator vanishes at these points, giving the same contradiction. Thus Q is constant.

Eliminating polynomial dependence on the Bessel coordinates

It remains to treat $P(\eta, \beta, C)$. Suppose that P has positive total degree D in (V, W), and write its highest homogeneous part as

$$P_D(U, V, W) = \sum_{b=0}^{D} p_b(U) V^{D-b} W^b, \qquad H_D(U, S) = \sum_{b=0}^{D} p_b(U) S^b \neq 0.$$

At the simple Bessel pole from Lemma 26.2, the exact line identity reads

$$D_j = C + s_j(B_j - \beta).$$

The coefficient of B_j^D in the polar expansion is therefore $H_D(e_j^{\text{pole}}, s_j)$. Let b_0 be the smallest index with $p_{b_0} \neq 0$. Since $s_j \to 0$ and $e_j^{\text{pole}} \to u_*$, we have

$$\frac{H_D(e_j^{\text{pole}}, s_j)}{s_j^{b_0}} \longrightarrow p_{b_0}(u_*) \neq 0.$$

The last inequality uses the transcendence of u_*. Hence, for all sufficiently large j, an order-D pole survives. Terms of lower Bessel total degree cannot cancel it, and the Ein image is holomorphic at the point. This contradicts the pole obstruction. Thus $D = 0$, so $P \in k[U]$, completing the proof. □

Corollary 26.4 (The matched-scale question)**.** *For every positive algebraic a,*

$$\overline{\mathbf{E}} \cap k(\mathrm{Ein}(a), \beta(2a), C(2a)) = k[\mathrm{Ein}(a)].$$

In particular, with $\eta = \mathrm{Ein}(1)$, the previously unresolved intersection $\overline{\mathbf{E}} \cap k(\eta, \beta(2), C(2))$ equals $k[\eta]$.

Remark 26.5 (Scope and review status). This argument concerns one Bessel block together with one Ein value; it does not assert the corresponding full rational-field statement for several rationally dependent Bessel scales. It also establishes neither the irrationality of γ nor full algebraic independence of the original four constants. The proof above is a newly derived, internally audited extension of the manuscript's established phase embeddings and joint independence theorem. Historical priority and external expert verification remain separate questions.

27 Adjoining the matching exponential

The full rational intersection remains exact after adjoining the exponential at the Ein argument. The inverse exponential is necessarily present: both e^a and e^{-a} are units of the E-value ring.

Theorem 27.1 (Exact exponential–Ein–Bessel intersection)**.** *Let a, t be arbitrary positive algebraic numbers, and put*

$$E = e^a, \qquad \eta = \mathrm{Ein}(a), \qquad \beta = \beta(t), \qquad C = C(t), \qquad k = \overline{\mathbb{Q}}.$$

Then

$$\boxed{\overline{\mathbf{E}} \cap k(E, \eta, \beta, C) = k[E, E^{-1}, \eta].} \tag{77}$$

No rational-independence or commensurability condition on a, t is required.

27.1 Limits with the exponential coordinate

Retain the notation $p = i\pi$, $\tau = 2\pi i$, $r = 2t/a$, h, $c_{a,j}$, $c_{t,j}$, d_j, v_j, s_j from Section 26. In particular,

$$u_* = \eta - p/r, \qquad q = e^{2t(1-X)}.$$

Write E_j, B_j, D_j for the continued images of η, β, C as in (68)–(70); the letter E_j continues to denote the Ein image, not the exponential image. For clarity, denote the latter by

$$\chi_j(X) := \Phi_j(e^a)(X) = e^{aX}. \tag{78}$$

This expression is independent of the winding index. Near $X = 1$, using the same logarithmic branch as before,

$$\chi_j = E q^{-1/r}. \tag{79}$$

The joint independence theorem over the field generated by algebraic exponentials, together with the transcendence of $E = e^a$, makes

$$E, \eta, \beta, C \quad \text{algebraically independent over } k. \tag{80}$$

We also have algebraic independence of E, u_*, C. Indeed, E, C are real and algebraically independent. If u_* were algebraic over $k(E, C)$, complex conjugation would make $\overline{u_*}$ algebraic over that same field, and their average η would be algebraic over $k(E, C)$, contradicting (80). In particular, E, u_* are algebraically independent.

The first two charts of Lemma 26.2 now give, uniformly on the fixed closed disks specified there,

$$B_j = x \text{ bounded}: \quad (\chi_j, E_j, D_j) \longrightarrow (E, \eta, C), \tag{81}$$

$$D_j = w \text{ away from } C: \quad (\chi_j, E_j) \longrightarrow (E, u_*), \qquad B_j = \beta + (w - C)/s_j. \tag{82}$$

The assertion for χ_j follows from $q_j \to 1$.

We also need the Ein chart with a variable target, rather than only an algebraic target. Prescribe $E_j = u$ and set

$$z_j(u) = \frac{u - c_{a,j}}{d_j} = 1 + \frac{u - \eta}{d_j}, \qquad q_j(u) = z_j(u)^r, \qquad \chi_j(u) = \frac{E}{z_j(u)}. \tag{83}$$

On every fixed closed disk avoiding u_*, the chosen branches are holomorphic for all large j, and, uniformly,

$$\chi_j(u) \longrightarrow E, \qquad D_j(u) \longrightarrow C, \tag{84}$$

$$\frac{B_j(u)}{j} \longrightarrow F(u) := \frac{\tau r(u-\eta)}{p + r(u-\eta)}. \tag{85}$$

This follows by the same expansion used for (74). On a disk also avoiding η, the limiting function F is bounded away from zero, so $|B_j(u)| \asymp j$. The local inverse has derivative $-1/(a(u - c_{a,j})) \neq 0$.

Finally, at the Bessel pole (75),

$$\chi_j \longrightarrow E, \qquad E_j \longrightarrow u_*, \qquad s_j \longrightarrow 0. \tag{86}$$

The exponential image is holomorphic and nonzero there.

27.2 Proof of the exact intersection

Proof of Theorem 27.1. The ring on the right of (77) is contained in $\mathbf{E}$, proving one inclusion. For the reverse inclusion, let

$$\xi = \frac{P(E,\eta,\beta,C)}{Q(E,\eta,\beta,C)} \in \overline{\mathbf{E}}, \qquad P, Q \in k[S,U,V,W], \quad (P,Q) = 1,$$

with $Q \neq 0$. The case $P = 0$ is immediate. Joint independence makes every nonzero denominator at the actual tuple nonzero. The continued embeddings extend to the resulting fraction field, and integrality excludes every finite pole of an image of ξ.

Denominators involving V. Let q_0 be an irreducible factor of Q of positive V-degree. Gauss's lemma and $q_0 \nmid P$ imply that

$$\mathrm{Res}_V(q_0, P) \in k[S,U,W] \setminus \{0\}.$$

Independence of E, η, C makes this resultant and the leading V-coefficient of q_0 nonzero at (E,η,C). Hence $q_0(E,\eta,V,C)$ is a nonconstant polynomial coprime to $P(E,\eta,V,C)$.

Choose a root and a small closed disk around it, with no boundary root and with the numerator nonzero throughout the disk. In the finite-B_j chart (81), the corresponding image polynomials converge uniformly to these two polynomials. Rouché's theorem gives a denominator zero while the numerator stays nonzero. The holomorphic local inverse transfers this uncancelled pole to a finite X, a contradiction. Thus $Q \in k[S,U,W]$.

Denominators involving W. Let $q_0(S,U,W)$ be an irreducible factor of Q of positive W-degree. Expand

$$P(S,U,V,W) = \sum_{\ell} P_\ell(S,U,W) V^\ell,$$

and choose the largest m with $q_0 \nmid P_m$. Every higher coefficient is divisible by q_0. The resultant

$$R(S,U) = \mathrm{Res}_W(q_0, P_m) \in k[S,U] \setminus \{0\}$$

remains nonzero at (E,u_*), by their algebraic independence. The leading W-coefficient remains nonzero as well. Thus $q_0(E,u_*,W)$ is nonconstant and coprime to $P_m(E,u_*,W)$. Independence of E, u_*, C also gives $q_0(E,u_*,C) \neq 0$.

Choose a root $w_0 \neq C$ and a sufficiently small closed disk avoiding C on which $P_m(E, u_*, w)$ is nonzero and whose boundary contains no root of $q_0(E, u_*, w)$. Use the finite-D_j chart (82). Rouché's theorem supplies roots w_j of $q_0(\chi_j(w), E_j(w), w)$ in the disk. At these roots, all coefficients with $\ell > m$ vanish exactly. The coefficient of B_j^m is bounded away from zero, while lower coefficients remain bounded and $|B_j| \asymp j^2$. The numerator is therefore nonzero for all large j. This is an uncancelled pole and a contradiction. It follows that $Q \in k[S, U]$.

Denominators involving U. Let $q_0(S, U)$ be an irreducible factor of Q of positive U-degree. Using the same expansion in V, choose the largest m for which $q_0 \nmid P_m(S, U, W)$. Irreducibility persists after adjoining the independent variable W, so Gauss's lemma gives

$$R(S, W) = \mathrm{Res}_U(q_0, P_m) \in k[S, W] \setminus \{0\}.$$

This resultant remains nonzero at (E, C). Also the leading U-coefficient of q_0 remains nonzero at E. Thus $q_0(E, U)$ is nonconstant and coprime to $P_m(E, U, C)$.

Every root u_0 of $q_0(E, U)$ differs from both η and u_*. Indeed, either equality would contradict respectively the independence of E, η or of E, u_*. Choose a small closed disk around one such root, avoiding both exceptional values, on which $P_m(E, u, C)$ stays nonzero and whose boundary contains no root of $q_0(E, u)$. Apply the Ein chart (83)–(85). Rouché's theorem gives zeros of $q_0(\chi_j(u), u)$ in the disk. At those zeros, all higher V-coefficients vanish exactly, the coefficient of B_j^m stays nonzero, and $|B_j| \asymp j$. The same leading-term argument gives a nonzero numerator at a denominator zero. This contradiction proves $Q \in k[S]$.

The nonzero exponential roots. Suppose $\alpha \in k^\times$ is a root of Q. Coprimality gives

$$P(\alpha, U, V, W) \neq 0.$$

Choose any finite X_0 satisfying $e^{aX_0} = \alpha$. At this point put

$$z_0 = e^{a(1-X_0)} = \frac{E}{\alpha}, \qquad q_0 = e^{2t(1-X_0)}.$$

Then $z_0 \neq 1$, since E is transcendental. Moreover $q_0 \neq 1$: the positive real number $|\alpha|$ is algebraic, whereas E is transcendental, and

$$|q_0| = \left(\frac{E}{|\alpha|}\right)^{2t/a} \neq 1.$$

For this fixed X_0, the exact phase formulas give

$$E_j(X_0) = \tau j(1 - E/\alpha) + O(1), \qquad B_j(X_0) = \tau j + O(1), \qquad D_j(X_0) \longrightarrow C. \tag{87}$$

For large j their Bessel denominators are nonzero, since $p + v_j(q_0 - 1)$ has a nonzero linear term in j.

Let D be the total degree in (U, V) of $P(\alpha, U, V, W)$, and let $H_D(U, V, W)$ be its nonzero highest homogeneous part in those two variables. The substitution $H_D(T, 1, W)$ is nonzero. Algebraic independence of E, C implies

$$H_D(1 - E/\alpha, 1, C) \neq 0.$$

Consequently (87) gives

$$\frac{P(\alpha, E_j(X_0), B_j(X_0), D_j(X_0))}{(\tau j)^D} \longrightarrow H_D(1 - E/\alpha, 1, C) \neq 0.$$

The numerator is nonzero for large j, while its denominator $Q(\chi_j(X_0)) = Q(\alpha)$ vanishes. Since $\chi_j'(X_0) = a\alpha \neq 0$, this produces a finite pole. Thus Q has no nonzero root. As k is algebraically closed, $Q(S)$ is a nonzero constant times a nonnegative power of S. We have proved

$$\xi \in k[E, E^{-1}, \eta, \beta, C].$$

The remaining Laurent polynomial. Write ξ as a polynomial in β, C with coefficients in $k[E, E^{-1}, \eta]$. Suppose its Bessel total degree is $D > 0$. Its highest homogeneous part can be encoded as

$$H_D(S, U, Z) = \sum_{b=0}^{D} p_b(S, U) Z^b, \qquad p_b \in k[S, S^{-1}, U],$$

where Z records the slope in the image line $W = s_j V + (C - s_j\beta)$, with β, C denoting the fixed original values. Let b_0 be the smallest index with $p_{b_0} \neq 0$. At the Bessel pole, (86) gives

$$\frac{H_D(\chi_j^{\text{pole}}, E_j^{\text{pole}}, s_j)}{s_j^{b_0}} \longrightarrow p_{b_0}(E, u_*) \neq 0.$$

The last inequality follows from independence of E, u_*; negative powers of E cause no difficulty because $E \neq 0$ and the image χ_j is holomorphic and nonzero. Therefore an order-D pole survives for large j, exactly as in the polynomial part of Theorem 26.1. Lower Bessel total degrees cannot cancel it. This contradicts integrality and forces $D = 0$. Thus $\xi \in k[E, E^{-1}, \eta]$, proving (77). □

Corollary 27.2 (Matched exponential and Bessel scales)**.** *For every positive algebraic a,*

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(2a), C(2a)) = k[e^a, e^{-a}, \mathrm{Ein}(a)].$$

In particular, at $a = 1$ the full resonant field obtained by adjoining e to $k(\eta, \beta(2), C(2))$ has exactly the Laurent-polynomial intersection $k[e, e^{-1}, \eta]$.

Remark 27.3 (What is and is not included). The exponential adjoined here is e^a, matching the Ein argument. The theorem permits an arbitrary positive algebraic Bessel argument t. It does not assert the analogous full rational-field theorem for several dependent Bessel blocks or for an arbitrary collection of additional exponential coordinates. Its proof uses the continued phase embeddings and the previously established joint independence theorem; it does not prove irrationality of γ.

28 Rational intersections at dependent logarithmic-derivative scales

Rational dependence between Bessel phases does not obstruct the exact intersection for their logarithmic derivatives. The next result allows, for example, the two scales 2 and 4, together with an exponential integral at an arbitrary positive algebraic point.

Theorem 28.1 (Arbitrary distinct positive logarithmic-derivative scales)**.** *Let $a, t_1, \ldots, t_m$ be positive algebraic numbers, with the t_i pairwise distinct, and put*

$$\eta_a = \mathrm{Ein}(a), \qquad C_i = \frac{I_1(t_i)}{I_0(t_i)}, \qquad k = \overline{\mathbb{Q}}.$$

Then

$$\boxed{\overline{\mathbf{E}} \cap k(\eta_a, C_1, \ldots, C_m) = k[\eta_a].} \tag{88}$$

No rational independence among $a, t_1, \ldots, t_m$ is required.

We first record three local forms of the continued phase images. They use the same winding for all coordinates, so they do not presume independent exponential variables.

Lemma 28.2 (Local charts for synchronized windings)**.** *Write $\eta = \eta_a$, $p = i\pi$, $\tau = 2\pi i$, and*

$$g_a = \gamma + \log a, \quad d_j = \eta - g_a - j\tau, \quad g_i = \gamma + \log(t_i/2), \quad v_{i,j} = \beta(t_i) - g_i - j\tau, \quad h_i = \frac{1}{t_i I_0(t_i)^2}.$$

For the continued embeddings $\Phi^{[j]} = \Phi^{[j]}_{-\pi/4}$, let $\eta_j(X) = \Phi^{[j]}(\eta)(X)$ and $\mathscr{C}_{i,j}(X) = \Phi^{[j]}(C_i)(X)$. Their exact formulas are

$$\eta_j(X) = g_a + j\tau + d_j e^{a(1-X)}, \tag{89}$$

$$\mathscr{C}_{i,j}(X) = C_i - h_i \frac{q_i - 1}{p + v_{i,j}(q_i - 1)}, \qquad q_i = e^{2t_i(1-X)}. \tag{90}$$

Put $r_i = 2t_i/a$ and $u_i^* = \eta - p/r_i$. As $j \to +\infty$:

1. *On each compact set avoiding $u_1^*, \dots, u_m^*$, there is an analytic local parametrization with $\eta_j(X_j(u)) = u$ and*
$$\mathscr{C}_{i,j}(X_j(u)) \longrightarrow C_i \qquad (1 \le i \le m)$$
uniformly on that compact set.

2. *Fix i. On each compact set avoiding C_i, there is an analytic local parametrization with $\mathscr{C}_{i,j}(X_{i,j}(w)) = w$ and*
$$\eta_j(X_{i,j}(w)) \longrightarrow u_i^*, \qquad \mathscr{C}_{l,j}(X_{i,j}(w)) \longrightarrow C_l \quad (l \ne i)$$
uniformly on that compact set. Both parametrizations have nonzero derivative for sufficiently large j.

3. *There is a simple pole $X_{i,j}^{\mathrm{pole}}$ of $\mathscr{C}_{i,j}$, tending to 1, at which all the other displayed images are holomorphic and satisfy*
$$\eta_j(X_{i,j}^{\mathrm{pole}}) \longrightarrow u_i^*, \qquad \mathscr{C}_{l,j}(X_{i,j}^{\mathrm{pole}}) \longrightarrow C_l \quad (l \ne i).$$

Proof. Formula (89) is the positive-point case of (53). Formula (90) follows by substituting the coefficients in (34)–(35) and using the Bessel Wronskian. In particular, its constants are coefficients of the phase images, not elements asserted to be fixed by the embedding.

For the first chart, choose the logarithm near 1 and set

$$X_j(u) = 1 - \frac{1}{a} \log\left(1 + \frac{u - \eta}{d_j}\right).$$

Then

$$q_i = \left(1 + \frac{u - \eta}{d_j}\right)^{r_i}, \qquad p + v_{i,j}(q_i - 1) \longrightarrow p + r_i(u - \eta).$$

The last expression is nonzero away from u_i^*, whereas $q_i - 1$ tends uniformly to zero. This proves the first assertion.

For the second chart solve (90) with value w:

$$q_i(w) = 1 + \frac{p(C_i - w)}{h_i - v_{i,j}(C_i - w)}, \qquad X_{i,j}(w) = 1 - \frac{1}{2t_i} \log q_i(w).$$

Uniformly away from C_i, $q_i(w) - 1 = -p/v_{i,j} + O(j^{-2})$. Consequently

$$\eta_j(X_{i,j}(w)) \longrightarrow \eta - \frac{pa}{2t_i} = u_i^*, \qquad p + v_{l,j}(q_i(w)^{t_l/t_i} - 1) \longrightarrow p(1 - t_l/t_i).$$

The latter limit is nonzero for $l \neq i$, which proves the claimed limits for the other logarithmic derivatives. The first chart has derivative $-1/(a(d_j + u - \eta))$. In the second chart, $q_i'(w) = -ph_i/(h_i - v_{i,j}(C_i - w))^2 \neq 0$ and $q_i(w) \neq 0$ for large j. Thus the inverse changes of parameter preserve poles.

Finally take $q_i = 1 - p/v_{i,j}$ and the logarithm near 1 to define $X_{i,j}^{\mathrm{pole}}$. The numerator and the derivative of the denominator in (90) are nonzero there for sufficiently large j, so this pole is simple. The same limits just computed show that the other images are holomorphic there and have the asserted limits. □

Proof of Theorem 28.1. By Theorem 3.3, the real numbers $\eta, C_1, \ldots, C_m$ are jointly algebraically independent over k. Let $F = k(C_1, \ldots, C_m)$. This field is stable under complex conjugation. For every i, the number u_i^* is transcendental over F: otherwise its conjugate would also be algebraic over F, and their average η would be algebraic over F, a contradiction. In particular, u_i^* and any subtuple of the C_l are jointly independent over k.

Take a nonzero rational function

$$R = \frac{P(U, V_1, \ldots, V_m)}{Q(U, V_1, \ldots, V_m)}, \qquad P, Q \in k[U, V_1, \ldots, V_m], \quad \gcd(P, Q) = 1.$$

All numerical denominators are nonzero by joint independence. We show that its value can be integral over $\mathbf{E}$ only when it is a polynomial in η.

Suppose first that $D = \deg_U Q > 0$. Over the rational function field $k(V_1, \ldots, V_m)$, the two polynomials P, Q remain coprime as polynomials in U. Their resultant in U is therefore nonzero. Specializing the V_i to the independent C_i preserves that nonvanishing. Hence $Q(U, C_1, \ldots, C_m)$ has degree D and shares no root with $P(U, C_1, \ldots, C_m)$.

Every root u_0 of this specialized denominator differs from all the u_i^*: a root is algebraic over F, whereas every u_i^* is transcendental over F. Choose a small disk about such a root u_0 avoiding every critical point, with no zero of the limiting denominator on its boundary. In the first chart of Lemma 28.2, the embedded denominator converges uniformly on this disk to $Q(u, C_1, \ldots, C_m)$. Rouché's theorem supplies a nearby denominator zero for all large j. The numerator is nonzero near u_0, by the resultant argument and uniform convergence. Thus the embedded rational function has a pole at a finite X.

Next suppose that Q is independent of U but is nonconstant. Choose an index i on which it depends and set $F_i = k(C_l : l \neq i)$. After specializing the other coordinates, Q is a nonconstant polynomial $Q_i(V) \in F_i[V]$, and P is a polynomial $P_i(U, V) \in F_i[U, V]$ coprime to Q_i. These statements follow by localization in the other independent variables. Let α be a root of Q_i. Then $\alpha \neq C_i$ because C_i is transcendental over F_i. Moreover,

$$P_i(u_i^*, \alpha) \neq 0.$$

Otherwise transcendence of u_i^* over the algebraic extension $F_i(\alpha)$ would give the polynomial identity $P_i(U, \alpha) = 0$. The minimal polynomial of α over F_i would divide both P_i and Q_i, contradicting their coprimality. Use the second chart of Lemma 28.2 on a small disk about α avoiding C_i. The denominator converges to $Q_i(w)$ and the numerator to $P_i(u_i^*, w)$. Rouché's theorem again gives an uncancelled denominator zero, hence a finite pole of the phase image.

It remains to treat a polynomial P depending on at least one V_i. Write its positive degree in this variable as d, with leading coefficient $P_d(U,(V_l)_{l\neq i}) \neq 0$. At the pole in the third chart of Lemma 28.2, this coefficient tends to

$$P_d(u_i^*,(C_l)_{l\neq i}) \neq 0,$$

by the independence established at the start of the proof. All the other coordinates are holomorphic there. Consequently the polynomial phase image has a pole of order d for sufficiently large j.

Every pole contradicts integrality by Lemma 18.2. Thus only polynomials in η can belong to the intersection. Conversely η is an E-value, so $k[\eta] \subseteq \mathbf{E}$. This proves (88). □

Corollary 28.3 (The full logarithmic-derivative field at dependent scales)**.** *For pairwise distinct positive algebraic $t_1,\dots,t_m$,*

$$\overline{\mathbf{E}} \cap k(C(t_1),\dots,C(t_m)) = k.$$

In particular

$$\overline{\mathbf{E}} \cap k(C(2),C(4)) = k, \qquad \overline{\mathbf{E}} \cap k(\mathrm{Ein}(1),C(2),C(4)) = k[\mathrm{Ein}(1)].$$

The same exclusion holds for the field generated by all $C(t)$ with positive algebraic t, since every rational expression uses only finitely many arguments.

Proof. Choose any positive algebraic a and apply the theorem. Independence of $\mathrm{Ein}(a)$ from the $C(t_i)$ gives $k[\mathrm{Ein}(a)] \cap k(C(t_1),\dots,C(t_m)) = k$. The infinite-family assertion follows by restricting any proposed element to its finite set of arguments. □

28.1 One harmonic coordinate and arbitrarily many derivative scales

The same local charts permit one harmonic Bessel coordinate to be adjoined, without imposing rational independence on any of the scales.

Theorem 28.4 (One harmonic coordinate at dependent derivative scales)**.** *Let $a,s,t_1,\dots,t_m$ be positive algebraic numbers, with the t_i pairwise distinct. Then*

$$\boxed{\overline{\mathbf{E}} \cap k(\mathrm{Ein}(a),\beta(s),C(t_1),\dots,C(t_m)) = k[\mathrm{Ein}(a)].} \tag{91}$$

In particular, the equality remains true when all positive algebraic derivative scales $C(t)$ are adjoined simultaneously.

Proof. Adjoining $C(s)$ if necessary only enlarges the field, so we may relabel the distinct scales such that $t_1 = s$. Abbreviate $\beta_s = \beta(s)$ and $\eta = \mathrm{Ein}(a)$, and retain the notation of Lemma 28.2. Write $\mathscr{B}_j(X) = \Phi^{[j]}(\beta_s)(X)$ and $b_j = v_{1,j}$. The exact formulas and the line identity (39) give

$$\mathscr{B}_j(X) = g_1 + j\tau + \frac{pb_jq_1}{p + b_j(q_1-1)}, \tag{92}$$

$$\mathscr{C}_{1,j}(X) = C_1 + \sigma_j(\mathscr{B}_j(X) - \beta_s), \qquad \sigma_j \sim -\frac{1}{4\pi^2 s I_0(s)^2 j^2}. \tag{93}$$

For large j the slope σ_j is nonzero.

We shall use one additional local chart. Prescribe a bounded value v of $\mathscr{B}_j$ and set

$$q_1(v) = 1 + \frac{p(v-\beta_s)}{b_j(p - v + g_1 + j\tau)}, \qquad X_j^{\beta}(v) = 1 - \frac{1}{2s}\log q_1(v),$$

with the logarithm near 1. These formulas give $\mathscr{B}_j(X_j^\beta(v)) = v$ exactly. Since $q_1(v) - 1 = O(j^{-2})$, uniformly on bounded sets,

$$\eta_j(X_j^\beta(v)) \longrightarrow \eta, \qquad \mathscr{C}_{l,j}(X_j^\beta(v)) \longrightarrow C_l \quad (1 \le l \le m). \tag{94}$$

The change of parameter has nonzero derivative for all sufficiently large j, as follows either from its displayed formula or from the nonconstant Möbius map (92).

Two growth estimates on the earlier charts will also be needed. On the first chart of Lemma 28.2, uniformly away from the critical points,

$$\frac{\mathscr{B}_j(X_j(u))}{j} \longrightarrow \frac{\tau r_1(u-\eta)}{p + r_1(u-\eta)}. \tag{95}$$

On the chart with $\mathscr{C}_{i,j} = w$, uniformly away from C_i,

$$\frac{\mathscr{B}_j(X_{i,j}(w))}{j} \longrightarrow \frac{\tau(s/t_i)}{s/t_i - 1} \qquad (i \ne 1), \tag{96}$$

$$\frac{\mathscr{B}_j(X_{1,j}(w))}{j^2} \longrightarrow -4\pi^2 s I_0(s)^2 (w - C_1) \qquad (i = 1). \tag{97}$$

To verify (95), insert $q_1 - 1 = r_1(u-\eta)/d_j + O(j^{-2})$ into (92). For (96), use $q_i - 1 = -p/v_{i,j} + O(j^{-2})$ and $q_1 = q_i^{s/t_i}$. The denominator in (92) then tends to $p(1 - s/t_i)$. Formula (97) follows directly from (93).

The numbers $\eta, \beta_s, C_1, \dots, C_m$ are jointly independent by Theorem 3.3. Let P/Q be a reduced nonzero rational function in the formal variables $U, B, V_1, \dots, V_m$, with coefficients in k, evaluated at that tuple.

If $\deg_B Q > 0$, coprimality over $k(U, V_1, \dots, V_m)$ gives a nonzero resultant in B. Independence of $\eta, C_1, \dots, C_m$ preserves it after specialization. Thus the nonconstant polynomial $Q(\eta, B, C_1, \dots, C_m)$ has a root where the specialized numerator is nonzero. The chart (94) and Rouché's theorem give a nearby uncancelled denominator zero of the phase image. This contradicts integrality by Lemma 18.2.

We may therefore suppose that Q is independent of B. If it is nonconstant, choose an irreducible factor $q(U, V_1, \dots, V_m)$ of Q. Write $P = \sum_b p_b(U, V_1, \dots, V_m) B^b$, and let d be the largest index for which q does not divide p_d. Such an index exists because P, Q are coprime. At an exact zero of q all terms with $b > d$ vanish.

If $\deg_U q > 0$, its resultant in U with p_d is nonzero over $k(V_1, \dots, V_m)$. After specializing the independent C_l, choose a root u_0 of $q(U, C_1, \dots, C_m)$. Then $p_d(u_0, C_1, \dots, C_m) \ne 0$. Furthermore u_0 differs from η and from every u_i^*, since it is algebraic over $F = k(C_1, \dots, C_m)$ whereas these numbers are transcendental over F. In the first chart of Lemma 28.2, Rouché's theorem gives zeros $u_j \to u_0$ of the embedded factor q. By (95), the corresponding β_s images grow linearly in j with a nonzero limiting coefficient. At these zeros the terms of P above degree d vanish exactly, its degree-d coefficient tends to a nonzero number, and all lower coefficients remain bounded. Hence the numerator is nonzero for large j.

If q is independent of U, choose a variable V_i on which it depends and put $F_i = k(C_l : l \ne i)$. Over F_i, the specialization $q_i(V)$ is a nonconstant polynomial coprime to the specialized polynomial $p_d(U, V)$. Let α be a root of q_i. As in the proof of Theorem 28.1, one has $\alpha \ne C_i$ and $p_d(u_i^*, \alpha, (C_l)_{l \ne i}) \ne 0$. Use the chart with $\mathscr{C}_{i,j} = w$ near α. Rouché's theorem gives nearby zeros of the embedded q, while (96)–(97) show that the β_s image tends to infinity, with order j or j^2 and nonzero leading coefficient. The same argument with the highest coefficient not divisible by q shows that the numerator is nonzero there. In either case an uncancelled pole contradicts integrality. Thus Q must be constant.

Finally let P be a polynomial of positive total degree D in the two variables B, V_1. Write its top homogeneous part in these two variables as

$$\sum_{b+n=D} p_{b,n}(U, V_2, \dots, V_m) B^b V_1^n, \qquad H_D(U, V_2, \dots, V_m, S) = \sum_{b+n=D} p_{b,n}(U, V_2, \dots, V_m) S^n.$$

At the pole $X_{1,j}^{\mathrm{pole}}$ of Lemma 28.2, both $\mathscr{B}_j$ and $\mathscr{C}_{1,j}$ have simple poles, related by (93). The coefficient of the possible pole of order D, measured in powers of $\mathscr{B}_j$, is

$$H_D\big(\eta_j(X_{1,j}^{\mathrm{pole}}), (\mathscr{C}_{l,j}(X_{1,j}^{\mathrm{pole}}))_{l\ge 2}, \sigma_j\big).$$

Divide this expression by the lowest power of σ_j occurring in H_D. The resulting expression tends to a nonzero coefficient $p_{b,n}(u_1^*, C_2, \dots, C_m)$, by the independence proved in Theorem 28.1. Consequently the pole of order D survives for all large j. If instead P is independent of both B, V_1, Theorem 28.1 already shows that it can be integral only when it is a polynomial in η.

The reverse inclusion holds because η is an E-value. This proves (91); the infinite-family assertion follows because each rational expression involves finitely many scales. □

Corollary 28.5 (A global derivative field with one harmonic coordinate)**.** *For every fixed positive algebraic s,*

$$\overline{\mathbf{E}} \cap k\big(\beta(s), C(t) : t > 0 \text{ algebraic}\big) = k.$$

Proof. Adjoin $\mathrm{Ein}(a)$ for any positive algebraic a and apply Theorem 28.4. Joint independence makes $\mathrm{Ein}(a)$ transcendental over the displayed field: any algebraic relation would involve only finitely many of its generators. Its polynomial ring therefore intersects that field in k. □

Remark 28.6 (The remaining dependent-scale question)*.* For example, the theorem gives

$$\overline{\mathbf{E}} \cap k\big(\mathrm{Ein}(1), \beta(2), C(2), C(4)\big) = k[\mathrm{Ein}(1)].$$

It does not classify the field with two harmonic coordinates $k(\beta(2), C(2), \beta(4), C(4))$. The unresolved part of the dependent-scale question therefore concerns simultaneous harmonic coordinates, rather than arbitrary derivative coordinates or one harmonic coordinate accompanied by them.

29 A global derivative field with an exponential and one harmonic value

The exponential extension and the dependent derivative-scale theorem can be combined. The result gives one exact intersection for the entire family of positive algebraic logarithmic-derivative arguments, while retaining one harmonic Bessel coordinate.

Theorem 29.1 (Exponential–Ein intersection over a derivative family)**.** *Let $a, s, t_1, \dots, t_m$ be positive algebraic numbers, with the t_i pairwise distinct. Put*

$$k = \overline{\mathbb{Q}}, \qquad E = e^a, \qquad \eta = \mathrm{Ein}(a), \qquad \beta_s = \beta(s), \qquad C_i = C(t_i).$$

Then

$$\boxed{\overline{\mathbf{E}} \cap k(E, \eta, \beta_s, C_1, \dots, C_m) = k[E, E^{-1}, \eta].} \tag{98}$$

No rational-independence hypothesis on the arguments is required.

Proof. Adding $C(s)$ enlarges the field, so it suffices to prove the result after doing so and relabelling the distinct arguments such that $t_1 = s$. Retain the notation and synchronized winding charts of Lemma 28.2, with

$$p = i\pi, \qquad \tau = 2\pi i, \qquad r_i = \frac{2t_i}{a}, \qquad u_i^* = \eta - \frac{p}{r_i}.$$

Write $\chi_j = \Phi^{[j]}(E)$, $\eta_j = \Phi^{[j]}(\eta)$, $\mathscr{B}_j = \Phi^{[j]}(\beta_s)$, and $\mathscr{C}_{i,j} = \Phi^{[j]}(C_i)$.

Independence and local limits. The joint theorem over the field generated by algebraic exponentials, and the transcendence of E, give joint independence over k of

$$E, \eta, \beta_s, C_1, \ldots, C_m. \tag{99}$$

Let $F = k(E, C_1, \ldots, C_m)$. All its displayed generators are real, so F is stable under conjugation. Each u_i^* is transcendental over F: algebraicity of u_i^* would imply algebraicity of its conjugate, and then of their average η, contrary to (99). In particular, for every subtuple of the C_l, the numbers E, u_i^* and that subtuple are jointly independent over k.

The exponential image is always

$$\chi_j(X) = e^{aX}.$$

All the local charts below have $X \to 1$ uniformly on their fixed closed disks. Consequently they all have $\chi_j \to E$ uniformly. Together with the previously proved chart formulas, this gives:

1. In the chart $\mathscr{B}_j = v$, with v bounded,

$$(\chi_j, \eta_j, \mathscr{C}_{1,j}, \ldots, \mathscr{C}_{m,j}) \longrightarrow (E, \eta, C_1, \ldots, C_m).$$

2. In the chart $\eta_j = u$, on disks avoiding every u_i^*,

$$\chi_j \longrightarrow E, \qquad \mathscr{C}_{i,j} \longrightarrow C_i, \qquad \frac{\mathscr{B}_j}{j} \longrightarrow \frac{\tau r_1 (u - \eta)}{p + r_1(u - \eta)}.$$

   If the disk also avoids η, the last limiting function is bounded away from zero.

3. In the chart $\mathscr{C}_{i,j} = w$, on disks avoiding C_i,

$$\chi_j \longrightarrow E, \qquad \eta_j \longrightarrow u_i^*, \qquad \mathscr{C}_{l,j} \longrightarrow C_l \quad (l \neq i).$$

   For $i \neq 1$ the additional limit is

$$\frac{\mathscr{B}_j}{j} \longrightarrow \frac{\tau(s/t_i)}{s/t_i - 1} \neq 0.$$

   For $i = 1$ it is

$$\frac{\mathscr{B}_j}{j^2} \longrightarrow -4\pi^2 s I_0(s)^2 (w - C_1),$$

   which is bounded away from zero on the chosen disk.

These are (94) and (95)–(97), with the additional exponential limit. All local inverse derivatives are nonzero for sufficiently large j, as proved with those charts, so poles in the target parameters are poles at finite values of X.

Reduction of a rational denominator. Take a reduced nonzero rational expression

$$\xi = \frac{P(E, \eta, \beta_s, C_1, \dots, C_m)}{Q(E, \eta, \beta_s, C_1, \dots, C_m)} \in \overline{\mathbf{E}}, \qquad P, Q \in k[S, U, B, V_1, \dots, V_m].$$

All nonzero denominators at the actual tuple are nonzero by (99). Continued phase embeddings therefore act on the field in question. Their images of ξ cannot have finite poles.

If Q has positive B-degree, its resultant in B with P is nonzero over $k(S, U, V_1, \dots, V_m)$, because the two polynomials are coprime. Clearing denominators gives a nonzero polynomial in the remaining variables. Independence of $E, \eta, C_1, \dots, C_m$ preserves this resultant and the leading coefficient after specialization. Thus the nonconstant polynomial $Q(E, \eta, B, C_1, \dots, C_m)$ has a root where the specialized numerator is nonzero. The finite-$\mathscr{B}_j$ chart and Rouché's theorem produce an actual denominator zero with nonzero numerator for every sufficiently large j. This is a forbidden pole. Hence Q is independent of B.

Choose an irreducible factor $q(S, U, V_1, \dots, V_m)$ of Q. Expand

$$P = \sum_b p_b(S, U, V_1, \dots, V_m) B^b,$$

and let d be the largest index such that $q \nmid p_d$. It exists by coprimality. At every exact zero of the embedded factor q, all terms with $b > d$ vanish exactly.

Suppose first that $\deg_U q > 0$. By Gauss's lemma, q and p_d are coprime over $k(S, V_1, \dots, V_m)$ as polynomials in U. Their resultant remains nonzero after the independent specialization $(S, V_1, \dots, V_m) = (E, C_1, \dots, C_m)$. The specialized $q(E, U, C_1, \dots, C_m)$ is nonconstant. Choose a root u_0; then

$$p_d(E, u_0, C_1, \dots, C_m) \neq 0.$$

The root u_0 is algebraic over F, whereas η and all u_i^* are transcendental over F. Thus it differs from those finitely many exceptional values. Choose a small closed disk around u_0 avoiding them, on which the displayed numerator coefficient stays nonzero and whose boundary contains no root of the limiting denominator.

On the Ein chart, uniform convergence and Rouché's theorem give exact zeros of the embedded q in this disk. At these zeros the terms above degree d vanish exactly, the degree-d coefficient stays bounded away from zero, all lower coefficients are bounded, and $|\mathscr{B}_j| \asymp j$. The full numerator is therefore nonzero for large j. This produces a forbidden pole. Hence no irreducible factor of Q can have positive U-degree, and $Q \in k[S, V_1, \dots, V_m]$.

Suppose next that an irreducible factor q of Q has positive degree in some V_i. Put

$$F_i = k(E, C_l : l \neq i).$$

After localizing in the other independent variables and specializing them, q becomes an irreducible nonconstant polynomial $q_i(V)$ over F_i, up to multiplication by a nonzero constant. The coefficient p_d becomes a polynomial $p_{d,i}(U, V)$ not divisible by q_i. Choose a root α of q_i. Then $\alpha \neq C_i$, since C_i is transcendental over F_i. Moreover

$$p_{d,i}(u_i^*, \alpha) \neq 0. \tag{100}$$

Indeed, u_i^* is transcendental over the algebraic extension $F_i(\alpha)$; vanishing in (100) would therefore imply the polynomial identity $p_{d,i}(U, \alpha) = 0$. The minimal polynomial q_i would then divide every coefficient in U, a contradiction.

Use the $\mathscr{C}_{i,j}$ chart in a small closed disk about α avoiding C_i. The denominator factor converges to $q_i(w)$, and its selected numerator coefficient converges to $p_{d,i}(u_i^*, w)$, which is nonzero near α. Rouché's theorem gives exact denominator-factor zeros in this disk. At those zeros the higher B-coefficients vanish, and $\mathscr{B}_j$ diverges at order j if $i \neq 1$, or at order j^2 if $i = 1$, with a nonzero limiting coefficient. The highest surviving coefficient therefore makes the numerator nonzero. This is another forbidden pole. We conclude that $Q \in k[S]$.

Only powers of the exponential may remain in the denominator. If $\alpha \in k^\times$ were a root of Q, coprimality would give $P(\alpha, U, B, V_1, \dots, V_m) \neq 0$. Choose a finite X_0 with $e^{aX_0} = \alpha$. For every l,

$$\left|e^{2t_l(1-X_0)}\right| = \left(\frac{E}{|\alpha|}\right)^{2t_l/a} \neq 1.$$

Here $|\alpha|$ is algebraic and E is transcendental. Thus none of the corresponding phase coordinates equals 1. The exact winding formulas at this fixed point give

$$\eta_j(X_0) = \tau j(1 - E/\alpha) + O(1), \qquad \mathscr{B}_j(X_0) = \tau j + O(1), \qquad \mathscr{C}_{l,j}(X_0) \longrightarrow C_l.$$

All Bessel images are finite at X_0 for large j. Let D be the total degree of $P(\alpha, U, B, V_1, \dots, V_m)$ in (U, B), and let H_D be its nonzero top homogeneous part in those variables. Independence of $E, C_1, \dots, C_m$ implies

$$H_D(1 - E/\alpha, 1, C_1, \dots, C_m) \neq 0.$$

Indeed $H_D(T, 1, V_1, \dots, V_m)$ is a nonzero polynomial. The embedded numerator, divided by $(\tau j)^D$, tends to this nonzero number. Its denominator vanishes at X_0, and the exponential image has derivative $a\alpha \neq 0$ there. This is an uncancelled finite pole. Therefore Q has no nonzero root, and Q is a constant times a power of S. We have reduced ξ to

$$k[E, E^{-1}, \eta, \beta_s, C_1, \dots, C_m].$$

Eliminating polynomial dependence on the Bessel coordinates. Suppose this Laurent polynomial has positive total degree D in the two coordinates β_s, C_1. Its top homogeneous part in formal variables B, V_1 has coefficients in $k[S, S^{-1}, U, V_2, \dots, V_m]$. At the common pole $X_{1,j}^{\mathrm{pole}}$ of the two images, all other images are holomorphic and

$$\chi_j \longrightarrow E, \qquad \eta_j \longrightarrow u_1^*, \qquad \mathscr{C}_{l,j} \longrightarrow C_l \quad (l \geq 2).$$

The exact line identity is

$$\mathscr{C}_{1,j} = C_1 + \sigma_j(\mathscr{B}_j - \beta_s), \qquad \sigma_j \neq 0, \quad \sigma_j \longrightarrow 0.$$

In the coefficient of the possible order-D pole, divide by the lowest power of σ_j occurring in the top homogeneous part. The quotient tends to a nonzero Laurent polynomial evaluated at $E, u_1^*, C_2, \dots, C_m$. It is nonzero by the independence established above. Thus an order-D pole survives, a contradiction. The Laurent polynomial must be independent of both β_s, C_1.

If it still depends on some C_i with $i \geq 2$, use the pole $X_{i,j}^{\mathrm{pole}}$ from Lemma 28.2. The leading coefficient in this coordinate tends to a nonzero Laurent polynomial in E, u_i^* and the remaining actual derivative values; these are jointly independent. The other coordinates are holomorphic at the point, and the exponential image is nonzero. Hence the positive degree in C_i again creates an uncancelled pole. This excludes every remaining derivative coordinate.

It follows that $\xi \in k[E, E^{-1}, \eta]$. The reverse inclusion holds because e^a, e^{-a} and $\mathrm{Ein}(a)$ are E-values. This proves (98). □

Corollary 29.2 (A single exact ring over all positive derivative scales)**.** *For every fixed pair of positive algebraic numbers a, s,*

$$\overline{\mathbf{E}} \cap k(e^a, \mathrm{Ein}(a), \beta(s), C(t) : t > 0 \text{ algebraic}) = k[e^a, e^{-a}, \mathrm{Ein}(a)].$$

Moreover

$$\overline{\mathbf{E}} \cap k(e^a, \beta(s), C(t) : t > 0 \text{ algebraic}) = k[e^a, e^{-a}].$$

Proof. Every rational expression involves only finitely many derivative arguments, so the first identity follows from the theorem. For the second, $\mathrm{Ein}(a)$ is transcendental over the indicated field by the finite-subset form of joint independence. Thus a polynomial in $\mathrm{Ein}(a)$ with coefficients in $k[e^a, e^{-a}]$ can belong to that field only if it is constant in $\mathrm{Ein}(a)$. □

Remark 29.3 (The remaining harmonic-coordinate restriction)*.* The theorem allows all positive algebraic derivative scales simultaneously and arbitrary rational expressions in the displayed generators. It retains only one harmonic coordinate $\beta(s)$. It does not settle the field obtained by adjoining two different harmonic coordinates such as $\beta(2)$ and $\beta(4)$. The result is an internally audited consequence of the continued phase embeddings and the established joint independence theorem; external expert verification and historical priority remain separate.

29.1 The polynomial consequence at a single positive scale

The full rational classification contains the earlier polynomial result. In the notation of the matched-scale section, $\eta_a = \mathrm{Ein}(a)$, $B_a = \beta(2a)$, and $C_a = C(2a)$.

Corollary 29.4 (One-scale polynomial classification)**.** *For every positive algebraic number a,*

$$\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}[\eta_a, B_a, C_a] = \overline{\mathbb{Q}}[\eta_a]. \tag{101}$$

Proof. Apply Theorem 26.1 with the Ein argument a and Bessel argument $2a$, and restrict its rational field to the displayed polynomial ring. The reverse inclusion follows from $\eta_a \in \mathbf{E}$. □

30 Integral classifications in the three proposed coordinates

The general positive-scale classification specializes at $a = 1$ to $\eta = \mathrm{Ein}(1)$, $\beta = \gamma + \kappa$, and $C = I_1(2)/I_0(2)$. It yields an exact statement in the original proposed coordinates.

Corollary 30.1 (Exact classification in the proposed coordinates)**.** *Let $\ell = \gamma - 2\kappa$ and*

$$A = \delta/e - \kappa = \eta - \beta, \qquad U = \ell + 3\delta/e = 3\eta - 2\beta, \qquad V = \ell + 3\kappa = \beta.$$

Then

$$U = V + 3A, \qquad \overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(A, U, V, C) = \overline{\mathbb{Q}}[U + 2V] = \overline{\mathbb{Q}}[A + V].$$

Equivalently, a rational expression in A, U, V, C is integral over the E-value ring exactly when, after imposing $U - V - 3A = 0$, it equals a polynomial in $U + 2V = 3\eta$. Every nonconstant rational function in A alone, U alone, or V alone is non-integral.

Proof. The invertible change $(\eta, \beta) \leftrightarrow (A, V)$ gives $U = 3A + V$ and $A + V = \eta$. Apply Theorem 26.1 at $(a, t) = (1, 2)$; multiplication of η by the nonzero algebraic scalar 3 does not change its polynomial ring. □

Remark 30.2. The coefficient ring in the theorem is $\overline{\mathbb{Q}}[\eta]$, not the whole E-value ring: allowing arbitrary E-value coefficients would make $I_0(2)\beta = Y(2) \in \mathbf{E}$ an immediate counterexample. The rational-field classification therefore keeps its specified algebraic coefficient field.

Theorem 30.3 (All nontrivial Laurent monomials are excluded)**.** *For integers a, u, v, not all zero,*

$$A^a U^u V^v \notin \overline{\mathbf{E}}.$$

In particular A, U, V are multiplicatively independent, and every nonzero positive or negative integer power of any one of them is non-integral over the E-value ring.

Proof. Independence of η, β identifies their polynomial ring with $k[X, Y]$. Its three linear factors $X - Y, 3X - 2Y, Y$ are pairwise nonassociate irreducibles. The valuations of $(X - Y)^a(3X - 2Y)^u Y^v$ at these factors are exactly a, u, v. Every nonzero polynomial in X has valuation zero at all three factors. Thus, unless all exponents vanish, the monomial does not belong to $k[\eta]$. Theorem 26.1 excludes its integrality. The same observation excludes equality to 1 and proves multiplicative independence. □

Corollary 30.4 (Rational-function exclusion for mixed linear forms)**.** *Let k_0 be any subfield of $\overline{\mathbb{Q}}$, let $a, b \in k_0^\times$, and put $Z = a\eta + b\beta$. Then*

$$\overline{\mathbf{E}} \cap k_0(Z) = k_0.$$

In particular,

$$\overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(A) = \overline{\mathbb{Q}}, \qquad \overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(U) = \overline{\mathbb{Q}}, \qquad \overline{\mathbf{E}} \cap \overline{\mathbb{Q}}(V, C) = \overline{\mathbb{Q}}.$$

Proof. The first assertion follows from Theorem 25.3 at the single scale 1, together with the fact that the algebraic constants of the purely transcendental field $k_0(Z)$ are precisely k_0. Take $(a, b) = (1, -1)$ and $(3, -2)$ for A and U. The last equality, which closes the previously missing field for $V = \beta(2)$, is Theorem 19.1 at $t = 2$. □

31 Units, radicals, and algebraic extensions of the exact intersections

The exact intersections determine more than membership in the integral closure. They also determine units and pure radicals inside the specified rational fields. Throughout this section $k = \overline{\mathbb{Q}}$, $\mathbf{E}$ is the ring of E-values, and $\overline{\mathbf{E}}$ is its integral closure inside $\mathbb{C}$. The notation $R^\times$ denotes the group of units of a ring R.

Proposition 31.1 (Units in the mixed rational field)**.** *Use the hypotheses and notation of Corollary 24.1, and set*

$$A = k[E_1^{\pm 1}, \dots, E_m^{\pm 1}, \eta_1, \dots, \eta_m, Z_1^{\pm 1}, \dots, Z_r^{\pm 1}].$$

Then

$$\mathbf{E}^\times \cap \mathcal{F} = \overline{\mathbf{E}}^\times \cap \mathcal{F} = A^\times = \left\{ \alpha \exp\left(\sum_{i=1}^{m} n_i a_i + \sum_{l=1}^{r} m_l b_l \right) : \alpha \in k^\times,\ n_i, m_l \in \mathbb{Z} \right\}. \tag{102}$$

For example, for $\mathcal{F} = k(e, \mathrm{Ein}(1), \beta(\sqrt{2}), C(\sqrt{2}))$, both intersections of unit groups are precisely $\{\alpha e^n : \alpha \in k^\times,\ n \in \mathbb{Z}\}$.

Proof. If $u \in \overline{\mathbf{E}}^{\times} \cap \mathcal{F}$, both u and u^{-1} belong to $\overline{\mathbf{E}} \cap \mathcal{F} = A$. Thus $u \in A^{\times}$. The generators of A are algebraically independent with the indicated Laurent variables, so its units are exactly scalar Laurent monomials in the E_i, Z_l. Each such monomial and its inverse are E-values. This proves all the equalities. The example satisfies the required phase independence because 1 and $2\sqrt{2}$ are linearly independent over $\mathbb{Q}$. □

Fischler–Rivoal conjecture that every unit of $\mathbf{E}$ has the form αe^{β} with $\alpha \in k^{\times}$ and $\beta \in k$ [3, Conjecture 1.2]. Proposition 31.1 verifies the predicted description after intersection with the particular fields specified there. It does not classify all units of $\mathbf{E}$.

Proposition 31.2 (Pure radicals and monic polynomials)**.** *Let $F \subseteq \mathbb{C}$ be a field containing k, and suppose that an exact intersection $\overline{\mathbf{E}} \cap F = A$ is known. For $y \in \mathbb{C}$ and $d \geq 1$ with $f = y^d \in F$, one has*

$$y \in \overline{\mathbf{E}} \quad \Longleftrightarrow \quad f \in A, \tag{103}$$

$$y \in \overline{\mathbf{E}}^{\times} \quad \Longleftrightarrow \quad f \in A^{\times} \qquad (f \neq 0). \tag{104}$$

More generally, let $P(T) \in F[T]$ be monic of positive degree. All its roots belong to $\overline{\mathbf{E}}$ if and only if all its coefficients belong to A. All its roots belong to $\overline{\mathbf{E}}^{\times}$ if and only if its coefficients belong to A and its constant coefficient belongs to $A^{\times}$.

Proof. If y is integral over $\mathbf{E}$, so is y^d, giving the forward implication in (103). Conversely, $f \in A$ is integral over $\mathbf{E}$, and y satisfies the monic equation $T^d - f = 0$ over $\overline{\mathbf{E}}$. Transitivity of integrality makes y integral over $\mathbf{E}$. Applying the same argument to y^{-1} and f^{-1} proves (104).

If all roots of P are integral, their elementary symmetric functions are integral and lie in F, hence in A. Conversely, coefficients in A make every root integral over $\overline{\mathbf{E}}$, and then over $\mathbf{E}$. If all roots are units, their product makes the constant coefficient a unit of A. For the reverse implication, the reciprocal roots satisfy the reversed polynomial divided by its constant coefficient; this is again monic with coefficients in A. Thus the roots and their inverses are integral, proving the unit assertion. □

For clarity, define the set

$$\mathrm{Rad}(F) = \{y \in \mathbb{C} : y^d \in F \text{ for some integer } d \geq 1\}.$$

It is not being called a field. For the mixed field $\mathcal{F}$ of Proposition 31.1, the radical-unit conclusion becomes

$$\mathbf{E}^{\times} \cap \mathrm{Rad}(\mathcal{F}) = \overline{\mathbf{E}}^{\times} \cap \mathrm{Rad}(\mathcal{F}) = \left\{\alpha e^{\lambda} : \alpha \in k^{\times}, \quad \lambda \in \mathrm{span}_{\mathbb{Q}}\{a_1, \ldots, a_m, b_1, \ldots, b_r\}\right\}. \tag{105}$$

Indeed, a root of a unit in (102) is an algebraic multiple of the exponential with the exponent divided by its root degree, and every such number is an E-unit. The converse follows by clearing the rational denominators in the exponent.

Corollary 31.3 (Units and radicals at resonant positive scales)**.** *For positive algebraic a, t, put $F = k(\mathrm{Ein}(a), \beta(t), C(t))$. Then*

$$\mathbf{E}^{\times} \cap F = \overline{\mathbf{E}}^{\times} \cap F = k^{\times}, \qquad \overline{\mathbf{E}}^{\times} \cap \mathrm{Rad}(F) = k^{\times}.$$

If $y^d = f \in F$, then $y \in \overline{\mathbf{E}}$ if and only if $f \in k[\mathrm{Ein}(a)]$. If instead an exact intersection $\overline{\mathbf{E}} \cap F = k$ is known, then

$$\overline{\mathbf{E}} \cap \mathrm{Rad}(F) = k.$$

Proof. Theorem 26.1 gives $A = k[\mathrm{Ein}(a)]$, whose units are $k^\times$. Apply Proposition 31.2; a complex root of a nonzero algebraic number is again in $k^\times$. In the last assertion the same argument applies to $y^d \in k$, including zero. □

Proposition 31.4 (Integral units which are not E-values)**.** *Let $\eta = \mathrm{Ein}(1)$ and let y_+, y_- be the two roots of*

$$T^2 - \eta T + 1 = 0.$$

Then both roots are transcendental, both belong to $\overline{\mathbf{E}}^\times$, and neither belongs to $\mathbf{E}$. Explicitly,

$$y_\pm = \frac{\eta \pm \sqrt{\eta^2 - 4}}{2}.$$

Proof. The monic equation proves integrality, and $y_+ y_- = 1$ proves that both are units in $\overline{\mathbf{E}}$. If a root were algebraic, then $\eta = y + y^{-1}$ would be algebraic, contrary to the joint theorem.

Suppose a root y belonged to $\mathbf{E}$. The other root is $\eta - y$, so $y^{-1} \in \mathbf{E}$ as well. Its image under the zero-winding phase embedding is therefore a unit of $\mathcal{R} = \mathbb{C}[k_{\mathrm{add}}]$. The units of this group algebra are precisely $ce^{\rho X}$, with $c \in \mathbb{C}^\times$ and $\rho \in k$: the supports of a unit and its inverse generate a finite-rank free subgroup, and units of its Laurent polynomial ring are scalar monomials.

On the positive real branch the exponential-integral decomposition gives

$$\Phi_0(\eta) = \gamma + \delta e^{-X}.$$

Consequently $y + y^{-1} = \eta$ would imply

$$ce^{\rho X} + c^{-1}e^{-\rho X} = \gamma + \delta e^{-X}.$$

The two coefficients on the right are nonzero, since $\gamma, \delta > 0$. For $\rho = 0$ the left side is constant; for $\rho \neq 0$ its phase support is $\{\rho, -\rho\}$, which cannot equal $\{0, -1\}$. Linear independence of distinct exponential functions gives a contradiction in either case. □

This example also specifies the limit of (105): its equality of the two unit groups is a statement about pure radicals of the prescribed rational field. It does not extend to an arbitrary algebraic extension of that field.

Corollary 31.5 (Units with the matching exponential)**.** *Let a, t be arbitrary positive algebraic numbers and put*

$$F = k(e^a, \mathrm{Ein}(a), \beta(t), C(t)), \qquad k = \overline{\mathbb{Q}}.$$

Then

$$\mathbf{E}^\times \cap F = \overline{\mathbf{E}}^\times \cap F = \{\alpha e^{na} : \alpha \in k^\times,\ n \in \mathbb{Z}\},$$
$$\mathbf{E}^\times \cap \mathrm{Rad}(F) = \overline{\mathbf{E}}^\times \cap \mathrm{Rad}(F) = \{\alpha e^{qa} : \alpha \in k^\times,\ q \in \mathbb{Q}\}.$$

These conclusions require no independence condition on a and t. In particular they apply to $a = 1$, $t = 2$.

Proof. Theorem 27.1 gives $\overline{\mathbf{E}} \cap F = A = k[e^a, e^{-a}, \mathrm{Ein}(a)]$. The joint independence theorem makes A a Laurent polynomial ring in e^a and a polynomial variable $\mathrm{Ein}(a)$, so $A^\times = k^\times e^{a\mathbb{Z}}$. A unit and its inverse lie in A, which proves the first equality; the displayed monomials and their inverses are E-values. If $y \in \mathrm{Rad}(F)$ is an $\overline{\mathbf{E}}$-unit, choose $d \geq 1$ with $y^d \in F$. Then $y^d = \alpha e^{na}$ by the first part, whence $y = \alpha' e^{na/d}$ with $\alpha' \in k^\times$. Conversely, clearing the denominator of q shows that every αe^{qa} belongs to $\mathrm{Rad}(F)$, and it and its inverse are E-values. □

Part IV
Critical asymptotics

32 Critical factorial and Newton–Laguerre limits

32.1 The factorial limit and its first correction

For integers $n, m \geq 1$, define the rational numbers

$$f_k = \frac{(m+k)!}{k!^2(n-k)!^2}, \qquad h_k = H_{m+k} - 2H_k + 2H_{n-k},$$
$$r_{n,m} = \frac{\sum_{k=0}^{n} f_k h_k}{\sum_{k=0}^{n} f_k}. \tag{106}$$

The leading critical limit appears in manuscript [019][18]; it was also recorded as an observation in the parent intersection manuscript. The following proof supplies a stronger expansion with a uniform first correction.

Theorem 32.1. *Fix $0 < a_- < a_+ < \infty$. Uniformly for integers n, m satisfying $a_- \leq a = m/n^2 \leq a_+$, put $s = 1/a = n^2/m$. As $n \to \infty$,*

$$r_{n,m} = \gamma + \frac{2K_0(2\sqrt{s})}{I_0(2\sqrt{s})} + \frac{s-1}{nI_0(2\sqrt{s})^2} + O(n^{-2}). \tag{107}$$

Consequently, if $m/n^2 \to a_0 > 0$, then

$$r_{n,m} \longrightarrow \gamma + \frac{2K_0(2/\sqrt{a_0})}{I_0(2/\sqrt{a_0})}.$$

On the exact square diagonal,

$$r_{n,n^2} = \gamma + 2\kappa + O(n^{-2}). \tag{108}$$

Proof. Write $j = n - k$ and normalize by f_n. The exact weights are

$$w_{n,j} = \frac{f_{n-j}}{f_n} = \frac{s^j}{(j!)^2} \prod_{r=0}^{j-1} \frac{(1-r/n)^2}{1 + s/n - sr/n^2}, \qquad 0 \leq j \leq n. \tag{109}$$

In particular $0 \leq w_{n,j} \leq s^j/(j!)^2$. Uniformly for $j \leq n^{1/4}$, expanding the logarithm of the product yields

$$w_{n,j} = \frac{s^j}{(j!)^2}\left(1 - \frac{\mathcal{B}_j}{n} + O\left(\frac{(j+1)^4}{n^2}\right)\right), \qquad \mathcal{B}_j = j^2 + (s-1)j. \tag{110}$$

Indeed the logarithmic first term is $-[2\sum_{r<j} r + sj]/n = -\mathcal{B}_j/n$, and its remainder is $O((j+1)^3/n^2)$; exponentiation gives (110). The expansion $H_N = \log N + \gamma + 1/(2N) + O(N^{-2})$ gives, on the same range,

$$h_{n-j} = \log a - \gamma + 2H_j + \frac{s - 1 + 2j}{n} + O\left(\frac{(j+1)^2}{n^2}\right). \tag{111}$$

All implied constants may depend on a_-, a_+ but not on m, n, j.

We justify summation before computing coefficients. The weights are dominated by $B^j/(j!)^2$ for a fixed $B > 0$. Also

$$h_{n-j} = A_n + \Delta_{n,j}, \qquad A_n = H_{m+n} - 2H_n = O(1), \qquad |\Delta_{n,j}| \le 5H_j.$$

For example, $H_N - H_{N-j} \le H_j$ for $N \ge j$, which bounds each harmonic difference. The tails beyond $n^{1/4}$ of the dominating factorial series, multiplied by any fixed polynomial in j and by $1 + H_j$, are $O(n^{-M})$ for every fixed M. Thus (110)–(111) may be summed with error $O(n^{-2})$.

Set $D(s) = I_0(2\sqrt{s})$ and introduce the probability distribution

$$\Pr(J = j) = \frac{s^j}{(j!)^2 D(s)}, \qquad j \ge 0.$$

Expanding the numerator and denominator of (106) now gives

$$r_{n,m} = \log a - \gamma + 2\mathbb{E}H_J + \frac{s - 1 + 2\mathbb{E}J - 2\,\mathrm{Cov}(H_J, B_J)}{n} + O(n^{-2}). \tag{112}$$

The relevant moments are

$$\mathbb{E}J^2 = s, \qquad \mathrm{Cov}(H_J, J^2) = \mathbb{E}J, \qquad \mathrm{Cov}(H_J, J) = \frac{1 - D(s)^{-2}}{2}. \tag{113}$$

The first two follow from $j^2 s^j/(j!)^2 = s\, s^{j-1}/((j-1)!)^2$ and $H_{j+1} = H_j + 1/(j+1)$. For the third, differentiating the normalized series gives

$$\mathrm{Cov}(H_J, J) = s\frac{d}{ds}\frac{Y(2\sqrt{s})}{I_0(2\sqrt{s})}.$$

By (3) and (4),

$$\frac{d}{dt}\left(\frac{Y(t)}{I_0(t)}\right) = \frac{1}{t} - \frac{1}{tI_0(t)^2},$$

which proves the third identity in (113). The coefficient in (112) therefore simplifies to

$$s - 1 + 2\mathbb{E}J - 2\left(\mathbb{E}J + \frac{s-1}{2}(1 - D^{-2})\right) = \frac{s-1}{D^2}.$$

Finally, $\mathbb{E}H_J = Y(2\sqrt{s})/I_0(2\sqrt{s})$ and $\log a = -\log s$ turn the constant term in (112) into the leading term of (107). For $m = n^2$, $s = 1$, so its entire $1/n$ correction vanishes. □

Remark 32.2. In the uniform expansion, $s = n^2/m$ is the current parameter. If only $m/n^2 \to a_0$ is known, substituting $1/a_0$ for s while retaining the $O(n^{-2})$ error is not justified without a rate of convergence. Equation (108) has no such parameter error.

32.2 Connecting two actual rational approximation limits

Let

$$L_+ = \lim_{n\to\infty} r_{n,n^2} = \gamma + 2\kappa.$$

The parent manuscript's critical Van Assche–Wolfs limit is $L_- = \gamma - 2\kappa$ [1, Section 3.5]. For precision, its approximants are

$$b_{n,p,k} = \frac{\binom{n}{k}^2\binom{n+k}{k}^p}{k!}, \qquad T_{n,p} = \frac{\sum_{k=0}^n b_{n,p,k}((p+3)H_k - 2H_{n-k} - pH_{n+k})}{\sum_{k=0}^n b_{n,p,k}},$$

and the input identification is $\lim_{q\to\infty} T_{2^q,3q} = L_-$. The arithmetic consequences of the two mirror values are Corollary 15.2 and the preceding one-defect statement.

32.3 A distinct critical Newton–Laguerre limit

The following finite sums are the Newton–Laguerre approximants of [14].

Proposition 32.3 (Exact critical Newton–Laguerre limit)**.** *Let*

$$L_n(x) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{x^j}{j!}, \qquad R_N(x) = -\sum_{n=1}^{N} \frac{L_n(x)}{n}.$$

For every $\lambda > 0$,

$$R_N(\lambda/N) - \log(\lambda/N) \longrightarrow \mathcal{L}(\lambda) := \sum_{j\geq 1} \frac{(-1)^{j-1}\lambda^j}{j(j!)^2} - \gamma - \log\lambda.$$

In particular

$$\mathcal{L} := \mathcal{L}(1) = B - \gamma = 0.30662313951466900068840664486588416 91\ldots.$$

Proof. Since $\binom{n}{k}/n = \binom{n-1}{k-1}/k$, the hockey-stick identity gives

$$\sum_{n=k}^{N} \frac{\binom{n}{k}}{n} = \frac{1}{k}\binom{N}{k}.$$

Consequently

$$R_N(x) = -H_N + \sum_{k=1}^{N} \frac{(-1)^{k-1}\binom{N}{k}x^k}{k\,k!}.$$

At $x = \lambda/N$, the absolute value of each summand is at most $\lambda^k/[k(k!)^2]$, a summable sequence, and it converges to the corresponding signed term. Dominated convergence and $H_N - \log N \to \gamma$ give the stated limit. □

Part V
Second-kind Bessel values and density

33 Affine multiplicative circuits and second-kind Bessel values

All results of this section are deductions from Theorem 3.3. They are not assertions of historical priority. Write $k = \overline{\mathbb{Q}}$ and $K_{\exp} = k(e^\alpha : \alpha \in k)$. For distinct positive algebraic numbers $t_1, \ldots, t_m$, the part of Theorem 3.3 used here says that

$$I_j = I_0(t_j), \qquad C_j = I_1(t_j)/I_0(t_j), \qquad B_j = \beta(t_j) \quad (1 \leq j \leq m)$$

are jointly algebraically independent over $K_{\exp}$. The connection identity is

$$B_j = \gamma + \log(t_j/2) + \frac{K_0(t_j)}{I_j}. \tag{114}$$

Only positive arguments and real logarithms occur in this section.

For a finite set $T = \{t_1, \dots, t_m\}$, define its space of affine multiplicative circuits by

$$W_T = \left\{ q \in \mathbb{Q}^m : \sum_j q_j = 0, \quad \sum_j q_j \log t_j = 0 \right\}, \qquad d_T = m - \dim_{\mathbb{Q}} W_T.$$

The logarithmic condition is equivalently $\prod_j t_j^{q_j} = 1$. If the logarithms of the t_j span a rational vector space of dimension r, then $d_T \le r + 1$; d_T records the potentially sharper affine rank.

Theorem 33.1 (Affine rank bound)**.** *For every finite set T as above,*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(K_0(t) : t \in T) \ge |T| - d_T. \tag{115}$$

More precisely, if $q^{(1)}, \dots, q^{(m-d_T)}$ form a rational basis of W_T, then the numbers

$$R_a = \sum_{j=1}^{m} q_j^{(a)} \frac{K_0(t_j)}{I_0(t_j)} \qquad (1 \le a \le m - d_T) \tag{116}$$

are algebraically independent over $K_{\exp}(I_1, C_1, \dots, I_m, C_m)$.

Proof. Put $F = K_{\exp}(I_1, C_1, \dots, I_m, C_m)$. Theorem 3.3 makes the B_j algebraically independent over F. A matrix whose rows are the $q^{(a)}$ has full row rank, so its linear combinations of the B_j are algebraically independent over F: extend its rows to an invertible matrix and change coordinates. Both the common γ term and the logarithms in (114) cancel along each row. Consequently these linear combinations are precisely (116). They belong to $F(K_0(t_1), \dots, K_0(t_m))$, whence this field has transcendence degree at least $m - d_T$ over F. Extending the base field cannot increase the transcendence degree of a fixed tuple, proving (115). □

Corollary 33.2 (Restrictions on algebraic values)**.** *If all $K_0(t_j)$ are algebraic, then $W_T = 0$. Thus the set of positive algebraic arguments at which K_0 is algebraic contains no affine multiplicative circuit. Its intersection with a multiplicative group whose real logarithms have finite rational dimension r has at most $r + 1$ elements.*

Proof. Algebraicity makes the left side of (115) zero, forcing $m = d_T$. Any $r + 2$ points in the indicated group would have $d_T \le r + 1 < m$. □

Corollary 33.3 (Geometric progressions)**.** *Let $a, r > 0$ be algebraic, with $r \ne 1$. Among the values*

$$K_0(ar^n), \qquad n \in \mathbb{Z},$$

at most two are algebraic. For any $m \ge 2$ distinct integer exponents $n_1, \dots, n_m$,

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(K_0(ar^{n_1}), \dots, K_0(ar^{n_m})) \ge m - 2.$$

In particular, at least one of $K_0(2), K_0(4), K_0(8)$ is transcendental. More strongly, one can delete at most two values from the entire family indexed by $n \in \mathbb{Z}$ so that all the remaining values are algebraically independent over $K_{\exp}$.

Proof. On the hyperplane $\sum q_j = 0$, the logarithmic condition becomes $(\log r) \sum n_j q_j = 0$. Since $\log r \ne 0$ and the exponents are distinct, these two rational equations are independent and $d_T = 2$. Theorem 33.1 proves the rank bound; three algebraic values would contradict it. For three consecutive exponents the circuit is $(1, -2, 1)$. No invocation of Hermite–Lindemann is needed for this deduction from Theorem 3.3.

To obtain the last assertion, enumerate the integer exponents and greedily retain a value precisely when it is transcendental over the field of previously retained values. At each finite stage with m values the retained tuple is a transcendence basis of the generated field. At most two values have been rejected, because its rank is at least $m-2$. The total number of rejections can therefore never exceed two. The retained infinite family is algebraically independent. □

Corollary 33.4 (A finite-rank multiplicative coset)**.** *Let $G \subset k \cap \mathbb{R}_{>0}$ be a multiplicative group such that $\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\log g : g \in G\} = r < \infty$, and let $a > 0$ be algebraic. From any family of distinct arguments in aG one can delete at most $r+1$ arguments so that the remaining K_0 values are algebraically independent over $K_{\exp}$.*

Proof. Every finite set $T \subset aG$ has $d_T \le r+1$: on the hyperplane $\sum q_j = 0$, the common term $\log a$ cancels. Thus its K_0 values have rank at least $|T| - r - 1$. The family is countable since its arguments are algebraic. Apply the same greedy construction as in Corollary 33.3; at most $r+1$ rejections are possible. □

Theorem 33.5 (Density one and an independent subsequence)**.** *Let $\pi(N)$ be the number of primes at most the positive integer N. Then*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}\big(K_0(1), \dots, K_0(N)\big) \ge N - \pi(N) - 1. \tag{117}$$

In particular,

$$\#\{1 \le n \le N : K_0(n) \in k\} \le \pi(N) + 1,$$

so $K_0(n)$ is transcendental for a set of positive integers of natural density one. There is, moreover, a single subset $S \subseteq \mathbb{N}$ such that

$$\#(S \cap [1, N]) \ge N - \pi(N) - 1 \quad (N \ge 1) \tag{118}$$

and the entire family $\{K_0(n) : n \in S\}$ is algebraically independent over $K_{\exp}$. Thus S has natural density one.

Proof. The primes up to N generate a multiplicative group containing $1, \dots, N$. The augmented exponent vectors of 1 and of these primes are linearly independent, and all remaining exponent vectors belong to their span. Hence $d_{\{1,\dots,N\}} = \pi(N) + 1$, and Theorem 33.1 proves (117). Corollary 33.2, applied to the algebraic values in this interval, proves the counting bound.

For the stronger assertion, process $n = 1, 2, \dots$ in order. Include n in S exactly when $K_0(n)$ is transcendental over the field generated over $K_{\exp}$ by the previously included values. Every finite selected tuple is independent. Every rejected value is algebraic over the previously selected tuple, so at every stage N the selected tuple is a transcendence basis for the field generated by the first N values. Its size is the left side of (117); this proves (118). Finally $\pi(N) = o(N)$. This argument proves existence of S; it does not provide an effective test for membership or identify an individual transcendental value such as $K_0(2)$. □

33.1 The value at two as an additional witness

Put

$$I = I_0(2), \quad K = K_0(2), \quad \kappa = K/I, \quad C = I_1(2)/I, \qquad H_4 = k(\gamma, \delta, \kappa, C).$$

The relevant part of Theorem 3.3 says that

$$e,\ I,\ \eta = \mathrm{Ein}(1) = \gamma + \delta/e,\ C,\ \beta(2) = \gamma + \kappa \tag{119}$$

are algebraically independent over k.

Proposition 33.6 (A second-kind value as a rank witness)**.** *If $K_0(2)$ is algebraic over H_4, then*

$$\operatorname{trdeg}_k H_4 = 4.$$

In particular, algebraicity of $K_0(2)$ over k would close the original four-constant independence problem. Conversely, if the original four constants are algebraically dependent, then e and $K_0(2)$ are jointly algebraically independent over H_4. In that case $K_0(2)$ is transcendental even over H_4.

Proof. All five independent numbers in (119) belong to $H_4(e, I)$. If K is algebraic over H_4, then so is $I = K/\kappa$, since $\kappa > 0$. Therefore

$$5 \le \operatorname{trdeg}_k H_4(e, I) \le \operatorname{trdeg}_k H_4 + 1 \le 5,$$

forcing $\operatorname{trdeg}_k H_4 = 4$.

More generally (119) gives $\operatorname{trdeg}_k H_4 \ge 3$. If the original four are dependent, this degree is exactly three. The field $H_4(e, I)$ then has transcendence degree exactly five over k, and hence degree two over H_4. Thus e, I are independent over H_4. Replacing I by the nonzero H_4-multiple $K = \kappa I$ proves the assertion. □

Corollary 33.7 (Six-constant consequence)**.** *Using the complex-scale conclusion of Theorem 3.3, put*

$$\nu = -\frac{\pi}{2}\frac{Y_0(2)}{J_0(2)}, \qquad C_J = \frac{J_1(2)}{J_0(2)}.$$

If $K_0(2)$ is algebraic over H_4, then $\gamma, \delta, \kappa, C, \nu, C_J$ are jointly algebraically independent over $K_{\exp}$.

Proof. Write $H_6 = k(\gamma, \delta, \kappa, C, \nu, C_J)$ and $J = J_0(2)$. The seven numbers

$$\eta, I, C, \gamma + \kappa, J, C_J, \gamma + \nu$$

are independent over $K_{\exp}$. They belong to $K_{\exp}H_6(I, J)$, in which I is algebraic over $K_{\exp}H_6$ by the hypothesis. Consequently

$$7 \le \operatorname{trdeg}_{K_{\exp}} K_{\exp}H_6(I, J) \le \operatorname{trdeg}_{K_{\exp}} K_{\exp}H_6 + 1 \le 7.$$

The six generators of H_6 are therefore independent over $K_{\exp}$. □

34 A smaller circuit defect if the original four are dependent

The affine circuit theorem has a sharper relative form in the rank-three alternative. The improvement comes from putting the common constant γ in the coefficient field: circuit rows then need to cancel only logarithms, not their coefficient sum.

Put

$$k = \overline{\mathbb{Q}}, \qquad K_{\exp} = k(e^{\alpha} : \alpha \in k), \qquad H_4 = k(\gamma, \delta, \kappa, C), \qquad F = K_{\exp}H_4,$$

where $\kappa = K_0(2)/I_0(2)$ and $C = I_1(2)/I_0(2)$. Throughout this section assume

$$\operatorname{trdeg}_k H_4 = 3. \tag{120}$$

Theorem 3.3 supplies the independence, over $K_{\exp}$, of $\eta = \mathrm{Ein}(1)$ together with $I_0(t), C(t), \beta(t)$ at any finite set of distinct positive algebraic arguments. Recall that

$$\eta = \gamma + \delta/e, \qquad \beta(t) = \gamma + \log(t/2) + \frac{K_0(t)}{I_0(t)}.$$

All logarithms in this section are real logarithms of positive numbers.

Lemma 34.1 (The relative joint input in the deficient case)**.** *Under* (120), $\operatorname{trdeg}_{K_{\exp}} F = 3$. *Moreover,*

$$I_0(2), \qquad I_0(t),\ C(t),\ \beta(t) \quad (t \in k \cap \mathbb{R}_{>0},\ t \neq 2)$$

form an algebraically independent family over F, in the finite-subset sense. The same assertion holds with $I_0(2)$ replaced by $K_0(2)$.

Proof. The three independent values $\eta, \beta(2), C$ belong to F, because $e \in K_{\exp}$. Thus $\operatorname{trdeg}_{K_{\exp}} F \geq 3$. Base extension cannot increase the transcendence degree of the fixed tuple generating H_4, so (120) gives equality. For m distinct additional scales $t_j \neq 2$, the joint input gives $4 + 3m$ independent values over $K_{\exp}$:

$$\eta, \beta(2), C, I_0(2), \qquad I_0(t_j), C(t_j), \beta(t_j) \quad (1 \leq j \leq m).$$

They lie in a field obtained from F, of relative transcendence degree three over $K_{\exp}$, by adjoining the displayed $1 + 3m$ additional generators. The tower formula forces their relative transcendence degree over F to be exactly $1 + 3m$. Finally $K_0(2) = \kappa I_0(2)$, with $0 \neq \kappa \in F$, is an invertible change of coordinate over F. □

Theorem 34.2 (Logarithmic rank bound over the target field)**.** *Let T be a finite set of distinct positive algebraic numbers, and put*

$$r_2(T) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\log(t/2) : t \in T\}.$$

Under (120)*,*

$$\operatorname{trdeg}_F F\big(K_0(t) : t \in T\big) \geq |T| - r_2(T). \tag{121}$$

Proof. Put $K_2 = K_0(2)$, $F_0 = F(K_2)$, and $T' = T \setminus \{2\}$. Write $m = |T'|$ and enumerate its points as $t_1, \ldots, t_m$. The logarithms for T' have rank $r_2(T)$, since the removed logarithm, if present, was zero. By Lemma 34.1, the m values $\beta(t_j)$ are independent over

$$G = F_0\big(I_0(t_j), C(t_j) : 1 \leq j \leq m\big).$$

Choose a rational basis of the kernel of

$$\mathbb{Q}^m \longrightarrow \mathbb{R}, \qquad (q_j) \longmapsto \sum_j q_j \log(t_j/2).$$

This basis has $m - r_2(T)$ rows. Its linear combinations of the independent $\beta(t_j)$ remain independent over G. Subtracting $\gamma \sum_j q_j \in F \subset G$ from each combination gives the same number of independent residuals

$$\sum_j q_j \frac{K_0(t_j)}{I_0(t_j)}.$$

All lie in $G(K_0(t_1), \ldots, K_0(t_m))$. Hence

$$\operatorname{trdeg}_{F_0} F_0(K_0(t) : t \in T') \geq m - r_2(T),$$

using base-extension monotonicity as in the affine circuit argument. If $2 \notin T$, extending the base from F to F_0 cannot increase the rank of this tuple, and this proves (121). If $2 \in T$, the element K_2 contributes one independent dimension over F by Lemma 34.1. The tower formula gives $1 + m - r_2(T) = |T| - r_2(T)$, as required. □

Corollary 34.3 (A one-deletion geometric family)**.** *Let $r > 0$ be algebraic with $r \neq 1$. Under (120), one can delete at most one value from*

$$\{K_0(2r^n) : n \in \mathbb{Z}\}$$

so that the remaining entire family is algebraically independent over $F = K_{\exp}H_4$. The retained family can be chosen to include $K_0(2)$. At most one member of the displayed family is algebraic over F. More generally, if a multiplicative subgroup $G_0 \subset k \cap \mathbb{R}_{>0}$ has logarithmic rational rank r, then any family of distinct arguments in $2G_0$ has a deletion budget of at most r over F.

Proof. For every finite set of exponents, $\log((2r^n)/2) = n \log r$ spans a rational vector space of dimension at most one. Theorem 34.2 therefore gives a rank deficit of at most one for every finite subfamily. Enumerate $\mathbb{Z}$ starting with zero, retaining a value exactly when it is transcendental over the previously retained field over F. The first value $K_0(2)$ is retained by Lemma 34.1. At each finite stage, the retained values form a transcendence basis of all values seen so far: rejected values remain algebraic over the enlarging retained field. Thus at most one rejection can ever occur. Every finite retained tuple is independent, proving the infinite assertion. Two values algebraic over F would contradict the rank bound for that pair. For $2G_0$, use $r_2(T) \leq r$ and the identical greedy argument. The families are countable because their arguments are algebraic. □

Corollary 34.4 (A sharper density-one family containing the point two)**.** *Under (120), for every integer $N \geq 2$,*

$$\operatorname{trdeg}_F F(K_0(1), \ldots, K_0(N)) \geq N - \pi(N). \tag{122}$$

There is a single set $S \subset \mathbb{N}$, with $2 \in S$, such that

$$\#(S \cap [1, N]) \geq N - \pi(N) \quad (N \geq 2),$$

and $\{K_0(n) : n \in S\}$ is algebraically independent over $F = K_{\exp}H_4$. In particular S has natural density one. At most $\pi(N)$ of the first N values can be algebraic over F.

Proof. For $N \geq 2$, the rational span of $\log(n/2)$, $1 \leq n \leq N$, is exactly the span of $\log p$ for primes $p \leq N$. One inclusion follows by factorization; for the other, $\log(1/2) = -\log 2$ is available and $\log p = \log(p/2) + \log 2$. Unique factorization makes the prime logarithms rationally independent, so the rank is $\pi(N)$. Theorem 34.2 proves (122). Process the arguments in the order $2, 1, 3, 4, \ldots$, using the same greedy rule as above. The first value is retained, and for every $N \geq 2$ the first N processed arguments form exactly $\{1, \ldots, N\}$. Their retained count equals their transcendence degree, giving the stated bound. Since $\pi(N) = o(N)$, the density is one. For the final assertion, apply (121) to the subset of arguments whose values are algebraic over F; its logarithmic rank is at most $\pi(N)$. □

These are conditional statements throughout. They improve the unconditional affine deletion budget $r + 1$ to r on the centered coset $2G_0$, and the unconditional integer defect $\pi(N) + 1$ to $\pi(N)$, while enlarging the coefficient field to $K_{\exp}H_4$. They do not establish (120) for the actual constants, do not identify an unconditional individual transcendental $K_0(n)$, and give no effective membership test for S. Equivalently, a failure of any stated rank bound over $K_{\exp}H_4$ would force $\operatorname{trdeg}_k H_4 = 4$, using the already known lower bound three.

35 A common logarithmic defect for six families

The modified- and ordinary-Bessel blocks share the same logarithmic coordinate. Combining them with the exponential integral and the critical primitive limit therefore costs one common affine

logarithmic rank, not a separate rank for each family. The results below are consequences of the joint-value theorems already proved; no assertion of historical priority is intended.

For a positive algebraic number λ, write

$$h_\lambda = E_1^{\mathrm{tail}}(\lambda) = \int_\lambda^\infty \frac{e^{-u}}{u}\,du, \qquad g_\lambda = \gamma + \log\lambda, \tag{123}$$

$$B(\lambda) = \sum_{n\ge 1} \frac{(-1)^{n-1}\lambda^n}{n(n!)^2}, \qquad \mathcal{L}(\lambda) = B(\lambda) - g_\lambda. \tag{124}$$

The tail h_λ is not the scaled quantity $e^\lambda E_1^{\mathrm{tail}}(\lambda)$. Put

$$Z_\lambda = \big(h_\lambda, K_0(2\lambda), K_1(2\lambda), \pi Y_0(2\lambda), \pi Y_1(2\lambda), \mathcal{L}(\lambda)\big). \tag{125}$$

Here Y_r is the classical Bessel function of the second kind. For a finite set $T = \{\lambda_1, \dots, \lambda_m\}$ of distinct positive algebraic numbers, use the affine logarithmic rank

$$W_T = \left\{ q \in \mathbb{Q}^m : \sum_j q_j = 0, \quad \sum_j q_j \log\lambda_j = 0 \right\}, \qquad d_T = m - \dim_{\mathbb{Q}} W_T. \tag{126}$$

This agrees with the notation in Section 33. Multiplying all arguments by one fixed positive algebraic number leaves W_T and d_T unchanged.

Theorem 35.1 (Six families with one affine defect)**.** *For every finite set T as above,*

$$\mathrm{trdeg}_{K_{\exp}} K_{\exp}(Z_{\lambda_j} : 1 \le j \le m) \ge 6m - d_T. \tag{127}$$

In particular, each of the following two $4m$-tuples has transcendence degree at least $4m - d_T$ over $K_{\exp}$:

$$(h_{\lambda_j}, K_0(2\lambda_j), \pi Y_0(2\lambda_j), \mathcal{L}(\lambda_j))_{j=1}^m, \tag{128}$$

$$(K_0(2\lambda_j), K_1(2\lambda_j), \pi Y_0(2\lambda_j), \pi Y_1(2\lambda_j))_{j=1}^m. \tag{129}$$

The tuple $(K_0(2\lambda_j), K_1(2\lambda_j))_{j=1}^m$ has transcendence degree at least $2m - d_T$, and the same bound holds for the ordinary-Bessel pair with the factor π.

Proof. Set $t_j = 2\lambda_j$, $I_{r,j} = I_r(t_j)$, $J_{r,j} = J_r(t_j)$, $P_{r,j} = \pi Y_r(t_j)$, and $g_j = g_{\lambda_j}$. Theorems 3.3 and 4.3 show that the following $8m$ coordinates are jointly algebraically independent over $K_{\exp}$:

$$\big(\mathrm{Ein}(\lambda_j), I_{0,j}, I_{1,j}, \beta(t_j), J_{0,j}, J_{1,j}, \beta(it_j), B(\lambda_j)\big)_{j=1}^m. \tag{130}$$

Indeed, the Bessel scales t_j, it_j are distinct modulo sign, and the primitive scales $2i\sqrt{\lambda_j}$ are distinct modulo sign. A primitive scale is allowed to coincide with an old Bessel scale: the primitive theorem gives independence over all the old coordinates. Replacing the quotients C by I_1 or J_1 is an invertible rational change since $I_{0,j}$ and $J_{0,j}$ are nonzero.

The connection identities are

$$\mathrm{Ein}(\lambda_j) = g_j + h_{\lambda_j}, \qquad B(\lambda_j) = g_j + \mathcal{L}(\lambda_j), \tag{131}$$

$$\beta(t_j) = g_j + K_0(t_j)/I_{0,j}, \qquad \beta(it_j) = g_j - P_{0,j}/(2J_{0,j}). \tag{132}$$

The Wronskians, with their signs fixed by [2, §§10.5, 10.28], give

$$I_{0,j}K_1(t_j) + I_{1,j}K_0(t_j) = 1/t_j, \qquad J_{1,j}P_{0,j} - J_{0,j}P_{1,j} = 2/t_j. \tag{133}$$

Since $K_0(t_j) > 0$, adjoining $I_{0,j}$ suffices to recover $I_{1,j}$ from the modified-Bessel pair. If $P_{0,j} \neq 0$, adjoining $J_{0,j}$ suffices to recover $J_{1,j}$. If $P_{0,j} = 0$, then $P_{1,j} \neq 0$ and $J_{0,j} = -2/(t_j P_{1,j})$; in this case adjoin $J_{1,j}$ instead. Thus exactly one freely adjoined first-kind coordinate per Bessel block suffices, including the possible zero case.

Every vector in W_T gives a rational linear relation among the g_j. Consequently $\operatorname{trdeg}_{K_{\exp}} K_{\exp}(g_1, \dots, g_m) \leq d_T$. The field generated by (130) is contained in the field generated by the Z_{λ_j}, the g_j, and at most $2m$ additional first-kind coordinates. Hence

$$8m \leq \operatorname{trdeg}_{K_{\exp}} K_{\exp}(Z_{\lambda_j} : 1 \leq j \leq m) + d_T + 2m,$$

which proves (127). Deleting r entries from a tuple can lower its transcendence degree by at most r. Deleting $2m$ or $4m$ entries proves all the stated projections. □

Corollary 35.2 (No exceptional positive algebraic point)**.** *For every positive algebraic λ, the six entries of Z_λ have transcendence degree at least five over $K_{\exp}$, and*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(E_1^{\mathrm{tail}}(\lambda), K_0(2\lambda), \pi Y_0(2\lambda), \mathcal{L}(\lambda)) \geq 3. \tag{134}$$

If g_λ is algebraic over $K_{\exp}$, all six entries are algebraically independent over $K_{\exp}$; in particular the four entries in (134) are independent.

Proof. A singleton has $d_T = 1$. If g_λ is algebraic over $K_{\exp}$, the recovery proof costs no transcendence degree for that coordinate, and yields the stronger lower bound six. □

These conclusions identify neither a prescribed independent triple nor a prescribed independent five-tuple. In particular, they assert that at most one of the four, or of the six, displayed entries is algebraic over $K_{\exp}$. There is no exceptional λ in this statement. An exception count for algebraic $\gamma + \log \lambda$ is relevant to independent tuples that include that logarithmic coordinate, but is unnecessary after projecting to the four scalar outputs.

Lemma 35.3 (Integer-order changes of coordinates)**.** *Fix a positive algebraic t. For every integer $r \geq 0$, each of $K_r(t)$ and $\pi Y_r(t)$ is a nonzero k-linear form in its respective order-zero and order-one pair. Every adjacent pair of orders is obtained from that basic pair by an invertible k-linear transformation.*

Proof. The standard recurrences [2, §§10.6, 10.29] read

$$K_{r+2}(t) = K_r(t) + \frac{2(r+1)}{t} K_{r+1}(t), \qquad Y_{r+2}(t) = -Y_r(t) + \frac{2(r+1)}{t} Y_{r+1}(t).$$

Their transition matrices on adjacent pairs have determinants -1 and 1, respectively. Starting with orders zero and one, every finite product is therefore invertible over k. Each row is nonzero, which proves the assertion about individual linear forms as well. □

Theorem 35.4 (A simultaneous density-one family)**.** *For every positive integer N,*

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(Z_{n/2} : 1 \leq n \leq N) \geq 6N - \pi(N) - 1. \tag{135}$$

There exists a single set $S \subseteq \mathbb{N}$ such that

$$\#(S \cap [1, N]) \geq N - \pi(N) - 1 \qquad (N \geq 1) \tag{136}$$

and all six coordinates of $Z_{n/2}$, as n ranges over S, are jointly algebraically independent over $K_{\exp}$. In particular, for every $n \in S$ all the numbers

$$E_1^{\mathrm{tail}}(n/2), \quad \mathcal{L}(n/2), \quad K_r(n), \quad \pi Y_s(n) \qquad (r, s \geq 0) \tag{137}$$

are individually transcendental over $K_{\exp}$. For each selected n, either basic Bessel pair may be replaced by any adjacent pair of nonnegative integer orders, and the resulting six-coordinate family is still jointly independent. These choices may vary with n.

Proof. The affine logarithmic rank of $\{1, \ldots, N\}$ is $\pi(N) + 1$, as proved in Theorem 33.5. The common scaling by $1/2$ preserves that rank, so Theorem 35.1 proves (135).

Process the six coordinates of $Z_{1/2}$, then those of $Z_{2/2}$, and so on, in any fixed order within each block. Retain a coordinate if and only if it is transcendental over the field generated over $K_{\exp}$ by previously retained coordinates. At stage N the retained coordinates form a transcendence basis of the field generated by the first N blocks. Thus at most $\pi(N) + 1$ coordinates have been rejected. Define S to consist of the indices of blocks in which no coordinate was rejected. Each omitted block accounts for at least one rejected coordinate, proving (136). Every finite subset of the retained coordinates is independent, so the entire six-coordinate family on S is independent.

Lemma 35.3 gives invertible changes for adjacent pairs. It also expresses each higher-order value as a nonzero algebraic linear form in an independent pair. Such a form is transcendental over $K_{\exp}$, proving (137). Finally $\pi(N) = o(N)$ implies that S has natural density one. □

Corollary 35.5 (Counting all exceptional orders at once)**.** *Let $\mathcal{A}_N$ be the set of integers $1 \leq n \leq N$ for which at least one of the following conditions holds:*

1. *$E_1^{\mathrm{tail}}(n/2)$ or $\mathcal{L}(n/2)$ is algebraic over $K_{\exp}$;*
2. *$K_r(n)$ is algebraic over $K_{\exp}$ for some integer $r \geq 0$;*
3. *$\pi Y_s(n)$ is algebraic over $K_{\exp}$ for some integer $s \geq 0$.*

Then $\#\mathcal{A}_N \leq \pi(N) + 1$.

Proof. Put $m = \#\mathcal{A}_N$ and $T = \{n/2 : n \in \mathcal{A}_N\}$. At every such point, the six entries of $Z_{n/2}$ satisfy a nonzero polynomial relation over $K_{\exp}$. For the higher-order cases this follows from Lemma 35.3: substitute its nonzero linear form into a minimal polynomial over $K_{\exp}$. Hence the transcendence degree of the union of these m blocks is at most $5m$. On the other hand, Theorem 35.1 gives a lower bound $6m - d_T$. Therefore $m \leq d_T \leq \pi(N) + 1$, as required. The case $m = 0$ is immediate. □

Corollary 35.6 (A common finite deletion on a geometric progression)**.** *Let $a, q > 0$ be algebraic with $q \neq 1$. From the entire family of six-coordinate blocks Z_{aq^n}, $n \in \mathbb{Z}$, one can delete at most two blocks so that every remaining coordinate is jointly algebraically independent over $K_{\exp}$. At every remaining argument, the corresponding tail, critical limit, and all nonnegative integer orders of both second-kind Bessel families are individually transcendental over $K_{\exp}$.*

Proof. Every finite set of at least two distinct arguments in this progression has affine rank two. Enumerate the integer exponents and apply the coordinate-by-coordinate construction of Theorem 35.4. Theorem 35.1 permits at most two rejected coordinates in any finite stage, and hence at most two rejected blocks in the entire family. Lemma 35.3 supplies the last assertion. □

The results subsume the separate paired K-value bound and the separate ordinary-Bessel density bound while retaining only one common logarithmic defect. The infinite-order conclusion is individual transcendence, not independence of all orders at one point: the recurrences impose relations. Even replacing a basic ordinary-Bessel pair by two arbitrary nonadjacent orders need not be invertible. For example, at $t = \sqrt{8}$ one has $Y_3(t) = -4Y_0(t)/t$. The density-one set is existential; these arguments do not provide an effective membership test or identify a particular integer at which a prescribed second-kind value is transcendental.

Part VI
Further applications

36 A joint classification of two different rank defects

The linear-rank classification in [9] can now be combined with the Bessel independence statement. Let

$$\rho = \dim_k \operatorname{span}_k\{1, e, \delta, e\eta\}, \qquad H = k(\gamma, \delta, \kappa, C).$$

The linear rank ρ and the transcendence degree of H are different invariants. Since e, η are algebraically independent, $1, e, e\eta$ are linearly independent, and therefore $\rho \in \{3, 4\}$.

Theorem 36.1 (Rank-three linear defect versus joint algebraic defect)**.** *Assume $\rho = 3$. Then precisely the following alternatives hold:*

1. *γ is algebraic; the triple δ, κ, C is algebraically independent, and $\operatorname{trdeg}_k H = 3$.*
2. *δ is algebraic; the triple γ, κ, C is algebraically independent, and $\operatorname{trdeg}_k H = 3$.*
3. *Both γ and δ are transcendental; the four numbers $\gamma, \delta, \kappa, C$ are algebraically independent, and $\operatorname{trdeg}_k H = 4$.*

Consequently, if H has a nonzero polynomial relation and both γ and δ are transcendental, then $\rho = 4$.

Proof. The condition $\rho = 3$ and the independence of $1, e, e\eta$ give a unique triple $a, b, c \in k$ such that

$$\gamma = a + b/e + c\eta, \qquad \delta = (1 - c)e\eta - ae - b. \tag{138}$$

Conjugation and uniqueness make a, b, c real. Independence of e, η shows that $\gamma \in k$ occurs exactly when $b = c = 0$, and $\delta \in k$ exactly when $a = 0, c = 1$. The two cases are disjoint.

Outside them, eliminate η in (138) to obtain

$$e\big((1 - c)\gamma - a\big) = c\delta + b.$$

The denominator $(1 - c)\gamma - a$ is nonzero: otherwise either $c = 1, a = 0$, or $\gamma = a/(1 - c)$ is algebraic, both excluded. Thus

$$e = \frac{c\delta + b}{(1 - c)\gamma - a}, \qquad \eta = \gamma + \delta/e,$$

and $k(\gamma, \delta) = k(e, \eta)$. Write $\beta = \beta(2) = \gamma + \kappa$. Theorem 3.3 makes e, η, β, C algebraically independent. Therefore in the nonexceptional case

$$H = k(e, \eta, \beta, C),$$

which proves independence of its four displayed original generators.

If $\gamma = a \in k$, the birational substitution $\delta = e(\eta - a)$ shows that e, δ, β, C are independent; replacing β by $\kappa = \beta - a$ proves the first case. If $\delta = -b \in k$, the substitution $\gamma = \eta + b/e$ shows that e, γ, β, C are independent; replacing β by $\kappa = \beta - \gamma$ proves the second case. Finally, the last assertion is the contrapositive of case three, since $\rho \geq 3$. □

This excludes simultaneous *generic* defects of the two ranks. It does not establish $\rho = 4$ for the actual constants, does not establish a polynomial relation in H, and does not establish the individual transcendence of γ or δ. A generic linear rank-three defect is compatible with, and indeed forces, maximal joint algebraic independence.

37 An exact coefficient-field obstruction for a moment integral

The positive-parameter Hamiltonian moment model in [10] provides a different application of the four-constant rank bound. For $a = (a_1, \dots, a_n) \in \mathbb{R}^n_{>0}$, put

$$F_a(t) = \int_{[0,1]^n} \exp\left(t \sum_{j=1}^{n} a_j x_j \right) dx = \prod_{j=1}^{n} \frac{e^{a_j t} - 1}{a_j t}, \qquad F_a(0) = 1. \tag{139}$$

The singularities in this product at $t = 0$ are removable. The following criterion concerns the field of coefficients of a homogeneous scalar linear differential equation, rather than its order.

Lemma 37.1 (The visible phases generate the lengths)**.** *After combining equal subset sums, write*

$$G_a(t) := \prod_{j=1}^{n} (e^{a_j t} - 1) = \sum_{\lambda \in \Sigma_a} c_\lambda e^{\lambda t}, \qquad c_\lambda \in \mathbb{Z} \setminus \{0\}.$$

Then

$$\operatorname{span}_{\mathbb{Z}} \Sigma_a = \operatorname{span}_{\mathbb{Z}} \{a_1, \dots, a_n\}.$$

Proof. Each visible phase is a subset sum, giving one inclusion. For the reverse inclusion, induct on the distinct lengths b in increasing order. If the coefficient at exponent b survives, then $b \in \Sigma_a$. Otherwise the singleton subsets with length b contribute a nonzero integer of one common sign. Their cancellation therefore requires a subset of at least two lengths whose sum is b. Positivity makes every length in that subset strictly smaller than b; the induction hypothesis expresses each of them in the integer span of Σ_a, and hence does the same for b. This also establishes the initial case, since the smallest length cannot be canceled in this way. □

Theorem 37.2 (Exact coefficient-field criterion)**.** *Let $K \subset \mathbb{C}$ be a field containing $\mathbb{Q}$. The function F_a satisfies a nonzero homogeneous scalar linear differential equation over $K(t)$ if and only if every a_j is algebraic over K.*

Proof. Multiplication by t^n preserves the existence of such an equation: the corresponding operators are conjugate by t^n. Multiplication by a nonzero constant also preserves a homogeneous equation. Since $G_a = (\prod_j a_j) t^n F_a$, it is therefore enough to treat G_a; this argument does not require $\prod_j a_j \in K$.

Suppose that an equation exists. Clearing its denominators gives a nonzero operator

$$L = \sum_{j=0}^{r} p_j(t) \partial_t^j, \qquad p_j(t) \in K[t], \qquad LG_a = 0.$$

Distinct real exponentials are linearly independent over $\mathbb{C}[t]$. Indeed, in a putative relation, divide by the exponential of largest rate and let $t \to +\infty$: its polynomial coefficient tends to zero and must vanish; induction removes the remaining terms. Consequently

$$P(t,\lambda) := \sum_{j=0}^{r} p_j(t)\lambda^j = 0 \quad \text{identically in } t \quad (\lambda \in \Sigma_a).$$

Write the nonzero polynomial $P(t,X)$ as $\sum_d t^d R_d(X)$. Any nonzero $R_d \in K[X]$ vanishes at every $\lambda \in \Sigma_a$. Thus every visible phase is algebraic over K; Lemma 37.1 shows that every length is algebraic over K as well.

Conversely, if all lengths are algebraic over K, so are the finitely many visible phases. A nonzero polynomial $R \in K[X]$ vanishing on Σ_a exists, for example the product of their minimal polynomials. Then $R(\partial_t)G_a = 0$, and the nonzero operator $t^{-n}R(\partial_t)t^n \in K(t)[\partial_t]$ annihilates F_a. □

Corollary 37.3 (The four-constant coefficient cost)**.** *Let*

$$H = \overline{\mathbb{Q}}(\gamma,\delta,\kappa,C), \qquad \delta = \int_0^\infty \frac{e^{-u}}{1+u}\,du, \quad \kappa = \frac{K_0(2)}{I_0(2)}, \quad C = \frac{I_1(2)}{I_0(2)},$$

where γ is Euler's constant, with the rank bound $\operatorname{trdeg}_{\overline{\mathbb{Q}}} H \geq 3$ established in Theorem 6.1. For $a = (\gamma,\delta,\kappa,C)$, let $\mathcal{K}_a$ denote the fields $\overline{\mathbb{Q}} \subset K \subset \mathbb{C}$ over which F_a has a nonzero homogeneous linear differential equation. Then

$$\min_{K \in \mathcal{K}_a} \operatorname{trdeg}_{\overline{\mathbb{Q}}} K = \operatorname{trdeg}_{\overline{\mathbb{Q}}} H \in \{3,4\}. \tag{140}$$

In particular, F_a has no such equation over $\overline{\mathbb{Q}}(e,\pi)(t)$.

Proof. The four lengths are positive. For $K \in \mathcal{K}_a$, Theorem 37.2 says that the compositum HK is algebraic over K. Therefore

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} H \leq \operatorname{trdeg}_{\overline{\mathbb{Q}}} HK = \operatorname{trdeg}_{\overline{\mathbb{Q}}} K.$$

Taking $K = H$ realizes equality. The rank input and the four generators give the two possibilities in (140). Finally, $\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e,\pi) \leq 2$, which is too small; no assertion about the algebraic independence of e and π is needed. □

The conclusion is a coefficient-field obstruction for an explicit entire moment integral. An equation over $H(t)$, hence over $\mathbb{C}(t)$, still exists. This application does not increase the rank bound from three to four or settle the individual arithmetic nature of γ or δ.

38 Scope, unresolved questions, and source concordance

38.1 What the enlarged families do and do not decide

The exact ring intersections and the simultaneous primitive theorem close specific questions about rational expressions and relative independence. They do not establish

$$\operatorname{trdeg}_k k(\gamma,\delta,\kappa,C) = 4.$$

Nor do they establish the irrationality of either γ or δ. The phrase "at most one dependence" is literal: no nonzero relation among the original four is constructed, and no proof that all such

relations are absent is given. A rank witness is a sufficient condition, not evidence that the condition is actually satisfied.

An elementary model explains one logical limitation of dimension counting. Start with algebraically independent formal coordinates $E, \eta, I_j, C_j, \beta_j$, put $\gamma_0 = 1/2$, $\delta_0 = E(\eta - 1/2)$ and $\mu_j = \beta_j - 1/2$. The indicated joint-coordinate independence survives, while $k(\gamma_0, \delta_0, \mu_1, C_1)$ has transcendence degree three. This is a model of the algebraic input alone, not a model of the analytic definitions of the actual constants. Additional analytic or arithmetic information can therefore matter, but coordinate changes alone cannot settle the four-number target.

The following questions remain beyond the proofs in this manuscript:

1. the individual arithmetic nature of γ, δ, and any specified member of $K_0(2), K_0(4), K_0(8)$;
2. full algebraic independence of $\gamma, \delta, \kappa, C$;
3. full rational integral-intersection classifications with two or more harmonic Bessel coordinates at dependent scales, for example $k(\beta(2), C(2), \beta(4), C(4))$;
4. an effective criterion selecting the existential density-one independent set of integer arguments.

The old matched-phase question is resolved by Theorems 26.1 and 27.1:

$$\overline{\mathbf{E}} \cap k(\eta, \beta(2), C(2)) = k[\eta], \qquad \overline{\mathbf{E}} \cap k(e, \eta, \beta(2), C(2)) = k[e, e^{-1}, \eta].$$

The derivative-coordinate part of the dependent-scale question is also resolved, even in the presence of one harmonic coordinate (Theorem 29.1), together with the matching exponential. These closures concern integrality over the E-value ring; they do not remove the arithmetic questions about γ and δ listed above. In particular, adjoining e to a coefficient field that already contains $K_{\exp}$ does not add an independent coordinate. Also $2\,\mathrm{Chi}(1) = \mathrm{Ei}(1) - E_1^{\mathrm{tail}}(1)$, so these three classical values must not be counted as three independent coordinates.

38.2 Attribution and the restricted common-logarithm obstruction

Mahler's theorem [16, Theorem 3 and p. 170] proves the transcendence, for nonzero algebraic α and compatible branches, of

$$\frac{\pi Y_0(\alpha)}{2J_0(\alpha)} - \log(\alpha/2) - \gamma.$$

This includes the classical one-point context of the combinations used above. The present synthesis concerns joint families, integral intersections, relative fields, and the additional primitive coordinates; it does not reclassify a known one-point result as a new discovery.

The common-logarithm statement in manuscript [030] [15] concerns meromorphic germs in a specified field with the coordinate $T = \gamma + \log z$. Its finite-monodromy conclusion is a theorem about that prescribed germ field. It explicitly does not assert the analogous numerical conclusion after specialization at $z = 1$. Thus it is not a proof that every possible E-value method has a rank-three ceiling. The numerical identity $\gamma = \mathrm{Ein}(1) - \delta/e$ is already a finite algebraic combination; what remains missing is the necessary arithmetic information about its terms. No universal impossibility theorem is used in this paper.

The approximation results also have a precise scope. A rational sequence can converge to a rational, algebraic irrational, or transcendental number. The uniform $O(n^{-2})$ error on the square diagonal supplies neither sufficiently small integer linear forms nor the denominator estimates needed to infer irrationality.

38.3 Concordance of the five predecessor notes

This is a mathematical synthesis of the five notes below, with one notation system and internal proofs. It is not a concatenation of their abstracts. Superseded proofs are replaced by stronger theorems; distinct hypotheses and applications are retained.

Predecessor	Material retained or strengthened
Intersection extensions [7]	Uniform finite and infinite relation ideals; affine-slope alternatives; arbitrary-distinct-scale linear nonintegrality; uniform factorial asymptotics and the first correction; mirror-limit consequences.
Joint closures [11]	All-order primitive towers and solvable PV closure; conjugation and Hankel reconstruction; relative coefficient fields; linear-rank classification; exact moment-integral coefficient cost. The positive joint theorem is replaced by its proved complex version, and the mixed polynomial exclusion by the full rational theorem under the same scale independence hypothesis.
A, U, V addendum [12]	Exact relation ideal and rational-shift classification; polynomial integrality; all nontrivial Laurent monomials; mixed rational subfields. The former open $k(V)$ intersection is closed by the stronger $k(V, C)$ theorem.
Primitive and multiscale note [17]	The single primitive is strengthened to simultaneous primitives at every scale and every depth. The distinct Newton–Laguerre limit, its independent combinations, matched multiscale rational exclusions, general one-point polynomial classification, and the two root witnesses are retained.
Density and monodromy note [20]	Affine circuits; finite-deletion and density-one independence; second-kind Bessel witnesses; continued embeddings and one-point rational closures; independent-block equivalences and relative six-tuples. The weaker factorial limit proof is replaced by the uniform expansion.

The additional proofs in this revision include the all-depth primitive extension, complex multiscale rational intersections, the full Ein–Bessel intersection with its matching exponential, arbitrary derivative scales with one harmonic coordinate, exact unit and radical criteria, the finite real-adjunction defect, and a simultaneous six-family density theorem valid at every algebraic point before selecting a common integer subsequence. The earlier multivariate exponential-integral classification, reconstruction, and critical asymptotics are retained. These statements are research results of this manuscript and remain subject to external mathematical review; numerical checks are not substitutes for their displayed proofs.

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[17] Keio, *Bessel Primitives, Critical Laguerre Limits, and Multiscale E-Value Exclusion*, unpublished research note, October 2026, 11 pp.

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[20] Keio, *Density-One Independence and Monodromy Closures for Bessel Values*, unpublished research note, October 2026, 15 pp.